
 $^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39**

$^{33}\text{S}+1\text{p}\rightarrow^{34}\text{Cl}$ from $J^\pi=3/2^+$ ^{33}S ground state.

- 1983Wa27:** 0.4-2.0-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at Potchefstroom University, South Africa. The targets were Ag_2S (92%-enriched in ^{33}S) on Ta backings. γ rays were detected using a NaI detector and two Ge(Li) detectors. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced reaction Q values, J, π , 41 bound levels, 41 resonance strengths, and lifetimes for 17 bound levels using DSAM.
- 1977Da02:** 1.0-2.0-MeV proton beams were provided from the 4-MV Van de Graaff accelerator at CENBG, France. The targets were Ag_2S (92%-enriched in ^{33}S) on Au backings. γ rays were detected using a NaI detector and two Ge(Li) detectors. Measured E_γ , I_γ , and radiative capture excitation functions. Deduced reaction Q values, levels, 45 resonance strengths, γ -ray branching ratios, and lifetimes using DSAM.
- 1977Da03:** similar setup as **1977Da02**. Protons were detected using four Si(Li) surface barrier detectors. Measured excitation functions for (p,p) elastic scattering. Deduced γ multipole mixing ratios, resonance energies, J, and π from an R-matrix fit on (p,p) excitation functions.
- 1992Ka39:** 1.8-3.5-MeV proton beams impinged on a Ag_2S target. γ rays were detected using Ge(Li) and NaI(Tl) detectors. Measured E_γ , I_γ , and $\gamma(\theta)$. Deduced J, π , and γ multipolarities and multipole mixing ratios.
- 2020Pa40:** a 1.072-MeV proton beam was produced from the 5U 5-MV single-ended vertical Pelletron at the Nuclear Science Laboratory of University of Notre Dame. The target was prepared by implanting 70-keV negative ^{33}S ions into a tantalum foil. γ rays were detected with the compact detector array GEORGINA, consisting of five Ge detectors. Measured E_γ , I_γ , $\gamma\gamma$ -coin. Deduced precise γ -ray branching ratios for the 666-keV level.
- 2013Fa10:** Direct measurement of the $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction in inverse kinematics using 194-514 keV/nucleon ^{33}S beams produced from the Isotope Separator and Accelerator (ISAC) radioactive beam facility at TRIUMF. The beams impinged on a windowless gas target of H_2 gas on the DRAGON (Detector of Recoils And Gammas Of Nuclear reactions) recoil separator. γ rays from radiative captures were detected by 30 BGO detectors surrounding the target. Recoil products were separated by the DRAGON separator and detected in a double-sided silicon strip detector (DSSSD) at the end. Measured E_γ , reaction yields. Discovered two new resonances. Deduced resonance energies (E_r) and resonance strengths ($\omega\gamma$). Summarized in Conference proceedings **2010PaZT**.
- 2011Fr04:** Direct measurement of the $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction in normal kinematics using 200-710 keV proton beams provided from a tandem Van de Graaff accelerator at the University of Washington. Targets were prepared by implanting 45-keV 10^{16} ^{33}S ions into oxygen-free high-conductivity copper substrates. γ rays were detected using two HPGe detectors at $\pm 55^\circ$, surrounded by plastic scintillators operating in anticoincidence mode. Measured E_γ and I_γ for three resonances at 432, 492, and 529 keV. Deduced precise γ -ray branching ratios for the three resonances and the percentage of decays producing ^{34}Cl g.s. and $^{34\text{m}}\text{Cl}$ at 147 keV. Summarized in Conference proceedings **2010PaZT**.
- 1994Li20:** 0.973-1.997-MeV proton beams were provided from the AURA2 accelerator at the University of Auckland, New Zealand and impinged on Ag_2S targets 98%-enriched in ^{33}S . γ rays were detected by a 22% Ge(Li) at 90° relative to the beam direction and a NaI detector. Positrons from ^{34}Cl decay were detected by a plastic scintillator ΔE -E telescope. A separate measurement was performed using the proton beams from the 3 MeV Van de Graaff laboratory at the University of Florence, Italy. γ rays were detected using an 80% intrinsic Ge detector. Measured E_γ , I_γ , and positron yields. Deduced $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction Q values, S(p) of ^{34}Cl , and precise excitation energies for three resonances.
- 1985La16:** 1.05-1.98-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a PGT 110-cm³ Ge(Li) detector. Deduced lifetimes for 35 bound levels using DSAM.
- 1984Ke01:** 0.82- and 1.43-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a PGT Ge(Li) detector. Deduced lifetimes for two bound levels using DSAM.
- 1983Ra04:** 0.9-1.4-MeV proton beams impinged on an 88.2%-enriched ^{33}S target (CdS on carbon backing). γ rays were detected using a NaI detector at 90° relative to the beam direction and protons were detected using a Si surface barrier detector at 135° relative to the beam direction. Measured E_γ , I_γ , reaction Q, and S(p).
- 1973An13:** 1.3-1.8-MeV proton beams were provided from the 2.5-MV van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a 40-cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced levels and lifetimes using DSAM.
- 1971Hy02:** 1058, 1098, 1121-keV proton beams were provided from the Aerospace Research Laboratories 2-MeV Van de Graaff accelerator and impinged on an 84%-enriched ^{33}S target (Ag backing soldered on brass disk). γ rays were detected using a Ge(Li) detector and a NaI detector. Measured E_γ , I_γ , $\text{py}(\theta)$, and $\text{p}\gamma\gamma(\theta)$. Deduced levels, J, π , resonance strengths for three resonances, γ multipolarities, and multipole mixing ratios.

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39 (continued)**

1969Gr29: 1.0-1.3-MeV proton beams impinged on a 25%-enriched ^{33}S target. γ rays were detected using a Ge(Li) detector at 0° and 115° or 120° relative to the beam direction. Measured E_γ , I_γ , $\text{p}\gamma(\theta)$, and $\text{p}\gamma\gamma(\theta)$. Deduced levels, J , π , γ multipolarities, multipole mixing ratios, and lifetimes using DSAM.

1964Gl04: 0.3-1.3-MeV proton beams impinged on a 22%-enriched ^{33}S target (CdS on Cu or Ta backings). γ rays were detected using NaI detectors at $+85^\circ$ and -85° relative to the beam direction. Measured E_γ , I_γ , $\gamma\gamma$ -coin for nine resonances ($E_{\text{p}}(\text{lab})=447$, 507, 546, 639, 662, 683, 731, 777, 822 keV). Deduced resonance strengths.

1959Ku79: protons were accelerated with the Utrecht 850-kV Cockcroft-Walton generator and impinged on a CdS target enriched in ^{33}S . Measured γ -ray yield curves. Deduced two resonances.

Standard definition of resonance strength $\omega\gamma=(2J_{\text{r}}+1)/[(2J_{\text{p}}+1)(2J_{\text{T}}+1)]\times\Gamma_{\gamma}\Gamma_{\text{p}}/\Gamma$, where J_{r} is the spin of the resonance, $J_{\text{p}}=1/2$ is the spin of the proton, and $J_{\text{T}}=3/2$ is the spin of the target ^{33}S ground state. The resonance strengths S from the original papers in this dataset are mainly obtained from resonance γ -ray yields and are defined without the $(2J_{\text{p}}+1)(2J_{\text{T}}+1)$ factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 8, except for $\omega\gamma$ values from **2013Fa10**, which are already in the standard definition.

 ^{34}Cl Levels

$E(\text{level})^{\dagger\ddagger}$	$J^{\pi\#}$	$T_{1/2}$	Comments
0	0^+		
146.4 1	3^+		E(level): weighted average of 146.5 5 (1977Da02), 146.4 1 (1973An13), and 146.8 10 (1969Gr29).
461.2 2	1^+		E(level): weighted average of 461.2 5 (1977Da02), 461.0 2 (1973An13), and 461.5 3 (1969Gr29).
665.3 2	1		E(level): weighted average of 665.7 5 (1977Da02), 665.6 2 (1973An13), and 664.6 3 (1969Gr29).
1230.33 8	2^+		J^π : from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1266$ in 1969Gr29 . E(level): weighted average of 1230.5 3 (1977Da02), 1230.33 8 (1983Wa27), 1230.5 5 (1973An13), and 1228.8 7 (1969Gr29).
1887.29 8	2^+		E(level): weighted average of 1887.2 3 (1977Da02), 1887.31 8 (1983Wa27), 1887.0 5 (1973An13), and 1885.9 16 (1969Gr29).
2157.9 1	2^+	32.7 fs 23	E(level): weighted average of 2158.1 3 (1977Da02), 2157.90 8 (1983Wa27), 2157.9 1 (1973An13), and 2158.4 12 (1969Gr29).
2181.10 8	3	0.45 ps 11	$T_{1/2}$: lifetime $\tau=47$ fs 3: weighted average of 36 fs 15 (1977Da02 , DSAM), 46 fs 10 (1983Wa27 , DSAM), 47 fs 5 (1984Ke01 , DSAM), and 49 fs 5 (1985La16 , DSAM). E(level): weighted average of 2181.2 5 (1977Da02), 2181.10 8 (1983Wa27), and 2179.9 14 (1973An13).
2375.7 2	4	139 fs 19	J^π : from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1545$ in 1977Da03 . Other: spin=(3) from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1098$ in 1971Hy02 . $T_{1/2}$: lifetime $\tau=649$ fs 154: weighted average of 600 fs 350 (1977Da02 , DSAM), 630 fs 190 (1983Wa27 , DSAM), and 800 fs 400 (1985La16 , DSAM). E(level): weighted average of 2375.6 5 (1977Da02), 2375.7 2 (1983Wa27), 2368 4 (1973An13), 2377.3 12 (1971Hy02), and 2375.6 7 (1969Gr29).
2580.3 2		2.1 fs 17	J^π : from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1165$ in 1969Gr29 . $T_{1/2}$: lifetime $\tau=201$ fs 27: weighted average of 0.15 ps 5 (1973An13 , DSAM), 210 fs 80 (1977Da02 , DSAM), 200 fs 50 (1983Wa27 , DSAM), and 250 fs 50 (1985La16 , DSAM). E(level): weighted average of 2579.9 5 (1977Da02), 2580.4 2 (1983Wa27), 2580.4 5 (1973An13), and 2580.6 13 (1969Gr29).
2611.03 11	3	1.6×10^2 fs 4	$T_{1/2}$: lifetime $\tau=3.0$ fs 25 (1977Da02 , DSAM). Others: <5 fs (1983Wa27 , DSAM) and <7 fs (1985La16 , DSAM). E(level): weighted average of 2610.6 5 (1977Da02), 2611.05 11 (1983Wa27), 2611.1 5 (1973An13), and 2611.2 13 (1971Hy02).
2721.3 2	2	>0.277 ps	J^π : from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1057$ in 1977Da03 . $T_{1/2}$: lifetime $\tau=230$ fs 55: weighted average of 210 fs 60 (1983Wa27 , DSAM) and 340 fs 140 (1985La16 , DSAM). Others: >450 fs (1977Da02 , DSAM). E(level): weighted average of 2721.4 7 (1977Da02), 2721.1 2 (1983Wa27), 2721 1 (1973An13), 2721.7 3 (1971Hy02), and 2720.4 16 (1969Gr29).
			J^π : from $\gamma\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1098$ in 1971Hy02 .

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	T _{1/2}	Comments
3129.12 12		1.9 fs 13	T _{1/2} : lifetime $\tau > 400$ fs (1977Da02, DSAM). Others: > 1.0 ps (1973An13, DSAM) and > 1500 fs (1985La16, DSAM). E(level): weighted average of 3128.9 5 (1977Da02), 3129.13 12 (1983Wa27), and 3129 2 (1973An13). T _{1/2} : lifetime $\tau = 2.4$ fs 19: weighted average of < 10 fs (1983Wa27, DSAM) and < 4 fs (1985La16, DSAM).
3334.0 2		37 fs 14	E(level): weighted average of 3333.9 10 (1977Da02) and 3334.0 2 (1983Wa27). T _{1/2} : lifetime $\tau = 53$ fs 20 (1985La16, DSAM).
3383.3 2		5.4 fs 11	E(level): weighted average of 3383.6 6 (1977Da02) and 3383.3 2 (1983Wa27). T _{1/2} : lifetime $\tau = 7.7$ fs 16: weighted average of 7.5 fs 17 (1984Ke01, DSAM) and 9 fs 4 (1985La16, DSAM). Others: < 3 fs (1977Da02, DSAM) and < 10 fs (1983Wa27, DSAM).
3545.08 15	3	102 fs 17	E(level): weighted average of 3545.2 8 (1977Da02), 3545.07 15 (1983Wa27), 3545 1 (1973An13), 3545.3 6 (1971Hy02), and 3544.6 13 (1969Gr29). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1057, 1545, and 1997 in 1977Da03 and from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1098 in 1971Hy02. T _{1/2} : lifetime $\tau = 147$ fs 25: weighted average of 140 fs 45 (1983Wa27, DSAM) and 150 fs 30 (1985La16, DSAM). Other: > 0.5 ps (1973An13, DSAM).
3600.28 11	4	> 0.485 ps	E(level): weighted average of 3600.3 8 (1977Da02), 3600.27 11 (1983Wa27), 3600.2 3 (1973An13), 3600.7 4 (1971Hy02), and 3598.4 12 (1969Gr29). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1545 and 1997 in 1977Da03 and $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1098 in 1971Hy02. T _{1/2} : lifetime $\tau > 0.7$ ps (1973An13, DSAM).
3631.7 3	5	> 0.485 ps	E(level): weighted average of 3631.1 8 (1977Da02), 3631.8 3 (1983Wa27), 3631.6 3 (1973An13), and 3632.5 14 (1971Hy02). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03. T _{1/2} : lifetime $\tau > 700$ fs (1977Da02, DSAM). Others: > 1100 fs (1985La16, DSAM).
3646.3 2		1.5×10^2 fs 6	E(level): weighted average of 3645.6 15 (1977Da02) and 3646.3 2 (1983Wa27). T _{1/2} : lifetime $\tau = 220$ fs 90 (1985La16, DSAM). Another I _γ =33 decays to unknown final levels (1983Wa27).
3660.0 3		< 3.5 fs	E(level): weighted average of 3659.4 15 (1977Da02) and 3660.0 3 (1983Wa27). T _{1/2} : lifetime $\tau < 5$ fs (1985La16, DSAM).
3773.72 41	1	52 fs 18	E(level): weighted average of 3773.9 10 (1977Da02) and 3773.84 15 (1983Wa27), and 3771.0 7 (1971Hy02). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02. T _{1/2} : lifetime $\tau = 75$ fs 26: weighted average of 55 fs 15 (1983Wa27, DSAM) and 110 fs 20 (1985La16, DSAM). Another I _γ =30 decays to unknown final levels (1983Wa27).
3791.7 2			E(level): from 1983Wa27. Another I _γ =18 decays to unknown final levels (1983Wa27).
3940.1 3		< 4 fs	E(level): weighted average of 3941.2 15 (1977Da02) and 3940.1 3 (1983Wa27). T _{1/2} : lifetime $\tau < 6$ fs (1985La16, DSAM).
3964.1 3		90 fs 32	E(level): weighted average of 3964.2 15 (1977Da02) and 3964.1 3 (1983Wa27). T _{1/2} : lifetime $\tau = 130$ fs 46: weighted average of 160 fs 60 (1977Da02, DSAM) and 90 fs 70 (1985La16, DSAM).
3983.0 3	3	131 fs 23	E(level): weighted average of 3982.2 5 (1977Da02), 3983.5 2 (1983Wa27), 3983 1 (1973An13), 3982.1 3 (1971Hy02), and 3982.0 11 (1969Gr29). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1058 and 1098 in 1971Hy02 and from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1057 and 1997 in 1977Da03. T _{1/2} : lifetime $\tau = 189$ fs 33: weighted average of 165 fs 60 (1983Wa27, DSAM) and 200 fs 40 (1985La16, DSAM). Others: > 0.1 ps (1973An13, DSAM).
4076.2 2	4	> 173 fs	E(level): weighted average of 4075.9 10 (1977Da02), 4076.3 2 (1983Wa27), 4074 12 (1973An13), and 4075.4 9 (1971Hy02). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1058 and 1098 in 1971Hy02 and from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1545 in 1977Da03. T _{1/2} : lifetime $\tau > 0.25$ ps (1973An13, DSAM). Other: > 500 fs (1977Da02, DSAM).
4139.7 2	2	90 fs 23	E(level): weighted average of 4139.9 15 (1977Da02) and 4139.8 2 (1983Wa27), and 4136.6

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	T _{1/2}	Comments
			12 (1971Hy02). J ^π : spin=2,3 from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02 and the authors ruled out J=3 by RUL. T _{1/2} : lifetime τ =129 fs 33: weighted average of 120 fs 40 (1983Wa27, DSAM) and 150 fs 60 (1985La16, DSAM).
4147.8 2 4325.89 15		5.1 fs 20	E(level): weighted average of 4325.6 5 (1977Da02) and 4325.91 15 (1983Wa27). T _{1/2} : lifetime τ =7.4 fs 28: weighted average of 11 fs 9 (1977Da02, DSAM) and 7 fs 3 (1985La16, DSAM). Another I γ =25 decays to unknown final levels (1983Wa27).
4354.2 2	(1)	30 fs 6	E(level): weighted average of 4352.9 10 (1977Da02) and 4354.3 2 (1983Wa27), and 4352.7 9 (1971Hy02). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02. T _{1/2} : lifetime τ =43 fs 8 (1985La16, DSAM). Other: <10 fs (1983Wa27, DSAM).
4417.3 2		24 fs 8	E(level): weighted average of 4416.4 5 (1977Da02), 4417.4 2 (1983Wa27), and 4415.8 23 (1971Hy02). T _{1/2} : lifetime τ =35 fs 11 (1985La16, DSAM).
4446.6 2	4	116 fs 19	E(level): weighted average of 4444.7 15 (1977Da02) and 4446.6 2 (1983Wa27). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03. T _{1/2} : lifetime τ =167 fs 27: weighted average of 160 fs 60 (1977Da02, DSAM), 180 fs 60 (1983Wa27, DSAM), and 165 fs 35 (1985La16, DSAM).
4461.4 3		0.26 ps 17	E(level): weighted average of 4462.2 15 (1977Da02) and 4461.4 3 (1983Wa27). T _{1/2} : lifetime τ =370 fs 250 (1985La16, DSAM).
4515.7 2		10 fs 3	E(level): weighted average of 4515.3 5 (1977Da02), 4515.8 2 (1983Wa27), and 4514.5 7 (1971Hy02). T _{1/2} : lifetime τ =15 fs 4: weighted average of 15 fs 7 (1983Wa27, DSAM) and 15 fs 5 (1985La16, DSAM).
4605.8 10		0.24 ps 17	E(level): from 1977Da02. T _{1/2} : lifetime τ =350 fs 240 (1985La16, DSAM).
4609.7 15 4638.9 4	1,2	31 fs 10	E(level): from 1977Da02. E(level): weighted average of 4638.3 15 (1977Da02) and 4638.9 4 (1983Wa27), and 4639.0 23 (1971Hy02). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02. T _{1/2} : lifetime τ =45 fs 14 (1985La16, DSAM).
4695.7 2		6 fs 3	E(level): weighted average of 4695 2 (1977Da02) and 4695.7 2 (1983Wa27). T _{1/2} : lifetime τ =9 fs 4 (1985La16, DSAM). Other: <10 fs (1983Wa27, DSAM).
4717.4 6		<9 fs	E(level): weighted average of 4717.5 10 (1977Da02) and 4717.4 6 (1983Wa27). T _{1/2} : lifetime τ <13 fs (1985La16, DSAM).
4824.41 18		0.33 ps 15	E(level): weighted average of 4823.4 10 (1977Da02), 4824.5 2 (1983Wa27), and 4824 10 (1973An13). T _{1/2} : lifetime τ =480 fs 220 (1983Wa27, DSAM). Other: >500 fs (1985La16, DSAM).
4941.9 4		<6 fs	E(level): weighted average of 4941.4 15 (1977Da02) and 4941.9 4 (1983Wa27). T _{1/2} : lifetime τ <8 fs (1985La16, DSAM).
4957.3 11		<12 fs	E(level): weighted average of 4959 2 (1977Da02) and 4957.3 11 (1983Wa27). T _{1/2} : lifetime τ <17 fs (1985La16, DSAM).
4995.6 3		5 fs 3	E(level): weighted average of 4993 2 (1977Da02) and 4995.6 3 (1983Wa27). T _{1/2} : lifetime τ =7 fs 4 (1985La16, DSAM). Other: <5 fs (1983Wa27, DSAM). Another I γ =43 decays to unknown final levels (1983Wa27).
5020? 5171.6 3	4	50 fs 16	E(level): from 1977Da02. E(level): weighted average of 5170 2 (1977Da02) and 5171.6 3 (1983Wa27). J ^π : from $\gamma\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03. T _{1/2} : lifetime τ =73 fs 23: weighted average of 60 fs 30 (1977Da02, DSAM) and 90 fs 35 (1985La16, DSAM).
5326			Observed in 2013Fa10 and E _r =183 is quoted. $\omega\gamma$ <1.42 μeV (2013Fa10).
5357			Observed in 2013Fa10 and E _r =214 is quoted.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	T _{1/2}	E _p (lab) (keV) [@]	Comments
5386.8 15				$\omega\gamma < 10.5 \mu\text{eV}$ (2013Fa10). E(level): weighted average of 5383 2 (1977Da02) and 5386.8 15 (1983Wa27). Also observed in 2013Fa10 and E _r =243.6 is quoted.
5424				$\omega\gamma < 21.8 \mu\text{eV}$ (2013Fa10). Observed in 2013Fa10 and E _r =281 is quoted.
5436? 2				$\omega\gamma < 10.1 \mu\text{eV}$ (2013Fa10). E _r =293 keV 2 (2013Fa10).
5454? 2				$\omega\gamma = 82 \mu\text{eV} + 18-21$ (2013Fa10). E _r =311 keV 2 (2013Fa10).
5485				$\omega\gamma = 28 \mu\text{eV} + 10-8$ (2013Fa10). Observed in 2013Fa10 and E _r =342 is quoted.
5540.8 11		>0.7 ps		$\omega\gamma < 14.2 \mu\text{eV}$ (2013Fa10). E(level): weighted average of 5540 2 (1977Da02) and 5540.8 11 (1983Wa27). T _{1/2} : lifetime $\tau > 1000$ fs (1977Da02, DSAM). Observed in 2013Fa10 and E _r =399 is quoted.
5577.6 10			447.0 10	$\omega\gamma < 3.99 \mu\text{eV}$ (2013Fa10). E(level): from E _{c.m.} =434.4 10: weighted average of E _{c.m.} =433.7 10 (1959Ku79,1964GI04) and E _{c.m.} =435 1 (2013Fa10). E _p (lab) (keV): weighted average of 446.5 35 (1959Ku79) and 447 1 (1964GI04), corresponding to E _{c.m.} =433.7 10. $\omega\gamma = 50 \text{ meV} + 10-7$ (2013Fa10). $\omega\gamma = 0.4 \text{ eV}$ 1 (1983Wa27).
5635.75 20	1,2		507.6 2	E(level): from E _{c.m.} =492.51 20: weighted average of 492.1 10 (1959Ku79), 492.0 10 (1964GI04), 492.55 20 (1983Wa27), and 491 keV 5 (2013Fa10). J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 507.1 10 (1959Ku79), 507 1 (1964GI04), and 507.6 2 (1983Wa27). $\omega\gamma = 0.16 \text{ eV}$ 8 (1964GI04). $\omega\gamma = 0.7 \text{ eV}$ 2 (1983Wa27). $\omega\gamma = 61 \text{ meV} + 11-8$ (2013Fa10).
5673.0 10			546 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.040 \text{ eV}$ 20 (1964GI04). $\omega\gamma = 0.7 \text{ eV}$ 3 (1983Wa27).
5763.3 10			639 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.011 \text{ eV}$ 6 (1964GI04). $\omega\gamma = 0.06 \text{ eV}$ 3 (1983Wa27).
5785.6 10			662 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.034 \text{ eV}$ 17 (1964GI04). $\omega\gamma = 0.4 \text{ eV}$ 2 (1983Wa27).
5805.9 10			683 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.05 \text{ eV}$ 3 (1964GI04). $\omega\gamma = 0.4 \text{ eV}$ 2 (1983Wa27).
5852.91 30			731.4 3	E _p (lab) (keV): weighted average of 731 1 (1964GI04) and 731.4 3 (1983Wa27). $\omega\gamma = 0.05 \text{ eV}$ 3 (1964GI04). $\omega\gamma = 0.5 \text{ eV}$ 2 (1983Wa27).
5868.0 10			747 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.010 \text{ eV}$ 5 (1964GI04).
5897.2 10			777 1	E _p (lab) (keV): from 1964GI04. $\omega\gamma = 0.05 \text{ eV}$ 3 (1964GI04). $\omega\gamma = 0.5 \text{ eV}$ 2 (1983Wa27).
5940.8 10	1 to 3		822 1	J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): from 1964GI04.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	T _{1/2}	E _p (lab) (keV) [@]	Comments
6030.1 20			914 2	$\omega\gamma=0.12$ eV 6 (1964Gl04). $\omega\gamma=0.8$ eV 2 (1983Wa27). E _p (lab) (keV): from 1964Gl04. $\omega\gamma=0.050$ eV 25 (1964Gl04). $\omega\gamma=0.4$ eV 2 (1983Wa27).
6088.94 6	1,2		974.62 4	E(level): weighted average of E(level)=6090.3 20 deduced at E _p (lab)=976 2 (1964Gl04), E(level)=6089.12 16 deduced at E _p (lab)=974.83 16 measured by 1983Ra04, E(level)=6088.90 6 deduced from E _p (lab)=974.61 4 measured by 1994Li20, and E(level)=6088.97 8 also by 1994Li20 based on γ data without reporting E γ . J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 976 2 (1964Gl04), 974.83 16 (1983Ra04), and 974.61 4 (1994Li20). $\omega\gamma=1.0$ eV 3 (1983Wa27). $\omega\gamma=0.13$ eV 7 (1964Gl04). J ^π : from $\gamma(\theta)$ in 1983Wa27.
6136.2 11	1 to 3		1023.3 11	E _p (lab) (keV): weighted average of 1023 2 (1964Gl04) and 1023.4 11 (1977Da02). $\omega\gamma=0.02$ eV 1 (1964Gl04). $\omega\gamma=0.7$ eV 2 (1983Wa27). $\omega\gamma=0.6$ eV 2 (1977Da02). J ^π : from $\gamma(\theta)$ in 1983Wa27.
6141.8 11	1,2		1029.1 11	E _p (lab) (keV): weighted average of 1029 2 (1964Gl04) and 1029.1 11 (1977Da02). $\omega\gamma=0.15$ eV 8 (1964Gl04). $\omega\gamma=1.1$ eV 3 (1983Wa27). $\omega\gamma=0.9$ eV 5 (1977Da02). J ^π : from $\gamma\gamma(\theta)$ in 1971Hy02 and 1977Da03. Other: spin=2:5 from $\gamma(\theta)$ in 1983Wa27.
6169.4 11	3		1057.6 5	E _p (lab) (keV): weighted average of 1057 2 (1964Gl04), 1057.3 11 (1977Da02), and 1057.7 5 (1971Hy02). $\omega\gamma=0.45$ eV 23 (1964Gl04). $\omega\gamma=1.8$ eV 5 (1983Wa27). $\omega\gamma=1.5$ eV 7 (1977Da02). $\omega\gamma=3.6$ eV 8 (1969Gr29). $\omega\gamma=2.8$ eV 9 (1971Hy02).
6181.27 20	1,2		1069.8 2	J ^π : from $\gamma(\theta)$ of E _p (lab)=1071→666 and E _p (lab)=1071→146 (1969Gr29). E _p (lab) (keV): weighted average of 1071 2 (1964Gl04), 1071.1 11 (1977Da02), and 1069.7 2 (1983Wa27). $\omega\gamma=0.60$ eV 30 (1964Gl04). $\omega\gamma=3.6$ eV 5 (1977Da02), used for absolute strength calibration by 1983Wa27. $\omega\gamma=3.5$ eV 6 (1969Gr29).
6208.2 5	4	<2 fs	1097.5 5	J ^π : from $\gamma\gamma(\theta)$ in 1971Hy02. T _{1/2} : lifetime $\tau < 0.002$ ps (1973An13, DSAM). E _p (lab) (keV): weighted average of 1096 2 (1964Gl04), 1096.5 12 (1977Da02), and 1097.7 5 (1971Hy02). $\omega\gamma=0.29$ eV 15 (1964Gl04). $\omega\gamma=1.4$ eV 3 (1983Wa27). $\omega\gamma=2.2$ eV 10 (1977Da02). $\omega\gamma=5.3$ eV 11 (1969Gr29). $\omega\gamma=3.1$ eV 9 (1971Hy02).
6219.3 12			1109.0 12	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.10$ eV 6 (1977Da02).
6229.5 5	1		1119.5 5	J ^π : from $\gamma\gamma(\theta)$ in 1971Hy02. E _p (lab) (keV): unweighted average of 1119 2 (1964Gl04), 1119.6 12 (1977Da02), 1118.5 3 (1983Wa27), and 1120.8 5 (1971Hy02).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) [@]	Comments
6266.6 13		1157.7 13	$\omega\gamma=1.2$ eV 3 (1983Wa27). $\omega\gamma=0.21$ eV 11 (1964Gl04). $\omega\gamma=0.9$ eV 3 (1977Da02). $\omega\gamma=1.0$ eV 3 (1969Gr29). $\omega\gamma=1.9$ eV 6 (1971Hy02). E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.4$ eV 2 (1983Wa27). $\omega\gamma=0.2$ eV 1 (1977Da02).
6273.3 13	3	1164.6 13	J ^π : from $\gamma\gamma(\theta)$ E _p (lab)=1165→2376→146 γ cascade (1969Gr29). Other: spin=2,3 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1165 2 (1964Gl04) and 1164.5 13 (1977Da02). $\omega\gamma=0.76$ eV 38 (1964Gl04). $\omega\gamma=3.3$ eV 7 (1983Wa27). $\omega\gamma=2.7$ eV 10 (1977Da02). $\omega\gamma=3.7$ eV 8 (1969Gr29).
6322.0 13	1,2	1214.8 13	J ^π : spin=1,2 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1214 2 (1964Gl04) and 1215.2 13 (1977Da02). $\omega\gamma=0.19$ eV 10 (1964Gl04). $\omega\gamma=2.2$ eV 9 (1983Wa27). $\omega\gamma=1.9$ eV 7 (1977Da02). $\omega\gamma=2.3$ eV 5 (1969Gr29).
6361.3 13		1255.3 13	E _p (lab) (keV): weighted average of 1255 2 (1964Gl04) and 1255.4 13 (1977Da02). $\omega\gamma=0.05$ eV 3 (1964Gl04). $\omega\gamma=0.4$ eV 2 (1977Da02).
6370.2 2	2	1264.5 2	J ^π : from $\gamma\gamma(\theta)$ E _p (lab)=1266→666→g.s. γ cascade (1969Gr29). E _p (lab) (keV): weighted average of 1266 2 (1964Gl04), 1266.4 13 (1977Da02), and 1264.4 2 (1983Wa27). $\omega\gamma=2.7$ eV 6 (1983Wa27). $\omega\gamma=0.23$ eV 12 (1964Gl04). $\omega\gamma=2.5$ eV 8 (1977Da02). $\omega\gamma=4.2$ eV 9 (1969Gr29).
6382.4 14		1277.1 14	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.3$ eV 1 (1977Da02).
6399.8 14		1295.0 14	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.2$ eV 1 (1977Da02).
6441.2 14		1337.7 14	E _p (lab) (keV): weighted average of 1336.8 15 (1973An13) and 1338.4 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.2$ eV 1 (1977Da02).
6450.54 20	3,4	1347.3 2	J ^π : spin=3,4 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1346.5 15 (1973An13), 1348.9 14 (1977Da02), and 1347.3 2 (1983Wa27). $\omega\gamma=0.9$ eV 3 (1983Wa27). $\omega\gamma=1.5$ eV (1973An13). $\omega\gamma=0.9$ eV 3 (1977Da02).
6477.5 14		1375.1 14	E _p (lab) (keV): weighted average of 1373.1 15 (1973An13) and 1376.9 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.2$ eV 1 (1977Da02).
6486.9 14	1 to 4	1384.8 14	J ^π : spin=1:4 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1383.0 15 (1973An13) and 1386.3 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.6$ eV 3 (1983Wa27). $\omega\gamma=0.8$ eV 3 (1977Da02).
6527.6 15		1426.7 15	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.2$ eV 1 (1977Da02).
6546.6 15		1446.3 15	E _p (lab) (keV): weighted average of 1445.0 15 (1973An13) and 1447.6 15 (1977Da02). $\omega\gamma=1.6$ eV (1973An13).

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$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#), [1977Da02](#), [1992Ka39](#) (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) [@]	Comments
			$\omega\gamma=1.4$ eV 4 (1983Wa27).
			$\omega\gamma=1.6$ eV 5 (1977Da02).
6574.7 15		1475.3 15	E _p (lab) (keV): weighted average of 1473.8 15 (1973An13) and 1476.8 15 (1977Da02).
			$\omega\gamma=0.9$ eV (1973An13).
			$\omega\gamma=0.7$ eV 3 (1983Wa27).
			$\omega\gamma=0.6$ eV 2 (1977Da02).
6583.7 15		1484.5 15	E _p (lab) (keV): from 1977Da02 .
			$\omega\gamma=0.4$ eV 2 (1977Da02).
6605? 2		1506.7 20	E _p (lab) (keV): from 1973An13 .
			$\omega\gamma=1.1$ eV (1973An13).
6624.7 15		1526.8 15	E(level): The presence of the primary transitions from this resonance to 1 ⁺ and 5 ⁻ levels suggests there may be a doublet with very closely spaced levels. A careful examination of the shape of the resonance peak showed no sign of a possible doublet (1983Wa27).
			E _p (lab) (keV): weighted average of 1524.0 20 (1973An13) and 1528.4 15 (1977Da02).
			$\omega\gamma=0.5$ eV (1973An13).
			$\omega\gamma=0.4$ eV 1 (1983Wa27).
			$\omega\gamma=0.2$ eV 1 (1977Da02).
6640.92 7	4	1543.50 5	E(level): unweighted average of 6640.85 9 (at E _p (lab)=1543.49 5 in 1994Li20) and 6640.97 3 reported by 1994Li20 based on γ data without reporting E _γ .
			J ^π : from $\gamma\gamma(\theta)$ in 1977Da03 . Other: spin=3,4 from $\gamma(\theta)$ in 1983Wa27 .
			E _p (lab) (keV): weighted average of 1542.0 20 (1973An13), 1545.4 15 (1977Da02), 1543.6 2 (1983Wa27), and 1543.49 5 (1994Li20). Other: 1542 (1975Ke11).
			$\omega\gamma=3.8$ eV 6 (1977Da02), used for absolute strength calibration by 1983Wa27 .
			$\omega\gamma=6.4$ eV (1973An13).
			$\omega\gamma=1.4$ eV 2 (1975Ke11).
6695.1 16		1599.3 16	E _p (lab) (keV): from 1977Da02 .
			$\omega\gamma=0.4$ eV 2 (1977Da02).
6702.8 16		1607.3 16	E _p (lab) (keV): from 1977Da02 .
			$\omega\gamma=0.3$ eV 2 (1977Da02).
6720.2 16		1625.2 16	E _p (lab) (keV): from 1977Da02 .
			$\omega\gamma=0.3$ eV 1 (1977Da02).
6723.7 16		1628.8 16	E _p (lab) (keV): weighted average of 1626.5 20 (1973An13) and 1630.3 16 (1977Da02).
			$\omega\gamma=1.0$ eV 4 (1983Wa27).
			$\omega\gamma=1.6$ eV (1973An13).
			$\omega\gamma=0.9$ eV 4 (1977Da02).
6737.0 16		1642.5 16	E _p (lab) (keV): weighted average of 1640.2 20 (1973An13) and 1644.0 16 (1977Da02).
			$\omega\gamma=1.2$ eV (1973An13).
			$\omega\gamma=0.7$ eV 3 (1983Wa27).
			$\omega\gamma=0.5$ eV 2 (1977Da02).
6790.8 16		1698.0 16	E _p (lab) (keV): from 1977Da02 .
			$\omega\gamma=0.2$ eV 1 (1983Wa27).
			$\omega\gamma=0.6$ eV 3 (1977Da02).
6797.4 16	1 to 5	1704.8 16	J ^π : spin=1:5 from $\gamma(\theta)$ in 1983Wa27 .
			E _p (lab) (keV): weighted average of 1703.2 20 (1973An13) and 1705.9 16 (1977Da02).
			$\omega\gamma=4.5$ eV (1973An13).
			$\omega\gamma=4.8$ eV 10 (1983Wa27).
			$\omega\gamma=4.0$ eV 25 (1977Da02).
6807.2 16		1714.9 16	E _p (lab) (keV): weighted average of 1713.8 20 (1973An13) and 1715.6 16 (1977Da02).
			$\omega\gamma=1.0$ eV (1973An13).
			$\omega\gamma=0.10$ eV 7 (1977Da02).
6828.6 14		1736.9 14	E _p (lab) (keV): weighted average of 1734.3 20 (1973An13) and 1738.2 14 (1977Da02).
			$\omega\gamma=1.2$ eV (1973An13).
			$\omega\gamma=0.4$ eV 1 (1983Wa27).
			$\omega\gamma=0.6$ eV 3 (1977Da02).
6842.8 14		1751.5 14	E _p (lab) (keV): from 1977Da02 .

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$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#), [1977Da02](#), [1992Ka39](#) (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	E _p (lab) (keV) [@]	Comments
			$\omega\gamma=4.7$ eV 20 (1983Wa27); much larger than $\omega\gamma=1.8$ 10 (1977Da02), and 1983Wa27 states that a new, clean and thin target was used and the resonance was nearly totally resolved from the nearby $^{13}\text{C}(\text{p},\gamma)$ resonance.
6853.0 14		1762.1 14	E _p (lab) (keV): from 1977Da02 . $\omega\gamma=2.1$ eV 5 (1983Wa27). $\omega\gamma=2.3$ eV 7 (1977Da02).
6871.18 30	5	1780.8 3	J ^π : spin=5 from $\gamma(\theta)$ in 1977Da03 . E _p (lab) (keV): weighted average of 1782.2 16 (1977Da02) and 1780.7 3 (1983Wa27). $\omega\gamma=1.6$ eV 8 (1977Da02). $\omega\gamma=0.4$ eV 2 (1983Wa27).
6887.97 30	2	1798.1 3	J ^π : spin=2 from $\gamma(\theta)$ in 1977Da03 . E _p (lab) (keV): weighted average of 1796.4 20 (1973An13), 1799.3 16 (1977Da02), and 1798.1 3 (1983Wa27). $\omega\gamma=2.9$ eV 10 (1983Wa27). $\omega\gamma=2.8$ eV (1973An13). $\omega\gamma=2.0$ eV 6 (1977Da02).
6901.75 30		1812.3 3	E _p (lab) (keV): weighted average of 1809.5 20 (1973An13), 1813.4 16 (1977Da02), and 1812.3 3 (1983Wa27). $\omega\gamma=2.4$ eV 6 (1983Wa27). $\omega\gamma=2.7$ eV (1973An13). $\omega\gamma=2.8$ eV 9 (1977Da02).
6918.0 17	3,4	1829.0 17	J ^π : from $\gamma(\theta)$ in 1983Wa27 . E _p (lab) (keV): from 1977Da02 . $\omega\gamma=11$ eV 2 (1977Da02), used for absolute strength calibration by 1983Wa27 .
6931.5 17		1843.0 17	E _p (lab) (keV): from 1977Da02 . $\omega\gamma=0.8$ eV 3 (1983Wa27). $\omega\gamma=0.8$ eV 5 (1977Da02).
6977.4 17		1890.3 17	E _p (lab) (keV): from 1977Da02 . $\omega\gamma=0.5$ eV 2 (1977Da02).
6984.9 10		1898 1	E _p (lab) (keV): other: 1898 2 component of a triplet (1977Da02). $\omega\gamma=0.7$ eV 3 (1992Ka39).
6986.85		1900 2	E _p (lab) (keV): component of a triplet (1977Da02). $\omega\gamma=1.7$ eV 10 for the triplet(1977Da02).
6990.7 20		1904 2	E _p (lab) (keV): component of a triplet (1977Da02).
6992.7 10		1906 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7018.9 10		1933 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7051.6 18		1966.7 18	E _p (lab) (keV): from 1977Da02 . $\omega\gamma=0.3$ eV 2 (1977Da02).
7059.04 30	1,2	1974.4 3	J ^π : spin=1,2 from $\gamma(\theta)$ in 1983Wa27 . E _p (lab) (keV): weighted average of 1975.3 18 (1977Da02), 1974.4 3 (1983Wa27), and 1974 1 (1992Ka39). $\omega\gamma=8$ eV 2 (1977Da02), used for absolute strength calibration by 1983Wa27 and 1992Ka39 .
7061.6 10		1977 1	$\omega\gamma=5$ eV 2 (1992Ka39).
7078.89 9	3	1994.86 7	E(level): unweighted average of 7078.83 10 (at E _p (lab)=1994.86 7 in 1994Li20) and 7079.00 6 reported by 1994Li20 based on γ data without reporting E γ . J ^π : from $\gamma\gamma(\theta)$ in 1977Da03 . E _p (lab) (keV): weighted average of 1997.2 18 (1977Da02) and 1994.86 7 (1994Li20). $\omega\gamma=1.7$ eV 4 (1983Wa27). $\omega\gamma=4.4$ eV 20 (1977Da02).
7115.9 10		2033 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7122.7 10		2040 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7126.6 10		2044 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7145.0 10		2063 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7165.4 10		2084 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7175.1 10		2094 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).

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$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#),[1977Da02](#),[1992Ka39](#) (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	E _p (lab) (keV) [@]	Comments
7183.8 10		2103 1	
7203.2 10		2123 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7218.8 10		2139 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7238.2 10		2159 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7249.8 10		2171 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7271.2 10		2193 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7284.7 10		2207 1	
7295.4 10		2218 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7321.6 10		2245 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7346.8 10		2271 1	$\omega\gamma=1.7$ eV 5 (1992Ka39).
7376.0 10		2301 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7382.7 10		2308 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7406.0 10		2332 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7424.5 10		2351 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7441.0 10		2368 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7449.7 10		2377 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7462.3 10		2390 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7470.1 10		2398 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7486.6 10		2415 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7496.3 10		2425 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7506.0 10		2435 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7511.8 10		2441 1	
7517.6 10		2447 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7527.3 10		2457 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7539.9 10		2470 1	
7543.8 10		2474 1	$\omega\gamma=0.5$ eV 3 (1992Ka39).
7562.3 10		2493 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7577.8 10		2509 1	
7582.6 10		2514 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7588.5 10		2520 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7609.8 10		2542 1	$\omega\gamma=4.5$ eV 10 (1992Ka39).
7624.4 10		2557 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7630.2 10		2563 1	
7643.8 10		2577 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7649.6 10		2583 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7666.1 10		2600 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7675.8 10	2,3	2610 1	J ^π : spin from $\gamma(\theta)$ in 1992Ka39 .
7690.3 10		2625 1	$\omega\gamma=12$ eV 3 (1992Ka39).
7708.8 10		2644 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7720.4 10		2656 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7730.1 10		2666 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7753.4 10		2690 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7784.5 10		2722 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7803.9 10		2742 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7809.7 10		2748 1	$\omega\gamma=5$ eV 2 (1992Ka39).
7831.0 10		2770 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7848.5 10		2788 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7853.4 10		2793 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7878.6 10		2819 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7886.4 10		2827 1	
7899.0 10		2840 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7904.8 10		2846 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7909.6 10		2851 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7934.9 10		2877 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7942.6 10		2885 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7966.9 10		2910 1	$\omega\gamma=4.5$ eV 10 (1992Ka39).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) [@]	Comments
7979.5 10		2923 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7990.2 10		2934 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7997.0 10		2941 1	$\omega\gamma=3.5$ eV 10 (1992Ka39).
8030.9 10		2976 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
8035.8 10		2981 1	$\omega\gamma=4$ eV 1 (1992Ka39).
8048.4 10		2994 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
8070.7 10		3017 1	$\omega\gamma=3$ eV 1 (1992Ka39).
8132.8 10		3081 1	
8141.6 10		3090 1	$\omega\gamma=7$ eV 3 (1992Ka39).
8154.2 10		3103 1	
8191.0 10		3141 1	$\omega\gamma=4$ eV 1 (1992Ka39).
8195.9 10		3146 1	
8214.3 10		3165 1	$\omega\gamma=9$ eV 3 (1992Ka39).
8235.7 10		3187 1	$\omega\gamma=4$ eV 2 (1992Ka39).
8248.3 10		3200 1	$\omega\gamma=6$ eV 2 (1992Ka39).
8259.0 10		3211 1	
8280.3 10		3233 1	$\omega\gamma=2$ eV 1 (1992Ka39).
8305.5 10	2,3	3259 1	J ^π : spin from $\gamma(\theta)$ in 1992Ka39. $\omega\gamma=27$ eV 5 (1992Ka39). Another I $\gamma=33$ decays to unknown levels (1992Ka39).
8319.1 10		3273 1	$\omega\gamma=7$ eV 2 (1992Ka39).
8337.6 10		3292 1	$\omega\gamma=12$ eV 3 (1992Ka39).
8361.8 10		3317 1	$\omega\gamma=5$ eV 3 (1992Ka39).
8372.5 10		3328 1	$\omega\gamma=9$ eV 4 (1992Ka39).
8390.0 10		3346 1	$\omega\gamma=4$ eV 2 (1992Ka39).

[†] For resonances above $\text{S}(\text{p})(^{34}\text{Cl})=5143.20$ 5 (2021Wa16), $\text{E}(\text{level})=\text{E}_{\text{c.m.}}+\text{S}(\text{p})(^{34}\text{Cl})$, where $\text{E}_{\text{c.m.}}=\text{E}_{\text{p}}(\text{lab})\times m(^{33}\text{S})/[m_{\text{p}}+m(^{33}\text{S})]$, unless otherwise noted. For levels below 7.08 MeV ($\text{E}_{\text{p}}(\text{lab})<2$ MeV), $\text{E}_{\text{p}}(\text{lab})$ is mainly from 1964GI04, 1977Da02, and 1983Wa27. The average is taken when multiple $\text{E}_{\text{p}}(\text{lab})$ values are available. For levels above 7.08 MeV ($\text{E}_{\text{p}}(\text{lab})>2$ MeV), $\text{E}_{\text{p}}(\text{lab})$ is mainly from 1992Ka39.

[‡] Most papers in this dataset report precise $\text{E}(\text{level})$ values based on E_{γ} data without reporting their measured E_{γ} values. The only exception is 1964GI04, which reports E_{γ} measured by a NaI detector with uncertainties ranging from 10 to 50 keV. In addition to the large uncertainties, 1964GI04 also used a much lower $\text{S}(\text{p})(^{34}\text{Cl})=5.12$ MeV in their analysis, which contributes to generally lower $\text{E}(\text{level})$ values. A least-squares fit to available E_{γ} values with uncertainties would not yield $\text{E}(\text{level})$ values that are more precise or complete than those already reported in the original papers. Therefore, the evaluators have adopted the precise $\text{E}(\text{level})$ values reported in papers and the average is taken when multiple $\text{E}(\text{level})$ values are available.

[#] For low-lying levels with J and π that are both given, J^π is from the Adopted Levels. For other levels, J is from the comparison of experimental angular distributions of γ -rays with theoretical predictions or deduced from γ -feedings, unless otherwise noted.

[@] From 1992Ka39, unless otherwise noted.

 $\gamma(^{34}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ [‡]	E _f	J _f ^π	Comments
146.4	3 ⁺	146.4		0	0 ⁺	
461.2	1 ⁺	314.8	<1	146.4	3 ⁺	I _γ : other: <5 (1969Gr29).
		461.2	100	0	0 ⁺	E _γ : other: 460 20, γ cascade at $\text{E}_{\text{p}}(\text{lab})=507, 546, 731, 822$ (1964GI04).
						I _γ : other: 100 (1969Gr29).
665.3	1	204.1	<0.3	461.2	1 ⁺	I _γ : from 2020Pa40. Other: <5 (1969Gr29).
		518.9	1.46 19	146.4	3 ⁺	E _γ : other: 530 10, γ cascade at $\text{E}_{\text{p}}(\text{lab})=507$ (1964GI04).
						I _γ : from 2020Pa40. Others: <1 (1983Wa27), <5 (1969Gr29).
		665.3	100	0	0 ⁺	E _γ : others: 670 20, γ cascade at $\text{E}_{\text{p}}(\text{lab})=447, 507, 546, 662, 683, 731, 777$,

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
								822 (1964GI04). I_γ : from 2020Pa40, 1983Wa27, 1977Da02, and 1969Gr29. $A_2=+0.28$ 4, A_4 set to zero; $A_2=+0.31$ 5, $A_4=-0.07$ 7 (1969Gr29). E_γ : other: 560 20, γ cascade at $E_p(\text{lab})=731$ (1964GI04). I_γ : others: 102.8 28 (1977Da02), 167 35 (1969Gr29). E_γ : others: 770 30, γ cascade at $E_p(\text{lab})=662$, 822; 770 20, γ cascade at $E_p(\text{lab})=731$ (1964GI04). I_γ : others: 100.0 28 (1977Da02), 100 18 (1969Gr29). E_γ : others: 1080 50, γ cascade at $E_p(\text{lab})=662$; 1090 20, γ cascade at $E_p(\text{lab})=731$; 1100 40, γ cascade at $E_p(\text{lab})=546$ (1964GI04). I_γ : others: 75.0 28 (1977Da02), 80 25 (1969Gr29). I_γ : others: <5.7 (1983Wa27), <7.0 (1969Gr29). I_γ : other: <16 (1969Gr29). E_γ : others: 1420 40, γ cascade at $E_p(\text{lab})=731$; 1350 60, γ cascade at $E_p(\text{lab})=546$ (1964GI04). I_γ : others: 100.0 33 (1977Da02), 100 36 (1969Gr29), 100 7 (1971Hy02). E_γ : other: 1750 30, γ cascade at $E_p(\text{lab})=546$, 683, 731 (1964GI04). I_γ : weighted average of 66.7 33 (1977Da02), 74.2 12 (1983Wa27), 100 40 (1969Gr29), and 67 7 (1971Hy02). I_γ : others: <1.7 (1983Wa27), <16 (1969Gr29). E_γ : others: 930 30, γ cascade at $E_p(\text{lab})=822$; 950 40, γ cascade at $E_p(\text{lab})=662$ (1964GI04). I_γ : weighted average of 9.1 15 (1977Da02), 8.54 31 (1983Wa27), 19 6 (1969Gr29), and 10.5 60 (1971Hy02). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=+0.14$ 10, $A_4=0.00$ 10 (1977Da03). I_γ : others: <3.1 (1983Wa27), <3.1 (1969Gr29), <1.5 (1971Hy02). E_γ : others: 1700 30, γ cascade at $E_p(\text{lab})=546$, 731; 1700 40, γ cascade at $E_p(\text{lab})=662$; 1710 30, γ cascade at $E_p(\text{lab})=639$ (1964GI04). I_γ : others: 100.0 30 (1977Da02), 100 17 (1969Gr29), 100 9 (1971Hy02). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=-0.12$ 1, $A_4=-0.01$ 1 (1977Da03). $A_2=-0.20$ 3 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.03$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$). E_γ : others: 2010 40, γ cascade at $E_p(\text{lab})=822$; 2020 50, γ cascade at $E_p(\text{lab})=639$, 662; 2030 30, γ cascade at $E_p(\text{lab})=546$ (1964GI04). I_γ : weighted average of 21.2 15 (1977Da02), 20.1
1230.33	2 ⁺	565.0	98.6 14	665.3	1			
		769.1	100.0 14	461.2	1 ⁺			
		1083.9	87.1 14	146.4	3 ⁺			
1887.29	2 ⁺	1230.31	<1.4 [#]	0	0 ⁺			
		656.95	<3.5	1230.33	2 ⁺			
		1222.0	<0.83	665.3	1			
		1426.1	100.0 12	461.2	1 ⁺			
		1740.8	73.2 12	146.4	3 ⁺			
2157.9	2 ⁺	1887.23	<0.83 [#]	0	0 ⁺			
		927.6	8.59 31	1230.33	2 ⁺	D(+Q)	+0.1 +6-3	
		1492.6	<0.76 [#]	665.3	1			
		1696.7	100.0 6	461.2	1 ⁺	D+Q	-0.05 3	
		2011.4	20.2 5	146.4	3 ⁺	D+Q	+0.22 10	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. $\&$	$\delta^\&$	Comments
								5 (1983Wa27), and 19 9 (1971Hy02). Other: <9.4 (1969Gr29). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=-0.11$ 7, $A_4=+0.02$ 7 (1977Da03). E_γ : others: 2160 30, γ cascade at $E_p(\text{lab})=639$, 822; 2160 50, γ cascade at $E_p(\text{lab})=662$; 2170 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : weighted average of 21.2 15 (1977Da02), 23.8 5 (1983Wa27), 38 14 (1969Gr29), and 19 5 (1971Hy02). I_γ : other: <3.9 (1983Wa27). I_γ : weighted average of 25.5 21 (1977Da02), 24.7 14 (1983Wa27), and 26 12 (1971Hy02). I_γ : weighted average of 87 4 (1977Da02), 69.8 27 (1983Wa27), and 49 9 (1971Hy02). $A_2=+0.03$ 12 (1971Hy02). I_γ : others: 100 4 (1977Da02), 100 14 (1971Hy02). $A_2=-0.02$ 6 (1971Hy02). I_γ : other: <3.9 (1983Wa27). I_γ : weighted average of 1.01 51 (1977Da02) and 0.81 20 (1983Wa27) Other: <2 (1969Gr29). I_γ : other: <4 (1969Gr29). I_γ : other: <2 (1969Gr29). I_γ : others: 100.0 5 (1977Da02), 100 (1969Gr29), 100 (1971Hy02). Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1165$ in 1969Gr29. $A_2=+0.11$ 3, $A_4=-0.29$ 4 (1969Gr29). I_γ : other: <3 (1969Gr29). I_γ : from 1969Gr29. I_γ : other: <6 (1969Gr29). I_γ : other: <6 (1969Gr29). I_γ : other: <15 (1969Gr29). I_γ : other: <8 (1969Gr29). E_γ : others: 2570 30, γ cascade at $E_p(\text{lab})=683$; 2580 30, γ cascade at $E_p(\text{lab})=507$; 2590 30, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : others: 100 (1977Da02), 100 (1969Gr29). I_γ : unweighted average of 65.7 29 (1977Da02) and 55.3 33 (1983Wa27). I_γ : weighted average of 69 6 (1977Da02) and 67.8 33 (1983Wa27) Other: 100 18 (1971Hy02). I_γ : weighted average of 51.4 29 (1977Da02), 38.3 28 (1983Wa27), and 10.0 80 (1971Hy02). I_γ : others: 100.0 33 (1983Wa27), 90 18 (1971Hy02). I_γ : weighted average of 16.3 21 (1977Da02), 14.5 11 (1983Wa27), and 17 7 (1971Hy02).
2157.9	2 ⁺	2157.8	23.5 5	0	0 ⁺			
2181.10	3	950.76 1515.8	<2.1 [#] 25.0 14	1230.33 665.3	2 ⁺ 1			
		1719.9	73.7 70	461.2	1 ⁺			
		2034.6	100.0 25	146.4	3 ⁺			
2375.7	4	2181.03 1145.4	<2.1 [#] 0.84 20	0 1230.33	0 ⁺ 2 ⁺			
		1710.4 1914.4 2229.2	<2.0 <4 100.00 20	665.3 461.2 146.4	1 1 ⁺ 3 ⁺	D+Q	+6.3 +32-20	
2580.3		2375.6 693.0 1349.9 1914.9 2119.0 2433.8 2580.2	<2.0 <8 <1 <2 <2 <2 100	0 1887.29 1230.33 665.3 461.2 146.4 0	0 ⁺ 2 ⁺ 2 ⁺ 1 1 ⁺ 3 ⁺ 0 ⁺			
2611.03	3	453.1	61 5	2157.9	2 ⁺			
		723.73 1380.67	<2.5 68.1 33	1887.29 1230.33	2 ⁺ 2 ⁺			
		1945.7	42.6 73	665.3	1			
		2149.8 2464.5	<5.1 100.0 [#] 29	461.2 146.4	1 ⁺ 3 ⁺			
2721.3	2	563.4	14.9 11	2157.9	2 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
2721.3	834.0	4.4 4	1887.29	2 ⁺			I_γ : weighted average of 6.1 21 (1977Da02), 4.3 4 (1983Wa27), and 4.3 21 (1971Hy02). $A_2=+0.1$ 3 (1971Hy02).
1490.9		4.6 4	1230.33	2 ⁺			I_γ : weighted average of 4.1 21 (1977Da02), 4.5 4 (1983Wa27), and 6.4 21 (1971Hy02). $A_2=-0.07$ 21 (1971Hy02).
2055.9		15.5 11	665.3	1			I_γ : weighted average of 12.2 21 (1977Da02), 16.4 11 (1983Wa27), and 17 7 (1971Hy02). $A_2=-0.20$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=-0.04$ 11 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
2260.0	100.0 15		461.2	1 ⁺	D(+Q)	-0.03 5	I_γ : others: 100 4 (1977Da02), 100 9 (1971Hy02). Mult., δ : or $\delta=-2.3$ 5 from $\gamma\gamma(\theta)$ in 1971Hy02. $A_2=-0.17$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$); $A_2=+0.11$ 3 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
2574.8		38.9 11	146.4	3 ⁺			I_γ : weighted average of 36.7 21 (1977Da02), 39.5 11 (1983Wa27), and 40 7 (1971Hy02). $A_2=+0.07$ 5 (1971Hy02).
2721.2		32.8 11	0	0 ⁺			I_γ : weighted average of 28.6 21 (1977Da02), 34.1 11 (1983Wa27), and 28 7 (1971Hy02). $A_2=+0.26$ 13 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.21$ 9 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$); $A_2=-0.20$ 6 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
3129.12	1898.73	<2	1230.33	2 ⁺			E_γ : other: 3150 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : other: 100 (1977Da02). I_γ : other: 100 18 (1977Da02).
	2667.8	<2	461.2	1 ⁺			
	2982.6	<2	146.4	3 ⁺			
	3128.97	100	0	0 ⁺			
3334.0	1176.1	100 15	2157.9	2 ⁺			I_γ : other: 33 9 (1977Da02). 1983Wa27 pointed out that it seems likely that 1977Da02 was unaware of a $^{34\text{m}}\text{Cl}$ contamination and therefore obtained too large $I_\gamma(1176)$. I_γ : other: 16.4 90 (1977Da02). Due to a contamination issue pointed out by 1983Wa27, $I_\gamma(1176)$ from 1977Da02 is too large, which makes other I_γ too small.
	1446.7	<4.9	1887.29	2 ⁺			
	2103.6	<4.9	1230.33	2 ⁺			
	2668.6	90 12	665.3	1			
	2872.7	54 12	461.2	1 ⁺			
	3187.4	<4.9	146.4	3 ⁺			I_γ : weighted average of 52 10 (1977Da02) and 52.0 25 (1983Wa27). E_γ : other: 2710 50, γ cascade at $E_p(\text{lab})=662$ (1964Gl04). I_γ : weighted average of 87 7 (1977Da02) and 85 4 (1983Wa27). E_γ : other: 2920 30, γ cascade at $E_p(\text{lab})=662$ (1964Gl04). I_γ : weighted average of 84 7 (1977Da02) and 72 4 (1983Wa27). E_γ : other: 3240 30, γ cascade at $E_p(\text{lab})=662$ (1964Gl04). I_γ : other: 100 7 (1977Da02).
	3333.8	<7.3	0	0 ⁺			
3383.3	2152.9	52.0 25	1230.33	2 ⁺			
	2717.9	86 4	665.3	1			
	2922.0	75 5	461.2	1 ⁺			
	3236.7	100.0 34	146.4	3 ⁺			
	3383.1	<3.1	0	0 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
3545.08	3	964.8	<3	2580.3				I_γ : from 1969Gr29.
		1169.4	<15	2375.7	4			I_γ : from 1969Gr29.
		1387.2	<3	2157.9	2 ⁺			I_γ : from 1969Gr29.
		1657.75	<5	1887.29	2 ⁺			I_γ : from 1969Gr29.
		2314.67	3.6 6	1230.33	2 ⁺			I_γ : weighted average of 4.2 21 (1977Da02) and 3.5 6 (1983Wa27) Other: <5 (1969Gr29).
		2879.7	<12	665.3	1			I_γ : from 1969Gr29.
		3083.7	<3.1	461.2	1 ⁺			I_γ : other: <8 (1969Gr29).
		3398.5	100.0 6	146.4	3 ⁺	D+Q	-0.09 4	E_γ : other: 3400 50, γ cascade at $E_p(\text{lab})=731$ (1964Gl04).
								I_γ : others: 100.0 21 (1977Da02), 100 (1969Gr29), 100 (1971Hy02).
								Mult.: from $\gamma(\theta)$ in 1971Hy02 and 1977Da03.
								δ : weighted average of -0.06 5 (1971Hy02) and -0.11 4 (1977Da03). 1977Da03 reports $\delta=-0.11$ 4 based on $\delta=+1.5$ +10-7 or -0.1 +3-2 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.14$ 4 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$; $\delta=-0.07$ +5-1 or +1.5 +3-2 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$.
								$A_2=+0.19$ 3, $A_4=+0.022$ 21 (1977Da03).
								$A_2=+0.18$ 9, $A_4=-0.02$ 9 (1977Da03, γ cascade at $E_p(\text{lab})=1057$); $A_2=+0.25$ 3, $A_4=+0.02$ 3 (1977Da03, γ cascade at $E_p(\text{lab})=1545$); $A_2=+0.17$ 2, $A_4=+0.03$ 3 (1977Da03, γ cascade at $E_p(\text{lab})=1997$); $A_2=+0.23$ 8 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.25$ 2 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$).
								I_γ : other: <4 (1969Gr29).
3600.28	4	3544.88	<3.1	0	0 ⁺			
		879.0	94.1 11	2721.3	2	Q+O	+0.08 5	I_γ : weighted average of 98 4 (1977Da02), 93.9 11 (1983Wa27), and 85 13 (1969Gr29), and 92 11 (1971Hy02).
								Mult.: from $\gamma\gamma(\theta)$ in 1977Da03 and 1971Hy02.
								δ : or +14 +43-5 (1977Da03). Other: $\delta=+0.01$ 4 (1971Hy02).
								$A_2=+0.30$ 3, $A_4=+0.02$ 3, $A_6=+0.01$ 4 (1977Da03).
								$A_2=+0.30$ 3, $A_4=+0.02$ 3, $A_6=+0.01$ 4 (1977Da03); $A_2=+0.24$ 6 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.25$ 1 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$).
		1020.0	<3.7	2580.3				I_γ : from 1969Gr29.
		1224.6	15.4 19	2375.7	4			I_γ : weighted average of 14.9 21 (1977Da02), 15.7 19 (1983Wa27), and 17 8 (1971Hy02). Others: <7.4 (1969Gr29).
								$A_2=-0.14$ 7 (1971Hy02).
		1419.15	<4.3	2181.10	3			
		1442.4	<5.6	2157.9	2 ⁺			I_γ : from 1969Gr29.
		1712.94	<7.4	1887.29	2 ⁺			I_γ : from 1969Gr29.
		2369.86	<7.4	1230.33	2 ⁺			I_γ : from 1969Gr29.
		2934.8	<4.2	665.3	1			I_γ : other: <3.7 (1969Gr29).
		3138.9	<4.2	461.2	1 ⁺			I_γ : other: <7.4 (1969Gr29).
		3453.7	100.0 11	146.4	3 ⁺	D+Q	+0.06 3	I_γ : others: 100 4 (1977Da02), 100 22 (1969Gr29), 100 11 (1971Hy02).
								Mult.: from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$ in 1977Da03.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								δ : weighted average of $\delta=+0.07$ 4 (1971Hy02) and $\delta=+0.05$ 3 (1977Da03). Other: $\delta=-7$ 2 (1977Da03). $A_2=-0.12$ 3, $A_4=-0.01$ 3 (1977Da03). $A_2=-0.12$ 3, $A_4=-0.01$ 3 (1977Da03); $A_2=-0.13$ 2 (1971Hy02). I_γ : other: <15 (1969Gr29). I_γ : others: 100 4 (1977Da02), 100 14 (1971Hy02). Mult., δ : $\delta=0.00$ 2 in Table 5 and $\delta=0.00 +2 -1$ in Table 3 (1977Da03). $A_2=-0.22$ 3, $A_4=-0.04$ 3 (1977Da03). $A_2=-0.22$ 3, $A_4=-0.04$ 3 (1977Da03); $A_2=-0.24$ 5 (1971Hy02).
3600.28	4	3600.08	<4.2	0	0 ⁺			
3631.7	5	1256.0	100.0 13	2375.7	4	D(+Q)	0.00 2	
		2401.3	<3.7	1230.33	2 ⁺			
		2966.3	<3.7	665.3	1			
		3170.3	<5.5	461.2	1 ⁺			
		3485.1	83.4 13	146.4	3 ⁺			I_γ : weighted average of 82 4 (1977Da02), 83.5 13 (1983Wa27), and 92 14 (1971Hy02). $A_2=+0.21$ 7 (1971Hy02).
		3631.5	<5.5	0	0 ⁺			
3646.3		2415.9	<2.7	1230.33	2 ⁺			
		3184.9	<4	461.2	1 ⁺			
		3499.7	100 7	146.4	3 ⁺			I_γ : other: 100 (1977Da02).
		3646.1	<4	0	0 ⁺			
3660.0		1502.1	100	2157.9	2 ⁺			I_γ : other: 100 (1977Da02).
		1772.7	<2	1887.29	2 ⁺			
		2429.6	<2	1230.33	2 ⁺			
		2994.6	<2	665.3	1			
		3198.6	<2	461.2	1 ⁺			
		3513.4	<5	146.4	3 ⁺			
		3659.8	<3	0	0 ⁺			
3773.72	1	3773.5	100	0	0 ⁺			I_γ : other: 100 (1971Hy02). $A_2=-0.20$ 5 (1971Hy02).
3791.7		2561.3	<3.5	1230.33	2 ⁺			
		3330.3	100 7	461.2	1 ⁺			
		3645.1	<8.2	146.4	3 ⁺			
		3791.5	<7.1	0	0 ⁺			
3940.1		2709.7	<3.5	1230.33	2 ⁺			
		3274.6	100 9	665.3	1			I_γ : other: 100 15 (1977Da02).
		3478.7	72 9	461.2	1 ⁺			I_γ : other: 133 15 (1977Da02).
		3939.9	<5.2	0	0 ⁺			
3964.1		1588.4 ^a	11 [#] 5	2375.7	4			I_γ : 1983Wa27 states that the existence of this transition observed by 1977Da02 is considered doubtful.
		1806.2	100 [#] 16	2157.9	2 ⁺			I_γ : other: 100 25 (1983Wa27).
		3817.5	67 25	146.4	3 ⁺			I_γ : other: 23 11 (1977Da02).
3983.0	3	1261.7	13.5 11	2721.3	2	D+Q	-0.8 3	I_γ : weighted average of 12.7 16 (1977Da02), 13.9 11 (1983Wa27), 12.3 62 (1971Hy02). Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ in 1977Da03 . $A_2=-0.59$ 7, $A_4=+0.05$ 7 (1977Da03). I_γ : from 1969Gr29 . I_γ : from 1969Gr29 .
		1402.7	<12	2580.3				
		1607.3	<12	2375.7	4			
		1825.1	43.2 17	2157.9	2 ⁺	D(+Q)	+0.02 4	I_γ : weighted average of 46.0 32 (1977Da02), 42.3 17 (1983Wa27), and 72 17 (1969Gr29), and 40 8 (1971Hy02). Mult.: from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ and

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	Comments
3983.0	3	2095.6	<26	1887.29	2 ⁺			1997 in 1977Da03 . δ : 1977Da03 reports +0.02 4 based on $\delta=+0.05$ 3 or $\delta=-5.1$ +6-12 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.02$ +5-4 or $\delta=-3.7$ +4-10 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$. $A_2=-0.11$ 3, $A_4=0.00$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1057$); $A_2=-0.17$ 4, $A_4=+0.04$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1997$); $A_2=-0.20$ 3 (1971Hy02). I_γ : from 1969Gr29 .
		2752.6	<6.4 [#]	1230.33	2 ⁺			I_γ : others: <7.8 (1983Wa27), <12 (1969Gr29).
		3317.5	<12	665.3	1			I_γ : from 1969Gr29 .
		3521.6	<1.6	461.2	1 ⁺			I_γ : other: <10 (1969Gr29).
		3836.4	100.0 31	146.4	3 ⁺	D(+Q)	-0.02 4	I_γ : others: 100.0 32 (1977Da02), 100 19 (1969Gr29), 100 9 (1971Hy02). Mult.: from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ and 1997 in 1977Da03 . δ : 1977Da03 reports -0.02 4 based on $\delta=0.0$ 1 or +1.3 3 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.04$ 10 or $\delta=+1.4$ 3 from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$. $A_2=+0.20$ 4, $A_4=0.00$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1057$); $A_2=+0.19$ 4, $A_4=+0.05$ 5 (1977Da03 , γ cascade at $E_p(\text{lab})=1997$); $A_2=+0.21$ 3 (1971Hy02 , γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.00$ 5 (1971Hy02 , γ cascade at $E_p(\text{lab})=1097$). I_γ : from 1969Gr29 .
4076.2	4	3982.8	<5.2	0	0 ⁺			I_γ : weighted average of 13.6 34 (1977Da02) and 8.8 8 (1983Wa27).
		1354.9	9.1 11	2721.3	2			I_γ : weighted average of 35.6 34 (1977Da02) and 35.5 13 (1983Wa27) Other: <15 (1971Hy02).
		1700.5	35.5 13	2375.7	4	D+Q	-0.27 +4-3	Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 . $A_2=+0.18$ 2, $A_4=0.00$ 2 (1977Da03).
		1895.0	22.0 13	2181.10	3	D+Q	+0.05 +3-1	I_γ : weighted average of 20.3 34 (1977Da02) and 22.3 13 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 . $A_2=-0.16$ 3, $A_4=+0.02$ 3 (1977Da03).
		2845.7	<5	1230.33	2 ⁺			I_γ : others: 100.0 34 (1977Da02), 100 (1971Hy02).
4139.7	2	3614.8	<3.3	461.2	1 ⁺			Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 . $A_2=-0.27$ 2, $A_4=+0.01$ 2 (1977Da03).
		3929.6	100.0 25	146.4	3 ⁺	D(+Q)	0.00 +1-2	$A_2=-0.27$ 2, $A_4=+0.01$ 2 (1977Da03); $A_2=-0.07$ 5 (1971Hy02).
		4075.9	<3.3	0	0 ⁺			I_γ : weighted average of 36 9 (1977Da02), 45 4 (1983Wa27), and 39 14 (1971Hy02).
		1559.4	37 6	2580.3	2 ⁺			
		1981.7	43 4	2157.9	2 ⁺			
4139.7	2	2909.2	14 4	1230.33	2 ⁺			
		3474.2	<9.8	665.3	1			
		3678.3	<3.9	461.2	1 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	Comments
4139.7	2	3993.1	100 6	146.4	3 ⁺			I_γ : others: 100 9 (1977Da02), 100 14 (1971Hy02). $A_2=+0.20$ 20 (1971Hy02).
		4139.4	<3.9	0	0 ⁺			
4147.8		1966.6	7.0 10	2181.10	3			
		1989.8	6.1 13	2157.9	2 ⁺			
		2917.3	100.0 23	1230.33	2 ⁺			
		3482.3	66.2 19	665.3	1			
		4001.2	<3.8	146.4	3 ⁺			
		4147.5	9.8 15	0	0 ⁺			E_γ : other: 4130 50, γ cascade at $E_p(\text{lab})=777$ (1964Gl04).
4325.89		2167.9	100 6	2157.9	2 ⁺			I_γ : other: 100 (1977Da02).
		2438.51	<1.3	1887.29	2 ⁺			
		3660.4	<2.5	665.3	1			
		3864.5	<3.8	461.2	1 ⁺			
		4179.2	<2.5	146.4	3 ⁺			
		4325.60	<5	0	0 ⁺			
4354.2	(1)	2196.2	33.2 28	2157.9	2 ⁺			I_γ : weighted average of 35.8 30 (1977Da02) and 31.0 28 (1983Wa27).
		3688.7	<4.2	665.3	1			
		3892.8	11.5 28	461.2	1 ⁺			I_γ : weighted average of 13.4 30 (1977Da02) and 9.9 28 (1983Wa27).
		4207.5	<2.8	146.4	3 ⁺			
		4353.9	100.0 28	0	0 ⁺			I_γ : others: 100.0 30 (1977Da02), 100 (1971Hy02). $A_2=-0.16$ 8 (1971Hy02).
4417.3		3186.8	<3.1	1230.33	2 ⁺			I_γ : weighted average of 64 8 (1977Da02) and 56 5 (1983Wa27).
		3751.8	58 5	665.3	1			I_γ : other: 100 8 (1977Da02).
		3955.9	100 5	461.2	1 ⁺			
		4270.6	<6.3	146.4	3 ⁺			
		4417.0	<7.8	0	0 ⁺			I_γ : other: 100 (1971Hy02).
4446.6	4	4299.9	100	146.4	3 ⁺	D(+Q)	-0.02 +2-1	Mult., δ : from $\gamma\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1782$ in 1977Da03. $A_2=-0.29$ 3, $A_4=0.00$ 4 (1977Da03).
		4446.3	<3	0	0 ⁺			
4461.4		4314.7	100	146.4	3 ⁺			
4515.7		3285.2	<3.8	1230.33	2 ⁺			I_γ : others: 100 4 (1977Da02), 100 12 (1971Hy02). $A_2=-0.00$ 8 (1971Hy02).
		3850.2	100.0 25	665.3	1			
		4054.2	<2.5	461.2	1 ⁺			E_γ : other: 4370 50, γ cascade at $E_p(\text{lab})=447$ (1964Gl04).
		4369.0	25.0 25	146.4	3 ⁺			I_γ : weighted average of 27 4 (1977Da02), 25.0 25 (1983Wa27), and 19 12 (1971Hy02). $A_2=-0.10$ 20 (1971Hy02).
		4515.4	<5	0	0 ⁺			
4605.8		1884.4	100 [#]	2721.3	2			
		2447.8	44 [#]	2157.9	2 ⁺			
		2718.4	35 [#]	1887.29	2 ⁺			
		3375.3	54 [#]	1230.33	2 ⁺			
		4459.1	<12	146.4	3 ⁺			
4609.7		1888.3	71 [#]	2721.3	2			
		3379.2	100 [#]	1230.33	2 ⁺			
		4463.0	73 [#]	146.4	3 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
4638.9	1,2	4177.4	100 [#] 8	461.2	1 ⁺	I_γ : other: 100 13 (1983Wa27).
		4492.2	58 10	146.4	3 ⁺	I_γ : weighted average of 64 8 (1977Da02) and 41 13 (1983Wa27).
4695.7		2537.7	100	2157.9	2 ⁺	I_γ : other: 100 (1977Da02).
		2808.3	<4	1887.29	2 ⁺	
		3465.2	<4	1230.33	2 ⁺	
		4030.1	<5	665.3	1	
		4234.2	<4	461.2	1 ⁺	
		4695.4	<2	0	0 ⁺	
4717.4		2106.3	16 5	2611.03	3	
		2341.6	11.1 32	2375.7	4	
		2536.2	35 5	2181.10	3	I_γ : weighted average of 40 5 (1977Da02) and 30 5 (1983Wa27).
		2559.4	11 5	2157.9	2 ⁺	I_γ : weighted average of 7.9 32 (1977Da02) and 18 5 (1983Wa27).
		2830.0	<3.3	1887.29	2 ⁺	
		3486.9	<4.9	1230.33	2 ⁺	
		4051.8	<4.9	665.3	1	
		4255.9	<4.9	461.2	1 ⁺	
		4570.7	100 5	146.4	3 ⁺	I_γ : other: 100 5 (1977Da02).
4824.41		1192.7	6.5 16	3631.7	5	
		1224.11	37 13	3600.28	4	I_γ : unweighted average of 24 5 (1977Da02) and 50.0 32 (1983Wa27).
		2643.20	4.8 16	2181.10	3	
		3593.88	<11	1230.33	2 ⁺	
		4362.9	<3.2	461.2	1 ⁺	
		4677.7	100.0 32	146.4	3 ⁺	I_γ : other: 100 5 (1977Da02).
		4824.04	<3.2	0	0 ⁺	
4941.9		2330.8	43 [#] 19	2611.03	3	
		2361.5	45 [#] 19	2580.3		
		4941.5	100 [#] 38	0	0 ⁺	I_γ : other: 100 and another $I_\gamma=43$ decays to unknown final levels (1983Wa27).
4957.3		3069.9	<3.9	1887.29	2 ⁺	
		3726.8	100 10	1230.33	2 ⁺	I_γ : other: 100 10 (1977Da02).
		4291.7	<5.9	665.3	1	
		4495.8	<5.9	461.2	1 ⁺	
		4810.5	94 10	146.4	3 ⁺	I_γ : weighted average of 92 10 (1977Da02) and 96 10 (1983Wa27).
		4956.9	<3.9	0	0 ⁺	
4995.6		2837.6	100	2157.9	2 ⁺	
5020?		2862 ^a	100 [#]	2157.9	2 ⁺	
5171.6	4	1571.3	100	3600.28	4	
		5024.8	100 [#]	146.4	3 ⁺	
5386.8		1786.5	100	3600.28	4	
		1841.7 ^a		3545.08	3	The only γ branch from 1977Da02 but is not present in the more complete data from 1983Wa27.
		4156.2	<2	1230.33	2 ⁺	
		4721.2	<2	665.3	1	
		4925.2	<4	461.2	1 ⁺	
		5240.0	<3	146.4	3 ⁺	
		5386.3	<3	0	0 ⁺	
5540.8		1940.5	100 8	3600.28	4	I_γ : other: 100 (1977Da02).
		1995.7	56 8	3545.08	3	
		4875.1	<9.4	665.3	1	
		5079.2	<4.7	461.2	1 ⁺	
		5393.9	<7.8	146.4	3 ⁺	
		5540.3	<6.3	0	0 ⁺	
5577.6		1061.9	9 3	4515.7		E_γ, I_γ : 1050 30 from 1964Gl04.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)						Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	
5577.6		1700 ^a	7 3			E_γ, I_γ : 1700 40 from 1964Gl04. The authors assigned a 3.87 MeV level as the final level, which has not been observed by any other work.
		1977.3	28.7 [@] 66	3600.28	4	I_γ : other: 38 (1983Wa27).
		2032.5	100 [@]	3545.08	3	E_γ : other: 2020 30 (1964Gl04). I_γ : others: 100 (1983Wa27), 100 14 (1964Gl04).
		2448.4	<3.9 [@]	3129.12		
		2856.2	22.6 [@] 23	2721.3	2	I_γ : other: 21 (1983Wa27).
		2966.4	0.71	2611.03	3	Observed in 1983Wa27 but not in 2011Fr04.
		2997.2	<1.2 [@]	2580.3		
		3201.7	3.0 [@] 12	2375.7	4	I_γ : other: 0.95 (1983Wa27).
		3690.1	5.5 [@] 10	1887.29	2 ⁺	I_γ : other: 4.8 (1983Wa27).
		4347.0	10.3 [@] 14	1230.33	2 ⁺	I_γ : other: 10 (1983Wa27).
		4911.9	<0.71	665.3	1	Observed in 1983Wa27 but not in 2011Fr04.
		5116.0	<1.4	461.2	1 ⁺	Observed in 1983Wa27 but not in 2011Fr04.
		5430.7	61.9 [@] 55	146.4	3 ⁺	I_γ : others: 62 (1983Wa27), 51 7 (1964Gl04).
		5577.1	<0.24	0	0 ⁺	I_γ : other: <8 (1964Gl04); not observed in 2011Fr04.
5635.75	1,2	996.8	0.44	4638.9	1,2	Observed in 1983Wa27 but not in 2011Fr04.
		1029.9 ^a	4 1	4605.8		E_γ, I_γ : 1040 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
		1218.4	0.44	4417.3		Observed in 1983Wa27 but not in 2011Fr04.
		1652.7 ^a	3 1	3983.0	3	E_γ, I_γ : 1650 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
		1844.0	3.7 [@] 7	3791.7		I_γ : other: 2.5 (1983Wa27).
		1862.0	1.0	3773.72	1	Observed in 1983Wa27 but not in 2011Fr04.
		1975.7	<0.3 [@]	3660.0		
		1989.4	<0.4 [@]	3646.3		
		2004.0	<0.3 [@]	3631.7	5	
		2506.53	3.9 [@] 4	3129.12		I_γ : other: 4.1 (1983Wa27).
		2914.3	1.0 [@] 3	2721.3	2	I_γ : other: 0.73 (1983Wa27).
		3024.58	<0.4 [@]	2611.03	3	
		3055.3	12.2 [@] 7	2580.3		I_γ : others: 15 (1983Wa27), 17 3 (1964Gl04). $A_2 = -0.13$ 6, $A_4 = +0.05$ 7 (1983Wa27).
		3259.9	<0.3 [@]	2375.7	4	
		3454.46	<0.5 [@]	2181.10	3	
		3477.7	0.6 [@] 3	2157.9	2 ⁺	
		3748.24	1.8 [@] 3	1887.29	2 ⁺	I_γ : other: 1.9 (1983Wa27).
		4405.11	0.6 [@] 4	1230.33	2 ⁺	I_γ : other: 0.58 (1983Wa27).
		4970.1	15.4 [@] 10	665.3	1	E_γ : other: 4960 40 (1964Gl04). I_γ : others: 16 (1983Wa27), 11 3 (1964Gl04).
		5174.1	2.2 [@] 7	461.2	1 ⁺	I_γ : others: 2.9 (1983Wa27), 2 1 (1964Gl04).
		5488.9	<0.65 [@]	146.4	3 ⁺	I_γ : others: <0.44 (1983Wa27).
		5635.25	100 [@]	0	0 ⁺	I_γ : others: 100 (1983Wa27), 100 13 (1964Gl04). $A_2 = +0.27$ 3, $A_4 = +0.05$ 4 (1983Wa27).
5673.0		1255.7	0.41	4417.3		Observed in 1983Wa27 but not in 2011Fr04.
		1318.8	0.81	4354.2	(1)	Observed in 1983Wa27 but not in 2011Fr04.
		1732.9	2.0	3940.1		Observed in 1983Wa27 but not in 2011Fr04.
		1881.2	<6.7 [@]	3791.7		I_γ : other: 1.9 (1983Wa27).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5673.0	1899.2	<4.6@	3773.72	1	I_γ : other: 0.54 (1983Wa27).
	2012.9	<3.4@	3660.0		
	2026.6	<3.3@	3646.3		
	2041.2	<4.0@	3631.7	5	
	2072.7	<5.5@	3600.28	4	
	2289.6	<1.3@	3383.3		
	2543.8	2.8	3129.12		
					E_γ : other: 2520 30 (1964GI04).
					I_γ : other: 4 2 (1964GI04).
					Observed in 1983Wa27 but not in 2011Fr04.
	2951.6	<3.8@	2721.3	2	E_γ : other: 3070 30 (1964GI04).
	3061.8	<1.5@	2611.03	3	
	3092.6	11.6@ 26	2580.3		
	3297.1	<3.0@	2375.7	4	I_γ : others: 6.6 (1983Wa27), 8 2 (1964GI04).
	3491.7	<3.2@	2181.10	3	
	3514.9	8.8@ 24	2157.9	2 ⁺	E_γ : other: 3500 30 (1964GI04).
					I_γ : others: 10 (1983Wa27), 14 3 (1964GI04).
	3785.5	<4.4@	1887.29	2 ⁺	E_γ : others: 3760 40 (1964GI04).
					I_γ : others: 1.8 (1983Wa27), 3 2 (1964GI04).
	4442.4 ^a	3 2	1230.33	2 ⁺	E_γ, I_γ : 4440 60 from 1964GI04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
	5007.3	6.4@ 34	665.3	1	E_γ : others: 4980 60 (1964GI04).
					I_γ : others: 8.2 (1983Wa27), 8 3 (1964GI04).
	5211.4	<3.8@	461.2	1 ⁺	I_γ : others: <0.54 (1983Wa27).
	5526.1	<1.4@	146.4	3 ⁺	I_γ : others: <0.41 (1983Wa27).
	5672.5	100@	0	0 ⁺	I_γ : others: 100 (1983Wa27), 100 20 (1964GI04).
5763.3	1823.2	2.2	3940.1		E_γ : other: 3600 40 (1964GI04).
	2379.9	17	3383.3		
	2634.1	1.5	3129.12		
	3605.2	100	2157.9	2 ⁺	
					I_γ : other: 100 18 (1964GI04).
	3875.8	1.3	1887.29	2 ⁺	I_γ : other: 56 11 (1964GI04).
	5097.6	<2.4	665.3	1	
	5301.7	3.0	461.2	1 ⁺	
	5616.4	16	146.4	3 ⁺	
	5762.8	44	0	0 ⁺	E_γ : other: 2390 30 (1964GI04).
5785.6	2011.8	0.8	3773.72	1	
	2402.2	52	3383.3		I_γ : other: 44 10 (1964GI04).
	2451.5	1.2	3334.0		E_γ : other: 3620 30 (1964GI04).
	3064.2	0.6	2721.3	2	
	3174.4	2.8	2611.03	3	
	3604.3	3	2181.10	3	
	3627.5	32	2157.9	2 ⁺	I_γ : other: 40 8 (1964GI04).
	3898.1	1.6	1887.29	2 ⁺	E_γ : other: 4550 50 (1964GI04).
	4554.9	6	1230.33	2 ⁺	
					I_γ : other: 26 8 (1964GI04).
	5119.9	<0.6	665.3	1	I_γ : other: <1 (1964GI04).
	5324.0	<1.2	461.2	1 ⁺	I_γ : other: 100 16 (1964GI04).
	5638.7	100	146.4	3 ⁺	

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5785.6	5785.1	<0.6	0	0^+	I_γ : other: <20 (1964Gl04).
5805.9	1200.1	9.4	4605.8		
	1658.1	3.2	4147.8		
	1822.9	1.9	3983.0	3	
	2205.5	1.6	3600.28	4	
	2260.7	1.3	3545.08	3	
	3225.4	58	2580.3		I_γ : other: 58 12 (1964Gl04).
	3430.0	3.9	2375.7	4	
	3918.4	1.3	1887.29	2^+	E_γ : other: 3920 40 (1964Gl04). I_γ : other: 16 5 (1964Gl04).
	4575.2	13	1230.33	2^+	
	5140.2	58	665.3	1	E_γ : other: 5120 50 (1964Gl04). I_γ : other: 38 10 (1964Gl04).
	5344.3	100	461.2	1^+	E_γ : other: 5330 40 (1964Gl04). I_γ : other: 100 21 (1964Gl04).
	5659.0	65	146.4	3^+	I_γ : other: 72 16 (1964Gl04).
	5805.4	6.1	0	0^+	I_γ : other: <30 (1964Gl04).
5852.91	1705.1	18	4147.8		
	1869.9	2.8	3983.0	3	
	2079.1	0.4	3773.72	1	
	2252.55	8.4	3600.28	4	
	2307.75	13	3545.08	3	E_γ : other: 2300 50 (1964Gl04). I_γ : other: 19 4 (1964Gl04).
	2469.5	3.6	3383.3		
	3131.5	7.6	2721.3	2	
	3241.71	0.6	2611.03	3	
	3671.60	2.4	2181.10	3	E_γ : other: 3680 50 (1964Gl04). I_γ : other: 16 7 (1964Gl04).
	3694.8	8.4	2157.9	2^+	
	3965.37	6.2	1887.29	2^+	E_γ : other: 3950 100 (1964Gl04). I_γ : other: 24 7 (1964Gl04).
	4622.24	28	1230.33	2^+	E_γ : other: 4650 50 (1964Gl04). I_γ : other: 36 11 (1964Gl04).
	5187.2	<1	665.3	1	I_γ : other: 7 3 (1964Gl04).
	5391.3	<1	461.2	1^+	
	5706.0	100	146.4	3^+	I_γ : other: 100 20 (1964Gl04).
	5852.37	<0.4	0	0^+	I_γ : other: <15 (1964Gl04).
5897.2	1291.4	5	4605.8		
	1749.4 ^a	24 8	4147.8		E_γ, I_γ : 1720 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27.
	2352.0	5	3545.08	3	
	2513.8	1.2	3383.3		
	2563.1	0.88	3334.0		
	3175.7	5	2721.3	2	
	3316.7	2.4	2580.3		
	4009.7	11	1887.29	2^+	
	4666.5	41	1230.33	2^+	
	5231.5	32	665.3	1	E_γ : other: 5200 50 (1964Gl04). I_γ : other: 43 18 (1964Gl04).
	5435.5	100	461.2	1^+	E_γ : other: 5400 50 (1964Gl04). I_γ : other: 100 30 (1964Gl04).
	5750.3	85	146.4	3^+	E_γ : other: 5700 50 (1964Gl04). I_γ : other: 73 20 (1964Gl04).
	5896.7	5	0	0^+	E_γ : other: 5850 60 (1964Gl04). I_γ : other: 28 10 (1964Gl04).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5940.8	1 to 3	983.5 ^a	9 3	4957.3		E_γ, I_γ : 980 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27.
		1245.1	0.85	4695.7		
		2557.4	6.3	3383.3		
		3329.6	0.56	2611.03	3	
		3360.3	1.6	2580.3		
		3564.9	0.56	2375.7	4	
		3782.7	100	2157.9	2 ⁺	I_γ : other: 100 18 (1964Gl04). $A_2=+0.05$ 2, $A_4=-0.05$ 3 (1983Wa27).
		4053.3	1.1	1887.29	2 ⁺	
		5275.1	0.99	665.3	1	
		5479.1	21	461.2	1 ⁺	E_γ : other: 5470 50 (1964Gl04). I_γ : other: 12 5 (1964Gl04).
		5793.9	6.8	146.4	3 ⁺	I_γ : other: <5 (1964Gl04).
		5940.2	0.99	0	0 ⁺	I_γ : other: <5 (1964Gl04).
6030.1		2646.7	4.7	3383.3		
		3872.0	<0.33	2157.9	2 ⁺	
		4142.5	<0.67	1887.29	2 ⁺	
		4799.4	<0.56	1230.33	2 ⁺	
		5364.4	<0.56	665.3	1	
		5568.4	<0.89	461.2	1 ⁺	
		5883.2	6.4	146.4	3 ⁺	
		6029.5	100	0	0 ⁺	I_γ : other: 100 (1964Gl04).
6088.94	1,2	2297.2	0.33	3791.7		
		2315.1	0.56	3773.72	1	
		3367.5	4.7	2721.3	2	
		3508.5	0.33	2580.3		
		3930.8	<0.33	2157.9	2 ⁺	
		4201.37	2.6	1887.29	2 ⁺	I_γ : other: 13 (1964Gl04).
		4858.24	<0.67	1230.33	2 ⁺	
		5423.2	<1.1	665.3	1	I_γ : other: 4.7 (1964Gl04).
		5627.2	2.7	461.2	1 ⁺	
		5942.0	<0.33	146.4	3 ⁺	
		6088.36	100	0	0 ⁺	I_γ : other: 100 (1964Gl04). $A_2=+0.53$ 7, $A_4=-0.00$ 8 (1983Wa27).
6136.2	1 to 3	1718.9	<1.8 [#]	4417.3		
		2153.1	<1.8 [#]	3983.0	3	
		2752.8	1.82 91	3383.3		I_γ : other: 1.4 (1983Wa27).
		3414.7	1.82 91	2721.3	2	I_γ : other: 1.9 (1983Wa27).
		3525.0	3.6 18	2611.03	3	I_γ : other: 3.8 (1983Wa27).
		3555.7	1.82 91	2580.3		I_γ : other: 1.2 (1983Wa27).
		3954.9	1.82 91	2181.10	3	I_γ : other: 1.7 (1983Wa27).
		3978.1	10.9 11	2157.9	2 ⁺	I_γ : other: 11 (1983Wa27).
		4905.5	40 4	1230.33	2 ⁺	I_γ : other: 35 (1983Wa27), 48 (1964Gl04). $A_2=-0.32$ 4, $A_4=+0.02$ 5 (1983Wa27).
		5470.4	100 10	665.3	1	I_γ : other: 100 (1983Wa27), 100 (1964Gl04). $A_2=+0.28$ 5, $A_4=-0.01$ 6 (1983Wa27).
		5674.5	<0.52	461.2	1 ⁺	I_γ : other: 21 (1964Gl04).
		5989.2	20 2	146.4	3 ⁺	I_γ : other: 17 (1983Wa27), 23 (1964Gl04).
6141.8	1,2	1626.1	1.16 [#] 58	4515.7		I_γ : other: 1.3 (1983Wa27).
		1724.5	0.58 [#] 29	4417.3		I_γ : other: 1.5 (1983Wa27).
		1787.6	1.74 [#] 93	4354.2	(1)	I_γ : other: 1.7 (1983Wa27).
		2368.0	2.3 [#] 12	3773.72	1	I_γ : other: 2.6 (1983Wa27).
		2596.6	0.58 [#] 29	3545.08	3	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	Comments
6141.8	1,2	2758.4	<0.58 [#]	3383.3				
		3420.3	6.4 [#] 7	2721.3	2			I_γ : other: 6.6 (1983Wa27).
		3561.3	1.16 [#] 58	2580.3				I_γ : other: 1.4 (1983Wa27).
		3983.7	0.58	2157.9	2 ⁺			
		4911.1	2.3 [#] 12	1230.33	2 ⁺			
		5476.0	<0.23	665.3	1			
		5680.1	<0.12	461.2	1 ⁺			
		5994.8	<0.35	146.4	3 ⁺			
		6141.2	100 [#] 10	0	0 ⁺			I_γ : other: 100 (1983Wa27), 100 (1964Gl04). $A_2=-0.14$ 3, $A_4=+0.08$ 3 (1983Wa27).
6169.4	3	1653.7	4.4 [#] 24	4515.7				I_γ : other: 20 (1983Wa27).
		2029.6	<2.9 [#]	4139.7	2			I_γ : other: 10 (1983Wa27).
		2093.1	15.5 60	4076.2	4	D(+Q)	+0.03 5	I_γ : weighted average of 14.7 74 (1977Da02) and 16 6 (1971Hy02). Other: 15 (1983Wa27).
								Mult., δ : or $\delta<-7$ from $\gamma\gamma(\theta)$ in 1971Hy02.
								$A_2=-0.12$ 5 (1971Hy02).
		2186.3	100 [#] 10	3983.0	3	D(+Q)	-0.03 +5-4	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 25 (1969Gr29), 100 19 (1971Hy02).
								Mult., δ : weighted average of -0.05 6 (1971Hy02) and -0.02 +5-4 (1977Da03).
								Other: $\delta=+1.3$ 2 (1971Hy02).
								$A_2=+0.27$ 3, $A_4=+0.01$ 3 (weighted average of values from 1983Wa27 and 1977Da03).
								$A_2=+0.25$ 6, $A_4=+0.07$ 7 (1983Wa27);
								$A_2=+0.27$ 3, $A_4=0.00$ 3 (1977Da03);
								$A_2=+0.25$ 3 (1971Hy02).
		2569.0	23.6 24	3600.28	4			I_γ : weighted average of 23.5 24 (1977Da02), 39 22 (1969Gr29), and 22 13 (1971Hy02).
								Other: 23 (1983Wa27).
		2624.2	16.5 60	3545.08	3	D(+Q)	-0.05 +19-14	I_γ : weighted average of 14.7 74 (1977Da02), 22 11 (1969Gr29), and 16 6 (1971Hy02).
								Other: 15 (1983Wa27), 28 (1964Gl04).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=+0.25$ 5, $A_4=+0.01$ 5 (1977Da03).
								$A_2=+0.25$ 5, $A_4=+0.01$ 5 (1977Da03);
								$A_2=+0.28$ 5 (1971Hy02).
		3447.9	20.7 21	2721.3	2			I_γ : weighted average of 20.6 21 (1977Da02) and 25 13 (1971Hy02). Other: 25 (1983Wa27).
		3558.2	7.3 30	2611.03	3			I_γ : weighted average of 5.9 30 (1977Da02) and 13 6 (1971Hy02). Other: 8.8 (1983Wa27).
								$A_2=+0.10$ 10 (1971Hy02).
		3588.9	<31	2580.3				I_γ : from 1969Gr29.
		3793.5	<19	2375.7	4			I_γ : from 1969Gr29.
		3988.1	15.5 60	2181.10	3			I_γ : weighted average of 14.7 74 (1977Da02) and 16 6 (1971Hy02). Other: 15 (1983Wa27).
								$A_2=+0.28$ 7 (1971Hy02).
		4011.3	1.47 [#] 74	2157.9	2 ⁺			I_γ : other: 2.1 (1983Wa27), <5.6 (1969Gr29).
		4281.8	7.1 31	1887.29	2 ⁺			I_γ : weighted average of 8.8 44 (1977Da02) and 6.3 31 (1971Hy02). Other: 6.4

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6169.4	3	4938.7	10.7 31	1230.33	2 ⁺	D(+Q)	0.00 1	(1983Wa27), <8.3 (1969Gr29). A ₂ =-0.21 12 (1971Hy02). I _γ : weighted average of 11.8 59 (1977Da02), 17 8 (1969Gr29), and 9.4 31 (1971Hy02). Other: 9.4 (1983Wa27). Mult.,δ: from γγ(θ) in 1971Hy02. A ₂ =-0.24 10 (1971Hy02). I _γ : other: <5.6 (1969Gr29). I _γ : other: 35 (1964Gl04), <5.6 (1969Gr29). I _γ : weighted average of 74 8 (1977Da02), 100 17 (1969Gr29), and 91 22 (1971Hy02). Other: 85 (1983Wa27), 70 (1964Gl04). Mult.: from γγ(θ) in 1971Hy02 and 1977Da03. δ: from 1971Hy02. Other: δ=-0.03 5 (1977Da03). A ₂ =+0.264 24, A ₄ =-0.01 3 (weighted average of values from 1983Wa27 and 1977Da03). A ₂ =+0.27 4, A ₄ =-0.03 5 (1983Wa27); A ₂ =+0.26 3, A ₄ =0.00 3 (1977Da03); A ₂ =+0.28 3 (1971Hy02). I _γ : other: <8.3 (1969Gr29). I _γ : other: 12 (1983Wa27). I _γ : other: 3.1 (1983Wa27). I _γ : other: 7.4 (1983Wa27). I _γ : other: 5.5 (1983Wa27). I _γ : other: 0.71 (1983Wa27). I _γ : other: 0.95 (1983Wa27). I _γ : other: 8.6 (1983Wa27). I _γ : weighted average of 18.2 18 (1977Da02) and 17 4 (1969Gr29). Other: 19 (1983Wa27), 22 (1964Gl04). I _γ : from 1969Gr29. I _γ : weighted average of 8.0 41 (1977Da02) and 5.7 38 (1969Gr29). Other: 10 (1983Wa27), 22 (1964Gl04). I _γ : weighted average of 9.1 46 (1977Da02) and 15 4 (1969Gr29). Other: 9.8 (1983Wa27), 14 (1964Gl04). I _γ : weighted average of 25.0 25 (1977Da02) and 25 4 (1969Gr29). Other: 23 (1983Wa27), 18 (1964Gl04). A ₂ =-0.07 4, A ₄ =-0.05 5 (1983Wa27); A ₂ =-0.06 13 with A ₄ =0; A ₂ =+0.09 19, A ₄ =-0.23 24 (1969Gr29). I _γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 8 (1969Gr29). Mult.: 1969Gr29 states that γ(θ) of E _p (lab)=1071, 6181 level to 666, 1 ⁺ can be fit with J(6181)=1,2 for a variety of mixing ratios. A ₂ =+0.00 3 A ₄ =+0.07 4 (1983Wa27); A ₂ =-0.06 5 with A ₄ =0; A ₂ =-0.04 7, A ₄ =-0.03 9 (1969Gr29). I _γ : weighted average of 10.2 52 (1977Da02) and 11 4 (1969Gr29). Other: 11 (1983Wa27). I _γ : weighted average of 20.5 21 (1977Da02) and
		5503.6	<0.61	665.3	1			
		5707.7	<0.91	461.2	1 ⁺			
		6022.4	80 8	146.4	3 ⁺	D(+Q)	-0.02 3	
6181.27	1,2	6168.8	<0.91	0	0 ⁺			
		1855.33	11.4 [#] 57	4325.89				
		2217.1	2.3 [#] 12	3964.1				
		2797.9	4.6 [#] 23	3383.3				
		2847.1	6.8 [#] 34	3334.0				
		3052.00	1.14 [#] 57	3129.12				
		3459.8	1.14 [#] 57	2721.3	2			
		3570.04	9.1 [#] 46	2611.03	3			
		3600.8	18.0 18	2580.3				
		3805.3	<0.94	2375.7	4			
		3999.92	1.4	2181.10	3			
		4023.1	6.8 38	2157.9	2 ⁺			
		4293.69	12.5 40	1887.29	2 ⁺			
		4950.55	25.0 25	1230.33	2 ⁺			
		5515.5	100 [#] 10	665.3	1	(D+Q)		
		5719.6	10.7 40	461.2	1 ⁺			
		6034.3	20.1 21	146.4	3 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								15.1 76 (1969Gr29). Other: 24 (1983Wa27), 28 (1964Gl04).
								Mult.: 1969Gr29 states that $\gamma(\theta)$ of $E_p(\text{lab})=1071$, 6181 level to 146, 3^+ can be fit with $J(6181)=1,2$ for a variety of mixing ratios.
								$A_2=-0.26$ 13 with $A_4=0$; $A_2=-0.24$ 23, $A_4=-0.02$ 25 (1969Gr29).
6181.27	1,2	6180.67	1.9	0	0^+			I_γ : other: <0.94 (1969Gr29).
6208.2	4	1383.8	$1.04^\#$ 52	4824.41				I_γ : other: 1.9 (1983Wa27).
		2131.9	18.6 19	4076.2	4			I_γ : weighted average of 18.8 19 (1977Da02) and 14.3 82 (1971Hy02). Other: 18 (1983Wa27).
		2225.1	10.1 38	3983.0	3			I_γ : weighted average of 7.3 38 (1977Da02), 26 15 (1969Gr29), and 12 4 (1971Hy02). Other: 13 (1983Wa27), 19 (1964Gl04).
		2561.8	$1.04^\#$ 52	3646.3				
		2576.4	20.8 21	3631.7	5			I_γ : weighted average of 20.8 21 (1977Da02) and 20 10 (1971Hy02). Other: 19 (1983Wa27).
		2607.8	$100^\#$ 10	3600.28	4	D(+Q)	-0.03 5	I_γ : other: 100 (1983Wa27), 100 20 (1969Gr29), 100 21 (1971Hy02).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1971Hy02.
								$A_2=+0.31$ 1 (1971Hy02).
		2663.0	49 5	3545.08	3	D(+Q)	0.00 2	I_γ : weighted average of 48 5 (1977Da02), 56 17 (1969Gr29), and 49 12 (1971Hy02). Other: 49 (1983Wa27), 100 (1964Gl04).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1971Hy02.
								$A_2=-0.22$ 2 (1971Hy02).
		3627.7	<7.3	2580.3				I_γ : from 1969Gr29.
		3832.3	$4.2^\#$ 21	2375.7	4			I_γ : other: 4.5 (1983Wa27), <7.3 (1969Gr29).
		4026.8	7.6 21	2181.10	3			I_γ : weighted average of 6.3 31 (1977Da02) and 8.2 21 (1971Hy02). Other: 5.7 (1983Wa27).
								$A_2=-0.24$ 7 (1971Hy02).
		4050.0	<7.3	2157.9	2^+			I_γ : from 1969Gr29.
		4320.6	0.43	1887.29	2^+			I_γ : other: 6 (1964Gl04), <3.6 (1969Gr29).
		4977.5	<1.8	1230.33	2^+			I_γ : from 1969Gr29.
		5542.4	<0.21	665.3	1			I_γ : other: <1.8 (1969Gr29).
		5746.5	<0.21	461.2	1^+			I_γ : other: <5.5 (1969Gr29).
		6061.2	$1.04^\#$ 52	146.4	3^+			I_γ : other: 1.7 (1983Wa27), <3.6 (1969Gr29).
		6207.6	<0.21	0	0^+			I_γ : other: <1.8 (1969Gr29).
6219.3		1501.9	$100^\#$ 10	4717.4				
		1893.4	$3.9^\#$ 21	4325.89				
		2572.9	$18.0^\#$ 18	3646.3				
		2587.5	$2.6^\#$ 13	3631.7	5			
		2885.2	$7.7^\#$ 39	3334.0				
		3608.1	$5.1^\#$ 26	2611.03	3			
		3843.4	$11.5^\#$ 59	2375.7	4			
		4037.9	$46^\#$ 5	2181.10	3			
		4988.6	$28.2^\#$ 28	1230.33	2^+			
		6072.3	$33.3^\#$ 33	146.4	3^+			
6229.5	1	1590.6	4.5 18	4638.9	1,2			I_γ : weighted average of 3.3 18 (1977Da02) and 6.0 20 (1971Hy02). Other: 3.9 (1983Wa27).
								$A_2=-0.18$ 15 (1971Hy02).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6229.5	1	1713.8	24.0 25	4515.7				I_γ : weighted average of 24.4 25 (1977Da02) and 22 6 (1971Hy02). Other: 29 (1983Wa27). $A_2=+0.05$ 3 (1971Hy02).
		1812.2	6.2 20	4417.3				I_γ : weighted average of 6.7 33 (1977Da02) and 6.0 20 (1971Hy02). Other: 6.3 (1983Wa27). $A_2=+0.32$ 16 (1971Hy02).
		1875.2	20 2	4354.2	(1)			I_γ : weighted average of 20 2 (1977Da02) and 16 6 (1971Hy02). Other: 23 (1983Wa27). $A_2=-0.24$ 5 (1971Hy02).
		1903.6	2.2 [#] 11	4325.89				I_γ : other: 2.2 (1983Wa27).
		2089.7	13.2 13	4139.7	2			I_γ : weighted average of 13.3 13 (1977Da02) and 12 4 (1971Hy02). Other: 12 (1983Wa27). $A_2=+0.01$ 9 (1971Hy02).
		2455.7	31.6 33	3773.72	1	D(+Q)	+0.03 5	I_γ : weighted average of 33.3 33 (1977Da02) and 26 6 (1971Hy02). Other: 44 (1983Wa27), 71 (1964Gl04). Mult., δ : from $\gamma\gamma(\theta)$ in 1971Hy02. $A_2=-0.23$ 3 (1971Hy02).
		3100.2	2.4	3129.12				
		3508.0	100 [#] 10	2721.3	2	D(+Q)	-0.05 5	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 12 (1971Hy02). Mult., δ : from $\gamma\gamma(\theta)$ in 1971Hy02. $A_2=+0.01$ 5 (1971Hy02).
		4071.3	10.4 40	2157.9	2 ⁺			I_γ : weighted average of 11.1 56 (1977Da02) and 10 4 (1971Hy02). Other: 9.5 (1983Wa27), 29 (1964Gl04). $A_2=-0.13$ 10 (1971Hy02).
		4341.9	39	1887.29	2 ⁺			I_γ : from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 1977Da02.
		5563.7	<0.49	665.3	1			I_γ : other: 24 (1964Gl04).
		5767.8	2.4 10	461.2	1 ⁺			I_γ : weighted average of 4.4 22 (1977Da02) and 2.0 10 (1971Hy02). Other: 5.1 (1983Wa27).
		6082.5	<0.49	146.4	3 ⁺			
		6228.9	1.9 8	0	0 ⁺			I_γ : weighted average of 3.3 18 (1977Da02) and 1.6 8 (1971Hy02). Other: 5.6 (1983Wa27).
6266.6		3545.1	5.8 [#] 29	2721.3	2			I_γ : other: 4.9 (1983Wa27).
		4085.2	5.8	2181.10	3			
		4108.4	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).
		4379.0	11.6 [#] 12	1887.29	2 ⁺			I_γ : other: 11 (1983Wa27).
		5035.9	8.6	1230.33	2 ⁺			
		5600.8	27.5 [#] 28	665.3	1			I_γ : other: 30 (1983Wa27).
		5804.9	<2.3	461.2	1 ⁺			
		6119.6	72	146.4	3 ⁺			I_γ : other: <33 (1977Da02).
		6266.0	<2.6	0	0 ⁺			
6273.3	3	1826.7	0.31	4446.6	4			
		2197.0	1.11 [#] 64	4076.2	4			I_γ : other: 0.62 (1983Wa27).
		2309.1	1.27 [#] 64	3964.1				I_γ : other: 1.4 (1983Wa27).
		2939.2	0.77	3334.0				
		3662.1	12.7 [#] 13	2611.03	3			I_γ : other: 10 (1983Wa27).
		3692.8	<0.99	2580.3				I_γ : from 1969Gr29.
		3897.4	100 [#] 10	2375.7	4	D+Q	-0.08 5	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 5 (1969Gr29). Mult., δ : from $\gamma\gamma(\theta)$ $E_p(\text{lab})=1165\rightarrow 2376\rightarrow 146$ γ

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)						Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	
6273.3	3	4091.9	25.4 [#] 26	2181.10	3	cascade (1969Gr29). $A_2=0.00$ 2 $A_4=-0.04$ 2 (1983Wa27); $A_2=-0.01$ 3 with $A_4=0$; $A_2=+0.02$ 3, $A_4=-0.06$ 4 (1969Gr29).
		4115.1	1.80 80	2157.9	2 ⁺	I_γ : other: 22 (1983Wa27). $A_2=+0.18$ 4, $A_4=-0.05$ 5 (1983Wa27).
		4385.7	11.1 11	1887.29	2 ⁺	I_γ : weighted average of 1.59 80 (1977Da02) and 6.2 37 (1969Gr29). Other: 1.4 (1983Wa27), 23 (1964Gl04).
		5042.6	1.2 12	1230.33	2 ⁺	I_γ : weighted average of 11.1 11 (1977Da02) and 11.1 13 (1969Gr29). Other: 9.9 (1983Wa27), 19 (1964Gl04). $A_2=0.00$ 7, $A_4=+0.05$ 9 (1983Wa27).
		5607.5	<0.15	665.3	1	I_γ : weighted average of 0.79 40 (1977Da02) and 4.9 13 (1969Gr29).
		5811.6	<0.15	461.2	1 ⁺	I_γ : other: 8 (1964Gl04), 1.24 62 (1969Gr29).
		6126.3	4.8 [#] 24	146.4	3 ⁺	I_γ : other: 6 (1964Gl04), <0.99 (1969Gr29).
		6272.7	<0.15	0	0 ⁺	I_γ : other: 7.7 (1983Wa27), <4.9 (1969Gr29).
		6322.0	1,2	4605.8		I_γ : other: <0.74 (1969Gr29).
		1967.7	2.3 [#] 12	4354.2	(1)	I_γ : other: 1.6 (1983Wa27).
		2245.7	6.8 [#] 34	4076.2	4	I_γ : other: 2.0 (1983Wa27).
		2690.2	2.3 [#] 12	3631.7	5	
		2776.8	9.1 [#] 46	3545.08	3	I_γ : other: 8.4 (1983Wa27).
		2938.6	1.14 [#] 57	3383.3		
6361.3	1,2	3600.5	12.5 [#] 14	2721.3	2	I_γ : other: 10 (1983Wa27).
		3710.8	4.6 [#] 23	2611.03	3	I_γ : other: 0.61 (1983Wa27).
		3741.5	20.4 21	2580.3		I_γ : weighted average of 20.5 21 (1977Da02) and 19 7 (1969Gr29). Other: 17 (1983Wa27).
		3946.1	<4.8	2375.7	4	$A_2=-0.23$ 5, $A_4=-0.07$ 6 (1983Wa27). I_γ : from 1969Gr29.
		4140.6	2.3 [#] 12	2181.10	3	I_γ : other: 1.6 (1983Wa27), <6.5 (1969Gr29).
		4434.4	2.3 [#] 12	1887.29	2 ⁺	I_γ : other: 1.4 (1983Wa27), <8.1 (1969Gr29).
		5091.3	4.6 [#] 23	1230.33	2 ⁺	I_γ : other: 3.5 (1983Wa27), <4.8 (1969Gr29).
		5656.2	34 4	665.3	1	I_γ : weighted average of 36 4 (1977Da02) and 31 5 (1969Gr29). Other: 41 (1983Wa27), 28 (1964Gl04).
		5860.3	16.5 18	461.2	1 ⁺	I_γ : weighted average of 18.2 18 (1977Da02) and 11.3 32 (1969Gr29). Other: 16 (1983Wa27).
		6175.0	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 11 (1969Gr29). $A_2=-0.04$ 3, $A_4=+0.01$ 3 (1983Wa27).
		6321.4	<0.61	0	0 ⁺	I_γ : other: <4.8 (1969Gr29).
		2977.9	71 [#] 7	3383.3		
		4203.1	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1964Gl04).
		5130.6	25.0 [#] 25	1230.33	2 ⁺	
6370.2	2	5899.6	14.8 [#] 16	461.2	1 ⁺	
		6214.3	17.1 [#] 18	146.4	3 ⁺	
		1412.9	3.5 [#] 17	4957.3		
		1731.3	1.72 [#] 86	4638.9	1,2	I_γ : other: 3.3 (1983Wa27).
		1854.5	3.5 [#] 17	4515.7		I_γ : other: 3.9 (1983Wa27).
		2015.9	8.6 [#] 45	4354.2	(1)	I_γ : other: 6.1 (1983Wa27).
		2387.1	1.72 [#] 86	3983.0	3	I_γ : other: 1.5 (1983Wa27).
		2406.0	1.72 [#] 86	3964.1		I_γ : other: 1.5 (1983Wa27).
		2825.0	6.9 [#] 35	3545.08	3	I_γ : other: 6.7 (1983Wa27).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#), [1977Da02](#), [1992Ka39](#) (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	Comments
6370.2	2986.8	1.72 [#] 86	3383.3			I_γ : other: 2.1 (1983Wa27).
	3036.1	2.1	3334.0			
	3240.9	3.5 [#] 17	3129.12			I_γ : other: 1.5 (1983Wa27).
	3648.7	20.7 [#] 21	2721.3	2		I_γ : other: 18 (1983Wa27).
	3759.0	3.5 [#] 17	2611.03	3		I_γ : other: 2.1 (1983Wa27).
	3789.7	23.5 24	2580.3			I_γ : weighted average of 24.1 24 (1977Da02) and 21 5 (1969Gr29). Other: 21 (1983Wa27), 40 (1964Gl04).
	3994.3	<2.3	2375.7	4		I_γ : from 1969Gr29 .
	4188.8	10.3 [#] 52	2181.10	3		I_γ : other: 7.3 (1983Wa27).
	4212.0	8.6 [#] 45	2157.9	2 ⁺		I_γ : other: 6.7 (1983Wa27), <4.6 (1969Gr29).
	4482.6	3.5 [#] 17	1887.29	2 ⁺		I_γ : other: 3.0 (1983Wa27), <2.3 (1969Gr29).
	5139.5	11.2 23	1230.33	2 ⁺		I_γ : weighted average of 10.3 52 (1977Da02) and 11.4 23 (1969Gr29). Other: 9.1 (1983Wa27).
	5704.4	100 [#] 10	665.3	1	D+Q	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 5 (1969Gr29). Mult.: from $\gamma\gamma(\theta)$ $E_p(\text{lab})=1266\rightarrow666\rightarrow\text{g.s.}$ γ cascade (1969Gr29). $A_2=-0.04$ 5 with $A_4=0$; $A_2=-0.04$ 6, $A_4=0.00$ 8 (1969Gr29).
	5908.5	25.6 23	461.2	1 ⁺		I_γ : weighted average of 31.0 31 (1977Da02) and 22.7 23 (1969Gr29). Other: 22 (1983Wa27).
	6223.2	36.5 53	146.4	3 ⁺		$A_2=-0.07$ 14 with $A_4=0$; $A_2=-0.11$ 16, $A_4=+0.09$ 21 (1969Gr29). I_γ : weighted average of 48 5 (1977Da02) and 34.1 23 (1969Gr29). Other: 42 (1983Wa27), 60 (1964Gl04).
	6369.6	46 5	0	0 ⁺		$A_2=+0.11$ 12 with $A_4=0$; $A_2=+0.11$ 13, $A_4=0.00$ 17 (1969Gr29). I_γ : weighted average of 52 5 (1977Da02) and 39 5 (1969Gr29). Other: 42 (1983Wa27).
						$A_2=-0.18$ 10 with $A_4=0$; $A_2=-0.21$ 13, $A_4=-0.03$ 15 (1969Gr29).
6382.4	1772.7	49 [#] 5	4609.7			
	1776.6	36 [#] 4	4605.8			
	2306.1	6.4 [#] 33	4076.2	4		
	2399.3	5.1 [#] 26	3983.0	3		
	2782.0	20.5 [#] 21	3600.28	4		
	2837.2	9.0 [#] 46	3545.08	3		
	3048.3	6.4 [#] 33	3334.0			
	3660.9	2.6 [#] 13	2721.3	2		
	3771.2	10.3 [#] 51	2611.03	3		
	4006.5	6.4 [#] 33	2375.7	4		
	5716.6	5.1 [#] 26	665.3	1		
	6235.4	100 [#] 10	146.4	3 ⁺		
6399.8	2799.4	5.5 [#] 27	3600.28	4		
	3016.4	58 [#] 6	3383.3			
	3678.3	3.6 [#] 18	2721.3	2		
	4023.8	5.5 [#] 27	2375.7	4		
	5169.1	9.1 [#] 46	1230.33	2 ⁺		
	6252.8	100 [#] 10	146.4	3 ⁺		
6441.2	2364.9	2.7 [#] 14	4076.2	4		
	3057.8	9.6 [#] 10	3383.3			
	4283.0	100 [#] 10	2157.9	2 ⁺		
	5775.4	5.5 [#] 28	665.3	1		

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)						
$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Comments
6441.2		5979.4	8.2 [#] 8	461.2	1 ⁺	
		6294.2	11.0 [#] 11	146.4	3 ⁺	
6450.54	3,4	1840.8	<4.2 [#]	4609.7		
		1844.7	8.3 [#] 42	4605.8		I_γ : other: 8.1 (1983Wa27).
		1989.1	6.9	4461.4		
		2003.9	7.3 [#] 38	4446.6	4	I_γ : other: 7.1 (1983Wa27).
		2374.3	4.2 [#] 21	4076.2	4	I_γ : other: 2.7 (1983Wa27).
		2818.7	10.4 [#] 52	3631.7	5	I_γ : other: 8.8 (1983Wa27).
		2850.13	38 [#] 4	3600.28	4	I_γ : other: 38 (1983Wa27). $A_2=+0.46$ 10, $A_4=-0.18$ 12 (1983Wa27).
		2905.33	7.3 [#] 38	3545.08	3	I_γ : other: 8.8 (1983Wa27).
		3116.4	1.0	3334.0		
		3839.28	29.2 [#] 29	2611.03	3	I_γ : other: 25 (1983Wa27).
		4269.15	4.2 [#] 21	2181.10	3	I_γ : other: 2.5 (1983Wa27).
		4292.4	<0.42	2157.9	2 ⁺	
		5784.7	<1.0	665.3	1	
		5988.8	<0.83	461.2	1 ⁺	
		6303.5	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27). $A_2=+0.03$ 7, $A_4=-0.06$ 9 (1983Wa27).
		6449.88	<1.7	0	0 ⁺	
6477.5		2123.2	26.2 [#] 26	4354.2	(1)	
		2703.7	<3.1 [#]	3773.72	1	
		3897.0	9.2 [#] 9	2580.3		
		5811.7	18.5 [#] 19	665.3	1	
		6015.7	100 [#] 10	461.2	1 ⁺	
6486.9	1 to 4	2546.7	16.7 [#] 17	3940.1		I_γ : other: 17 (1983Wa27).
		3357.6	1.5	3129.12		
		4328.7	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27). $A_2=+0.15$ 7, $A_4=+0.07$ 8 (1983Wa27).
		5821.1	3.3 [#] 17	665.3	1	I_γ : other: 2.0 (1983Wa27).
		6025.1	13.3 [#] 13	461.2	1 ⁺	I_γ : other: 13 (1983Wa27).
		6339.9	<0.85	146.4	3 ⁺	
		6486.2	33.3 [#] 33	0	0 ⁺	I_γ : other: 36 (1983Wa27).
6527.6		2927.2	21.0 [#] 21	3600.28	4	
		3144.1	100 [#] 10	3383.3		
		4369.4	9.7 [#] 10	2157.9	2 ⁺	
		4640.0	6.5 [#] 32	1887.29	2 ⁺	
		5861.8	11.3 [#] 11	665.3	1	
		6380.6	12.9 [#] 13	146.4	3 ⁺	
6546.6		1375.0	1.6	5171.6	4	
		3163.1	0.95	3383.3		
		3966.1	7.7 [#] 8	2580.3		I_γ : other: 5.5 (1983Wa27).
		5315.8	20.5 [#] 21	1230.33	2 ⁺	I_γ : other: 11 (1983Wa27).
		5880.8	<0.36	665.3	1	
		6084.8	<0.48	461.2	1 ⁺	
		6399.6	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27).
		6545.9	<0.48	0	0 ⁺	

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. $\&$	$\delta^\&$	Comments
6574.7		4198.7	1.9	2375.7	4			
		4416.5	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).
		5908.9	<0.65	665.3	1			
		6427.7	5.3 [#] 26	146.4	3 ⁺			I_γ : other: 5.6 (1983Wa27).
		6574.0	<0.65	0	0 ⁺			
6583.7		3038.5	53 [#] 5	3545.08	3			
		4425.5	21.3 [#] 21	2157.9	2 ⁺			
		5917.9	21.3 [#] 21	665.3	1			
		6121.9	100 [#] 10	461.2	1 ⁺			
		6583.0	17.0 [#] 17	0	0 ⁺			
6624.7		1237.9	30	5386.8				
		1907.2	32	4717.4				
		2484.9	19	4139.7	2			
		2548.4	75	4076.2	4			I_γ : other: 34.3 34 (1977Da02).
		2992.9	28	3631.7	5			
		3024.3	100	3600.28	4			I_γ : other: 100 10 (1977Da02).
		3079.5	52	3545.08	3			I_γ : other: 34.3 34 (1977Da02).
		3290.5	52	3334.0				
		4013.4	23	2611.03	3			
		4443.3	69	2181.10	3			
		4466.5	77	2157.9	2 ⁺			
		5393.9	21	1230.33	2 ⁺			
		5958.8	<3.9	665.3	1			
		6162.9	100	461.2	1 ⁺			I_γ : other: 74 8 (1977Da02).
		6477.6	92	146.4	3 ⁺			I_γ : other: 43 4 (1977Da02).
6640.92	4	2314.95	1.32 [#] 66	4325.89				
		2564.6	100 [#] 10	4076.2	4	D+Q	-0.14 3	I_γ : other: 100 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=+0.31$ 2, $A_4=+0.03$ 2 (1977Da03). $A_2=+0.44$ 5, $A_4=+0.01$ 6 (1983Wa27); $A_2=+0.31$ 2, $A_4=+0.03$ 2 (1977Da03).
		2657.8	7.9 [#] 40	3983.0	3			I_γ : other: 4.1 (1983Wa27).
		2994.5	1.32 [#] 66	3646.3				
		3009.1	1.32 [#] 66	3631.7	5			
		3040.49	13.2 [#] 66	3600.28	4	D+Q	-0.12 3	I_γ : other: 6.4 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=+0.32$ 3, $A_4=+0.08$ 3 (1977Da03).
		3095.69	47 [#] 5	3545.08	3	D+Q	+0.04 2	I_γ : other: 36 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=-0.25$ 2, $A_4=0.00$ 2 (1977Da03). $A_2=-0.27$ 9, $A_4=0.00$ 11 (1983Wa27); $A_2=-0.25$ 2, $A_4=0.00$ 2 (1977Da03).
		4029.63	15.8 [#] 16	2611.03	3	D(+Q)	+0.02 +1-2	I_γ : other: 8.6 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=-0.27$ 3, $A_4=-0.01$ 3 (1977Da03).
		4264.9	15.8 [#] 16	2375.7	4	D+Q	-0.14 4	I_γ : other: 11 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=+0.30$ 3, $A_4=+0.06$ 3 (1977Da03).
		4459.51	1.32 [#] 66	2181.10	3			I_γ : other: 1.4 (1983Wa27).
		5975.1	<0.68	665.3	1			
		6179.1	<0.46	461.2	1 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6640.92	6493.9	58 [#] 6	146.4	3 ⁺	D(+Q)	+0.02 +I-2	I_γ : other: 59 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03. $A_2=-0.28$ 2, $A_4=+0.02$ 2 (1977Da03). $A_2=-0.27$ 9, $A_4=-0.04$ 9 (1983Wa27); $A_2=-0.28$ 2, $A_4=+0.02$ 2 (1977Da03).
	6640.22	<0.91	0	0 ⁺			
6695.1	1870.6	50 [#] 5	4824.41				
	1999.3	13 [#] 6	4695.7				
	3048.7	31.3 [#] 31	3646.3				
	3094.7	47 [#] 5	3600.28	4			
	4083.8	72 [#] 7	2611.03	3			
	4319.1	100 [#] 10	2375.7	4			
6702.8	2626.5	2.4 [#] 12	4076.2	4			
	3102.4	4.8 [#] 24	3600.28	4			
	3157.6	3.6 [#] 18	3545.08	3			
	3368.6	2.4 [#] 12	3334.0				
	4326.8	6 [#] 3	2375.7	4			
	6555.7	100 [#] 10	146.4	3 ⁺			
6720.2	2110.4	31.3 [#] 31	4609.7				
	2114.3	22.9 [#] 23	4605.8				
	2737.1	8.3 [#] 42	3983.0	3			
	3336.7	9.4 [#] 48	3383.3				
	4832.5	11.5 [#] 13	1887.29	2 ⁺			
	5489.4	25.0 [#] 25	1230.33	2 ⁺			
	6573.1	100 [#] 10	146.4	3 ⁺			
6723.7	1899.2	100	4824.41				I_γ : other: 100 11 (1977Da02).
	2027.9	20	4695.7				I_γ : other: 18 9 (1977Da02).
	2759.5	4.8	3964.1				
	3077.3	33	3646.3				I_γ : other: 31.6 32 (1977Da02).
	3340.2	1.6	3383.3				
	4112.4	72	2611.03	3			I_γ : other: 91 9 (1977Da02).
	4347.7	92	2375.7	4			I_γ : other: 118 12 (1977Da02).
	4542.3	5.6	2181.10	3			I_γ : other: 9 5 (1977Da02).
	4565.5	7.6	2157.9	2 ⁺			I_γ : other: 14 7 (1977Da02).
	6057.8	<1.2	665.3	1			
	6261.9	<2	461.2	1 ⁺			
	6576.6	64	146.4	3 ⁺			I_γ : other: 73 7 (1977Da02).
	6723.0	<0.8	0	0 ⁺			
6737.0	3607.7	5.1 [#] 26	3129.12				I_γ : other: 6.7 (1983Wa27).
	4578.8	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).
	4849.3	3.6	1887.29	2 ⁺			
	5506.2	12.8 [#] 64	1230.33	2 ⁺			I_γ : other: 12 (1983Wa27).
	6071.1	36 [#] 4	665.3	1			I_γ : other: 42 (1983Wa27).
	6275.2	28.2 [#] 28	461.2	1 ⁺			I_γ : other: 31 (1983Wa27).
	6589.9	74 [#] 8	146.4	3 ⁺			I_γ : other: 83 (1983Wa27).
	6736.3	<1.7	0	0 ⁺			
6790.8	4903.1	5.3 [#] 26	1887.29	2 ⁺			I_γ : other: <3 (1983Wa27).
	6124.9	<3	665.3	1			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6790.8		6329.0	<4	461.2	1 ⁺	
		6643.7	<6	146.4	3 ⁺	
		6790.1	100 [#] 10	0	0 ⁺	I_γ : other: 100 (1983Wa27).
6797.4	1 to 5	2833.2	1.10 [#] 55	3964.1		I_γ : other: 1.1 (1983Wa27).
		3413.9	0.55 [#] 28	3383.3		
		4186.1	2.2 [#] 11	2611.03	3	I_γ : other: 2.2 (1983Wa27).
		4421.4	1.10 [#] 55	2375.7	4	I_γ : other: 1.1 (1983Wa27).
		4639.2	0.55 [#] 28	2157.9	2 ⁺	I_γ : other: 0.44 (1983Wa27).
		4909.7	4.4 [#] 22	1887.29	2 ⁺	I_γ : other: 3.9 (1983Wa27).
		6131.5	<0.22	665.3	1	
		6335.6	<0.33	461.2	1 ⁺	
		6650.3	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27).
						$A_2=+0.16$ 5, $A_4=-0.04$ 7 (1983Wa27).
		6796.7	<0.44	0	0 ⁺	
6807.2		3423.7	12.7 [#] 13	3383.3		
		4226.6	16.9 [#] 17	2580.3		
		4649.0	11.3 [#] 11	2157.9	2 ⁺	
		6141.3	100 [#] 10	665.3	1	
6828.6		3283.4	7.3	3545.08	3	
		3445.1	15.5 [#] 16	3383.3		I_γ : other: 14 (1983Wa27).
		4670.4	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27).
		5597.8	<1.2	1230.33	2 ⁺	
		6162.7	5.2 [#] 26	665.3	1	I_γ : other: 12 (1983Wa27).
		6366.8	19.0 [#] 19	461.2	1 ⁺	I_γ : other: 33 (1983Wa27).
		6681.5	32.8 [#] 33	146.4	3 ⁺	I_γ : other: 27 (1983Wa27).
		6827.9	<1.4	0	0 ⁺	
6842.8		2233.0	<10 [#]	4609.7		I_γ : other: 6.9 (1983Wa27).
		2236.9	5.0 [#] 25	4605.8		I_γ : other: 18 (1983Wa27).
		2327.0	25 [#] 13	4515.7		I_γ : other: 35 (1983Wa27).
		2488.5	45 [#] 5	4354.2	(1)	I_γ : other: 49 (1983Wa27).
		2694.9	34	4147.8		
		2703.0	16	4139.7	2	
		2859.7	48 [#] 5	3983.0	3	I_γ : other: 54 (1983Wa27).
		3068.9	6.3	3773.72	1	
		3459.3	23 [#] 12	3383.3		I_γ : other: 18 (1983Wa27).
		3508.6	10 [#] 5	3334.0		I_γ : other: 11 (1983Wa27).
		3713.5	9.4	3129.12		
		4231.5	12.5 [#] 65	2611.03	3	I_γ : other: 16 (1983Wa27).
		4262.2	45 [#] 5	2580.3		I_γ : other: 46 (1983Wa27).
		4661.4	6.9	2181.10	3	
		4684.6	30 [#] 3	2157.9	2 ⁺	I_γ : other: 29 (1983Wa27).
		4955.1	35.0 [#] 35	1887.29	2 ⁺	I_γ : other: 38 (1983Wa27).
		5612.0	65 [#] 7	1230.33	2 ⁺	I_γ : other: 75 (1983Wa27).
		6176.9	35.0 [#] 35	665.3	1	I_γ : other: 44 (1983Wa27).
		6381.0	100 [#] 10	461.2	1 ⁺	I_γ : other: 100 (1983Wa27).
		6695.7	12.5 [#] 65	146.4	3 ⁺	I_γ : other: 13 (1983Wa27).
		6842.1	<1.9	0	0 ⁺	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
6853.0		1857.4	2.5 [#] 13	4995.6				I_γ : other: 2.3 (1983Wa27).
		2157.2	2.5 [#] 13	4695.7				I_γ : other: 2.3 (1983Wa27).
		2498.7	2.5 [#] 13	4354.2	(1)			
		3307.8	5.1	3545.08	3			
		3723.7	7.5 [#] 38	3129.12				I_γ : other: 8.2 (1983Wa27).
		4272.4	7.5 [#] 38	2580.3				I_γ : other: 9 (1983Wa27).
		4965.3	58 [#] 6	1887.29	2 ⁺			I_γ : other: 56 (1983Wa27).
		5622.2	32.5 [#] 33	1230.33	2 ⁺			I_γ : other: 33 (1983Wa27).
		6187.1	15.0 [#] 15	665.3	1			I_γ : other: 18 (1983Wa27).
		6391.2	100 [#] 10	461.2	1 ⁺			I_γ : other: 100 (1983Wa27).
		6705.9	10 [#] 5	146.4	3 ⁺			I_γ : other: 10 (1983Wa27).
6871.18	5	6852.3	13 [#] 6	0	0 ⁺			I_γ : other: 12 (1983Wa27).
		1330.4	13.6 [#] 14	5540.8				I_γ : other: 10 (1983Wa27).
								$A_2=+0.49$ 9, $A_4=-0.17$ 9 (1977Da03).
		1484.4	6.8 [#] 34	5386.8				I_γ : other: 2.8 (1983Wa27).
		1699.5	25.0 [#] 25	5171.6	4	D(+Q)	+0.02 3	I_γ : other: 24 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.24$ 4, $A_4=+0.02$ 4 (1977Da03).
		2424.5	64 [#] 7	4446.6	4	D+Q	+0.05 +1-3	I_γ : other: 61 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.19$ 3, $A_4=0.00$ 3 (1977Da03).
		2794.9	5.7	4076.2	4			
6887.97	2	3239.3	100 [#] 10	3631.7	5	D(+Q)	+0.04 7	I_γ : other: 100 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=+0.41$ 3, $A_4=+0.05$ 4 (1977Da03).
		4495.2	18.2 [#] 18	2375.7	4	D(+Q)	+0.04 +3-5	I_γ : other: 14 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.21$ 7, $A_4=-0.06$ 7 (1977Da03).
		4712.9	<0.87	2157.9	2 ⁺			
		6205.3	<1.3	665.3	1			
		6409.3	<1.5	461.2	1 ⁺			
		6724.1	<1.3	146.4	3 ⁺			
		6870.44	<1.3	0	0 ⁺			
6901.75		1930.6	1.16 [#] 58	4957.3				I_γ : other: 1.2 (1983Wa27).
		2170.5	2.3 [#] 12	4717.4				I_γ : other: 4.3 (1983Wa27).
		2278.2	1.16 [#] 58	4609.7				
		2282.1	1.16 [#] 58	4605.8				I_γ : other: 1.3 (1983Wa27).
		3504.5	2.3 [#] 12	3383.3				I_γ : other: 1.7 (1983Wa27).
		4706.52	1.16 [#] 58	2181.10	3			I_γ : other: 3.5 (1983Wa27).
		4729.7	100 [#] 10	2157.9	2 ⁺	D+Q	-0.52 +3-2	I_γ : other: 100 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.13$ 2, $A_4=+0.01$ 3 (1977Da03).
		5000.29	1.16 [#] 58	1887.29	2 ⁺			I_γ : other: 1.7 (1983Wa27).
		6222.1	<0.37	665.3	1			
		6426.1	<0.37	461.2	1 ⁺			
6901.75		6740.9	5.8 [#] 29	146.4	3 ⁺			I_γ : other: 8.2 (1983Wa27).
		6887.22	<0.49	0	0 ⁺			
		3518.3	21.1 [#] 21	3383.3				I_γ : other: 20 (1983Wa27).
		4743.5	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6901.75		5670.91	0.81	1230.33	2 ⁺	
		6235.8	<0.27	665.3	1	
		6439.9	1.2	461.2	1 ⁺	
		6754.6	2.6 [#] 13	146.4	3 ⁺	I_γ : other: 2.8 (1983Wa27).
		6901.00	7.9 [#] 8	0	0 ⁺	I_γ : other: 10 (1983Wa27).
6918.0	3,4	1377.2	0.48	5540.8		
		3271.5	2.4 [#] 13	3646.3		I_γ : other: 1.9 (1983Wa27).
		3286.1	3.2 [#] 16	3631.7	5	I_γ : other: 3.6 (1983Wa27).
		3372.7	0.81	3545.08	3	
		3583.8	2.4 [#] 13	3334.0		I_γ : other: 2.3 (1983Wa27).
		4306.7	4.0 [#] 21	2611.03	3	I_γ : other: 3.9 (1983Wa27).
		4542.0	40 [#] 4	2375.7	4	I_γ : other: 40 (1983Wa27). $A_2=+0.50$ 4, $A_4=+0.04$ 5 (1983Wa27).
		4736.6	8.9 [#] 10	2181.10	3	I_γ : other: 8.1 (1983Wa27).
		6252.1	<0.16	665.3	1	
		6456.1	<0.32	461.2	1 ⁺	
		6770.9	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27). $A_2=+0.06$ 2, $A_4=-0.04$ 2 (1983Wa27).
6931.5		6917.2	<0.65	0	0 ⁺	
		2214.0	24	4717.4		I_γ : other: 14 7 (1977Da02).
		2321.7	100	4609.7		I_γ : other: 100 10 (1977Da02).
		2325.6	64	4605.8		I_γ : other: 110 11 (1977Da02).
		2415.7	8.2	4515.7		I_γ : other: 4.8 24 (1977Da02).
		2470.0	9.1	4461.4		I_γ : other: <9.6 (1977Da02).
		2577.2	9.1	4354.2	(1)	I_γ : other: 10 5 (1977Da02).
		2948.4	15	3983.0	3	I_γ : other: 19 10 (1977Da02).
		3331.0	4.6	3600.28	4	I_γ : other: 10 5 (1977Da02).
		3386.2	21	3545.08	3	I_γ : other: 14 7 (1977Da02).
		3548.0	55	3383.3		I_γ : other: 67 7 (1977Da02).
		4209.9	13	2721.3	2	I_γ : other: 14 7 (1977Da02).
		4350.9	19	2580.3		I_γ : other: 10 5 (1977Da02).
		4773.2	19	2157.9	2 ⁺	I_γ : other: 10 5 (1977Da02).
		5043.8	10 [#] 5	1887.29	2 ⁺	
		6265.6	46	665.3	1	I_γ : other: 19 10 (1977Da02).
		6469.6	<9.1	461.2	1 ⁺	
		6784.4	50	146.4	3 ⁺	I_γ : other: 67 7 (1977Da02).
		6930.7	<6.8	0	0 ⁺	
6977.4		3593.9	11.2 [#] 12	3383.3		
		4819.1	100 [#] 10	2157.9	2 ⁺	
		5746.6	8.2 [#] 9	1230.33	2 ⁺	
		6830.3	29.9 [#] 30	146.4	3 ⁺	
6986.85		2381.0	<4.9 [#]	4605.8		
		3003.708	18.3 [#] 20	3983.0	3	
		3386.39	18.3 [#] 20	3600.28	4	
		3441.58	19.5 [#] 20	3545.08	3	
		4265.263	11.0 [#] 56	2721.3	2	
		4828.582	100 [#] 10	2157.9	2 ⁺	
		5756.00	59 [#] 6	1230.33	2 ⁺	
		6839.711	18.3 [#] 20	146.4	3 ⁺	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
7051.6		2094.2	39 [#] 4	4957.3				
		3506.3	17.9 [#] 89	3545.08	3			
		3668.1	36 [#] 4	3383.3				
		4893.3	93 [#] 9	2157.9	2 ⁺			
		6385.7	71 [#] 7	665.3	1			
		6589.7	100 [#] 10	461.2	1 ⁺			
7059.04	1,2	2063.4	2.6 [#] 14	4995.6				I_γ : other: 2.2 (1983Wa27).
		2117.1	1.75 [#] 88	4941.9				I_γ : other: 2.1 (1983Wa27).
		2363.3	0.35	4695.7				
		2449.3	0.86	4609.7				
		2733.03	0.88 [#] 44	4325.89				I_γ : other: 0.69 (1983Wa27).
		3285.2	<0.88 [#]	3773.72	1			I_γ : other: 0.17 (1983Wa27).
		3398.9	1.75 [#] 88	3660.0				I_γ : other: 1.4 (1983Wa27).
		3929.68	4.4 [#] 23	3129.12				I_γ : other: 5.3 (1983Wa27).
		4337.4	1.75 [#] 88	2721.3	2			I_γ : other: 1.2 (1983Wa27).
		4447.70	1.2	2611.03	3			
		4478.4	0.88 [#] 44	2580.3				I_γ : other: 1.4 (1983Wa27).
		4877.56	2.6 [#] 14	2181.10	3			I_γ : other: 1.4 (1983Wa27).
		4900.8	0.88 [#] 44	2157.9	2 ⁺			I_γ : other: 0.86 (1983Wa27).
		5171.33	8.8 [#] 44	1887.29	2 ⁺			I_γ : other: 7.1 (1983Wa27).
		5828.17	5.3 [#] 26	1230.33	2 ⁺			I_γ : other: 5.7 (1983Wa27).
		6393.1	44 [#] 5	665.3	1			I_γ : other: 40 (1983Wa27).
								$A_2=+0.03$ 6, $A_4=-0.13$ 7 (1983Wa27).
		6597.2	100 [#] 10	461.2	1 ⁺			I_γ : other: 100 (1983Wa27).
								$A_2=+0.04$ 2, $A_4=-0.03$ 3 (1983Wa27).
		7078.89	3	6911.9	<1.8 [#]	146.4	3 ⁺	
7058.25	<0.35			0	0 ⁺			
2473.0	6.3			4605.8				
2752.88	2.2			4325.89				
3002.6	3.6 [#] 18			4076.2	4			
3095.7	57 [#] 6			3983.0	3	D(+Q)	0.00 +6-4	I_γ : other: 59 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=+0.28$ 3, $A_4=+0.01$ 3 (1977Da03).
3478.42	61 [#] 6			3600.28	4	D(+Q)	0.00 3	I_γ : other: 56 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.09$ 3, $A_4=+0.01$ 3 (1977Da03).
3533.61	54 [#] 6			3545.08	3	D(+Q)	-0.07 +7-4	I_γ : other: 56 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=+0.24$ 3, $A_4=+0.04$ 4 (1977Da03).
3744.7	<3.6 [#]			3334.0				
4357.3	39 [#] 4			2721.3	2	D(+Q)	-0.03 +2-3	I_γ : other: 41 (1983Wa27).
								Mult., δ : or $\delta=-3.5$ 5 from $\gamma\gamma(\theta)$ in 1977Da03.
						$A_2=-0.26$ 3, $A_4=0.00$ 3 (1977Da03).		
		4897.41	3.6 [#] 18	2181.10	3			I_γ : other: 8.5 (1983Wa27).
		5191.18	10.7 [#] 54	1887.29	2 ⁺			I_γ : other: 8.9 (1983Wa27).
		5848.02	100 [#] 10	1230.33	2 ⁺	D(+Q)	-0.04 +4-2	I_γ : other: 100 (1983Wa27).
								Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03.
								$A_2=-0.25$ 3, $A_4=+0.02$ 3 (1977Da03).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#), [1977Da02](#), [1992Ka39](#) (continued) $\gamma(^{34}\text{Cl})$ (continued)

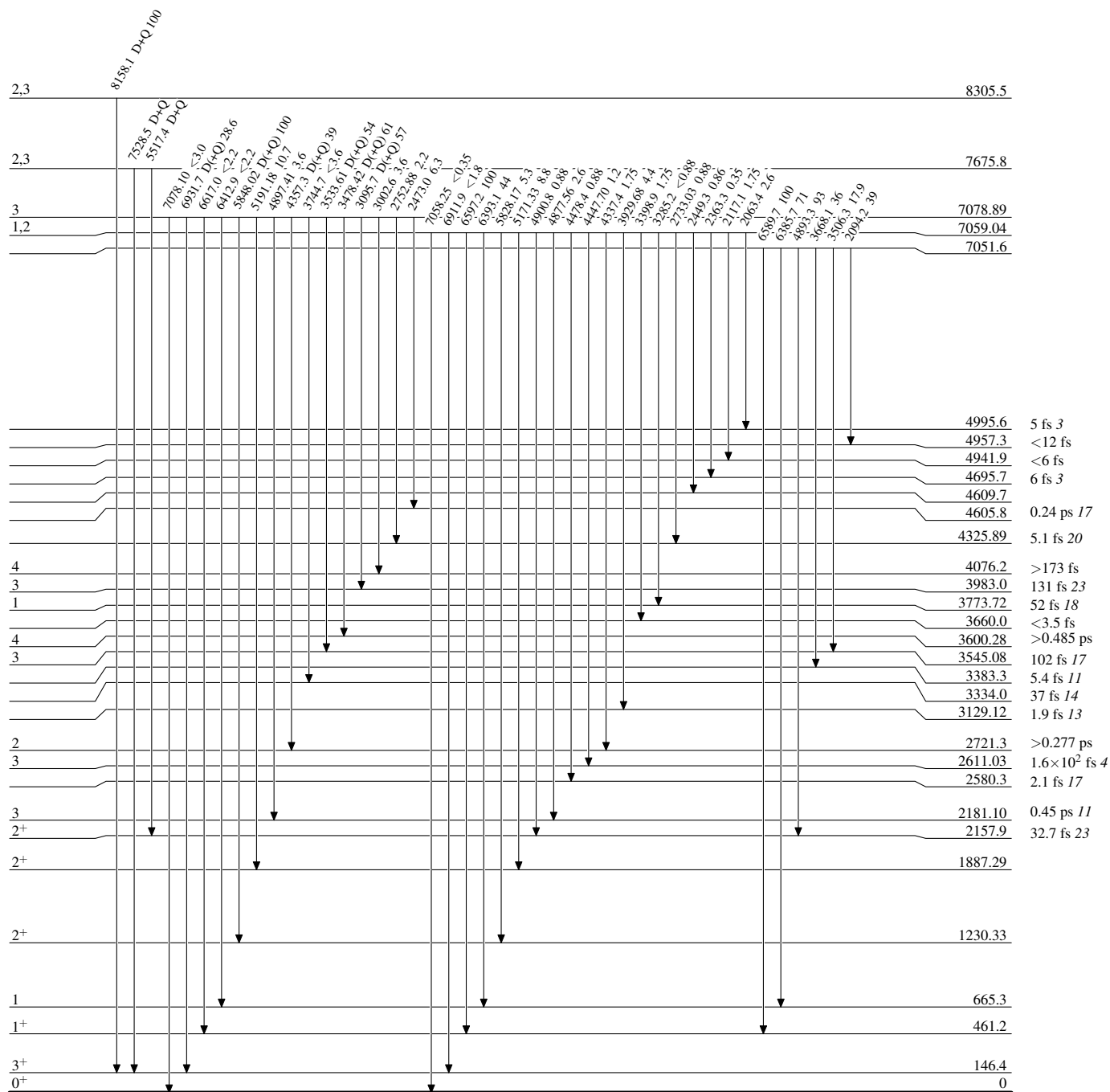
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	Comments
7078.89	3	6412.9	<2.2	665.3	1			
		6617.0	<2.2	461.2	1 ⁺			
		6931.7	28.6 [#] 29	146.4	3 ⁺	D(+Q)	-0.05 7	I_γ : other: 33 (1983Wa27). Mult., δ : from $\gamma\gamma(\theta)$ in 1977Da03 . $A_2=+0.24$ 4, $A_4=0.00$ 5 (1977Da03).
7675.8	2,3	7078.10 5517.4	<3.0	0 2157.9	0 ⁺ 2 ⁺	D+Q	0.12 8	Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given). $A_2=-0.14$ 9, $A_4=-0.18$ 12, $A_6=-0.005$ 12 (1992Ka39).
		7528.5		146.4	3 ⁺	D+Q	0.32 25	Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given). $A_2=-0.51$ 7, $A_4=+0.23$ 7, $A_6=-0.08$ 8 (1992Ka39).
8305.5	2,3	8158.1	100	146.4	3 ⁺	D+Q	0.27 18	I_γ : from 1992Ka39 . Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given). $A_2=-0.41$ 5, $A_4=-0.12$ 6, $A_6=+0.05$ 6 (1992Ka39).

[†] From level-energy differences, unless otherwise noted.[‡] From [1983Wa27](#), unless otherwise noted.[#] From [1977Da02](#).[@] From [2011Fr04](#).[&] From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ data given under comments.^a Placement of transition in the level scheme is uncertain.

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme

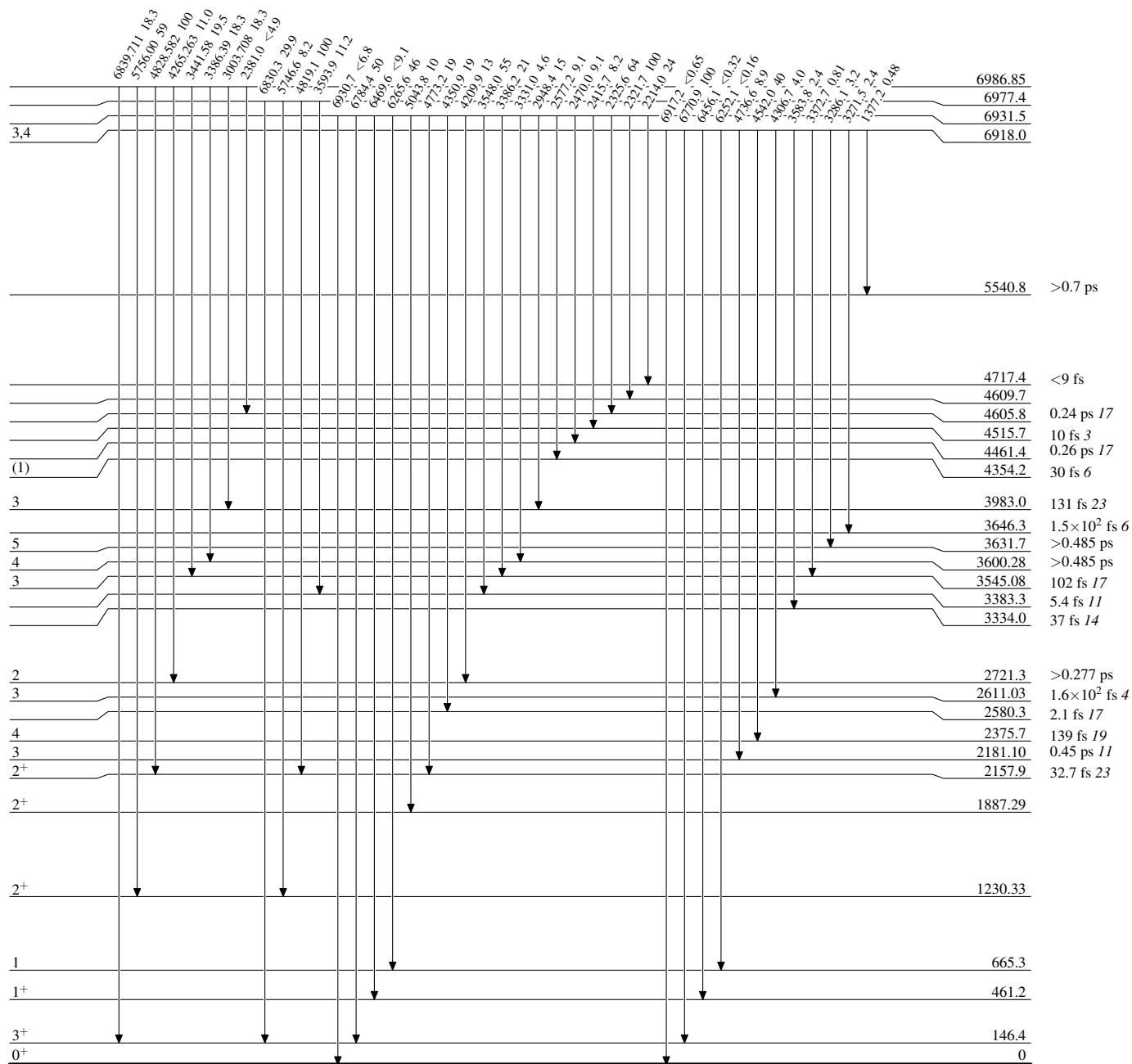
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

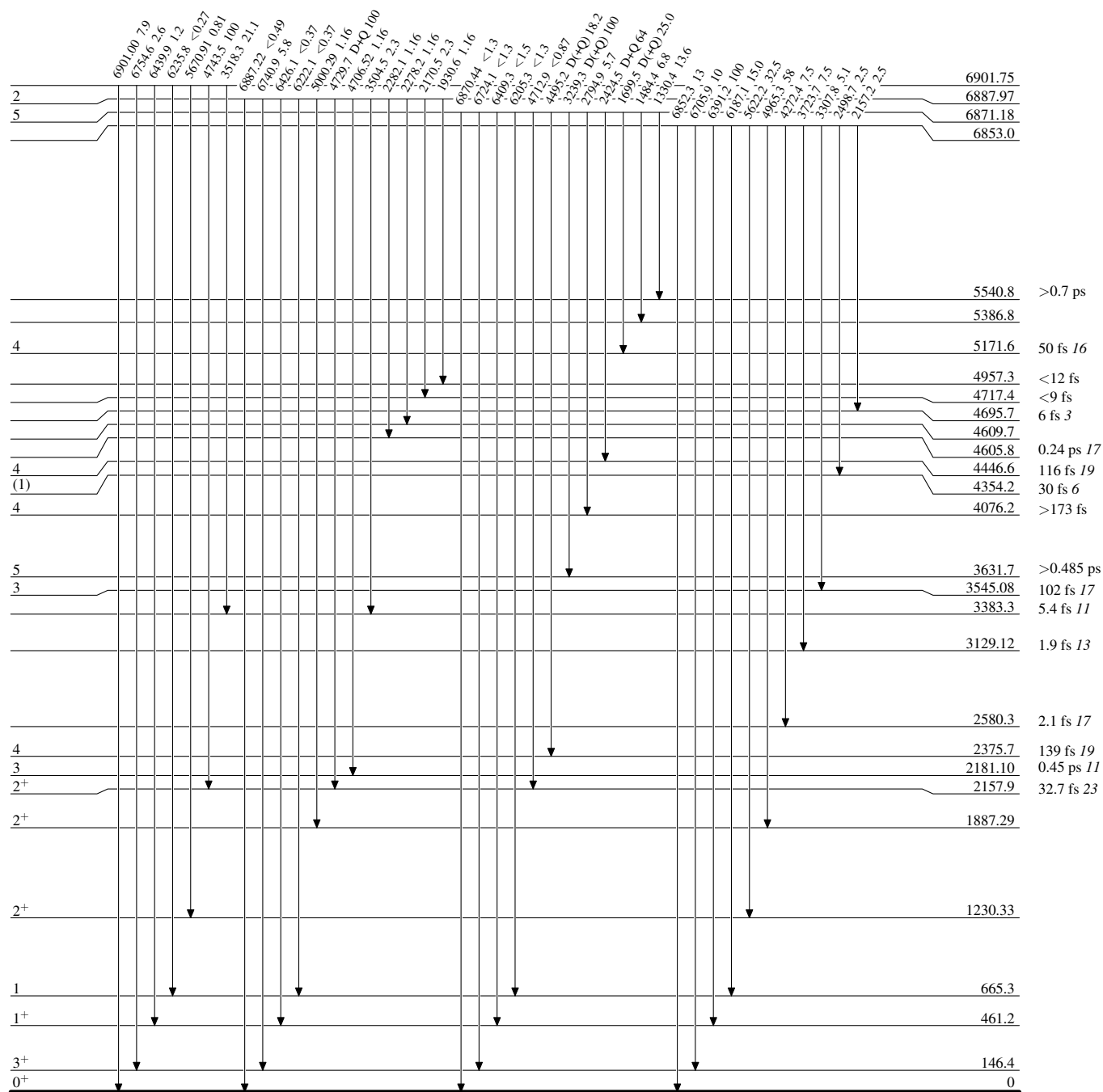
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

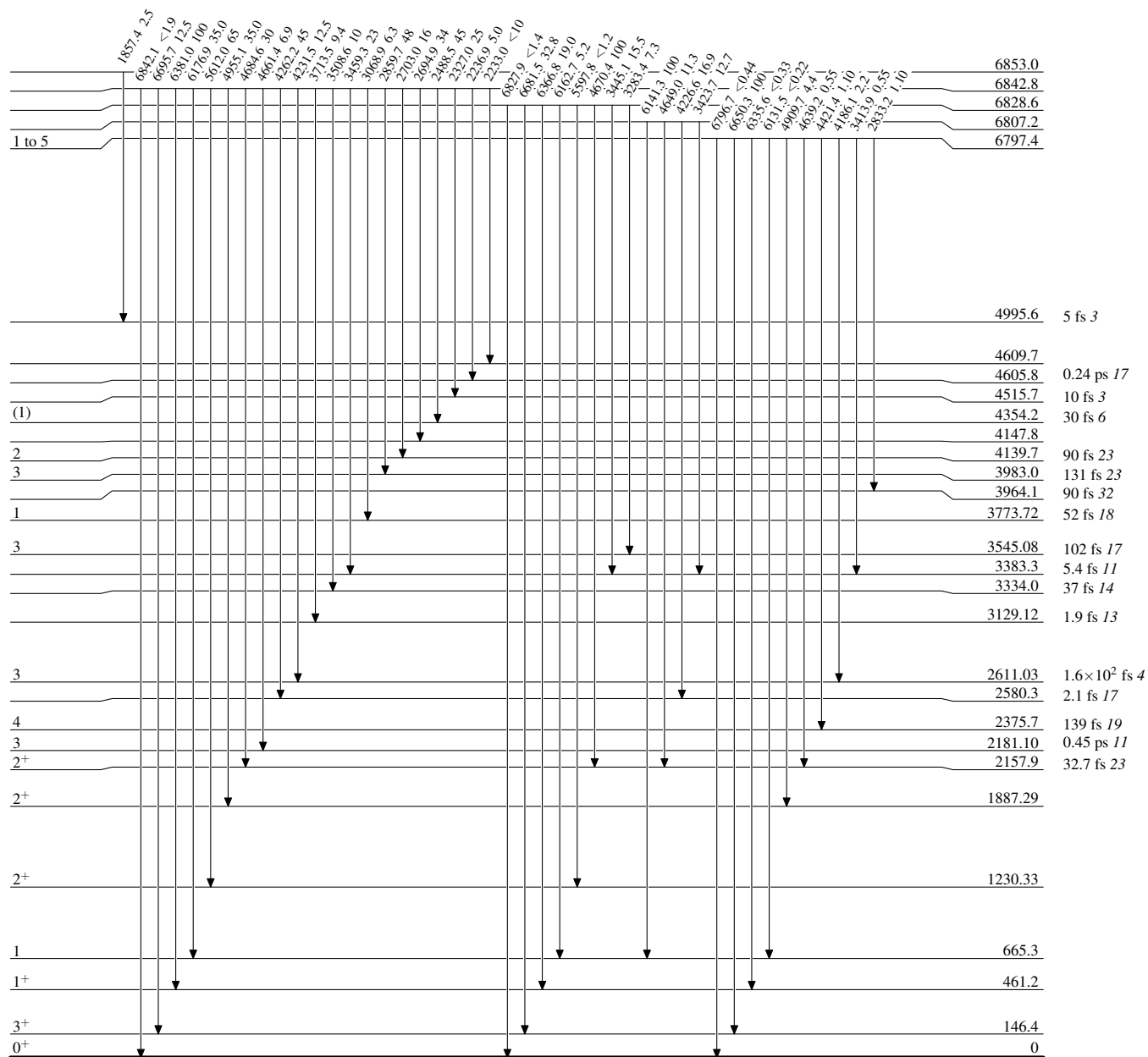
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

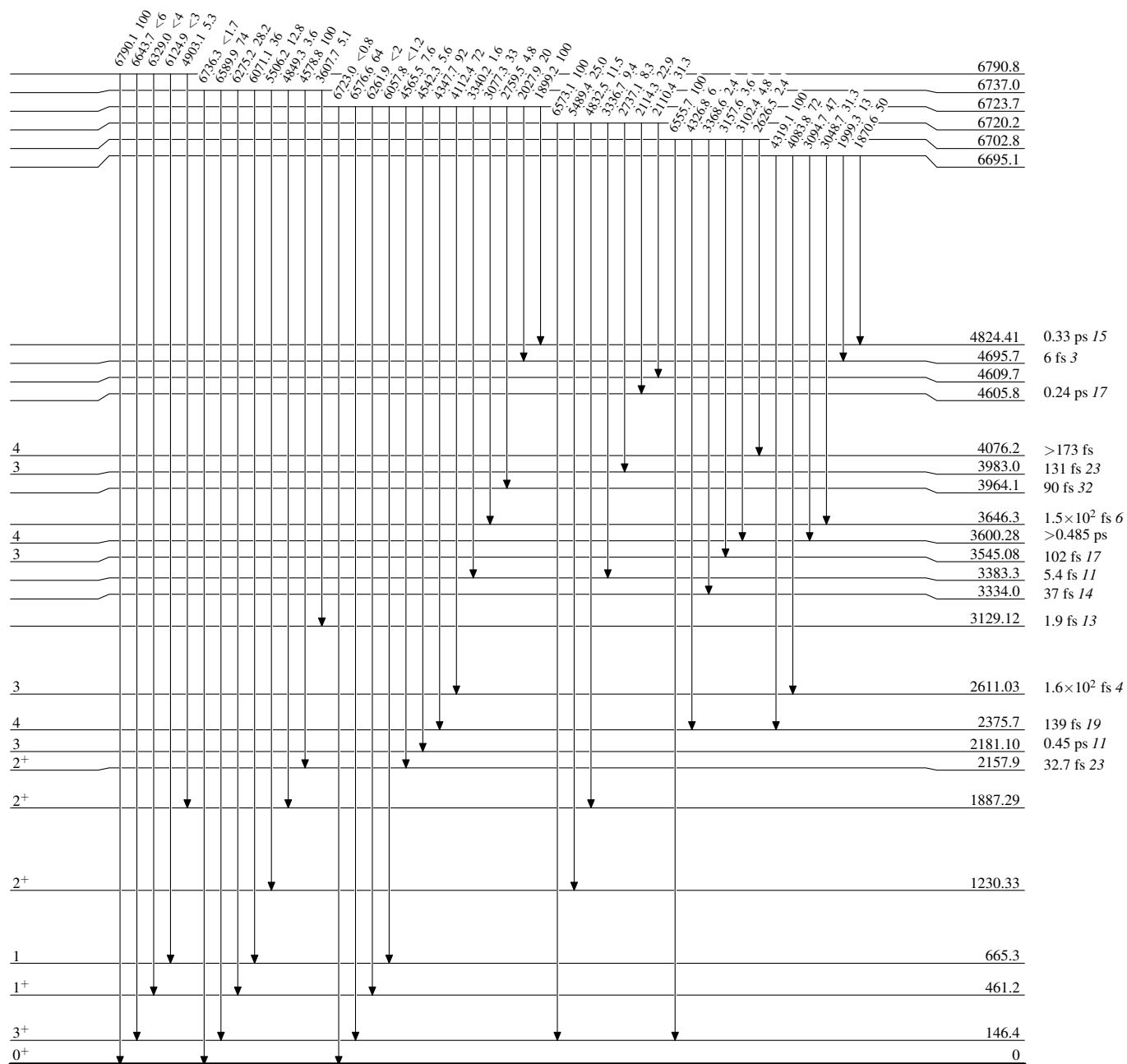
Intensities: Relative photon branching from each level



$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

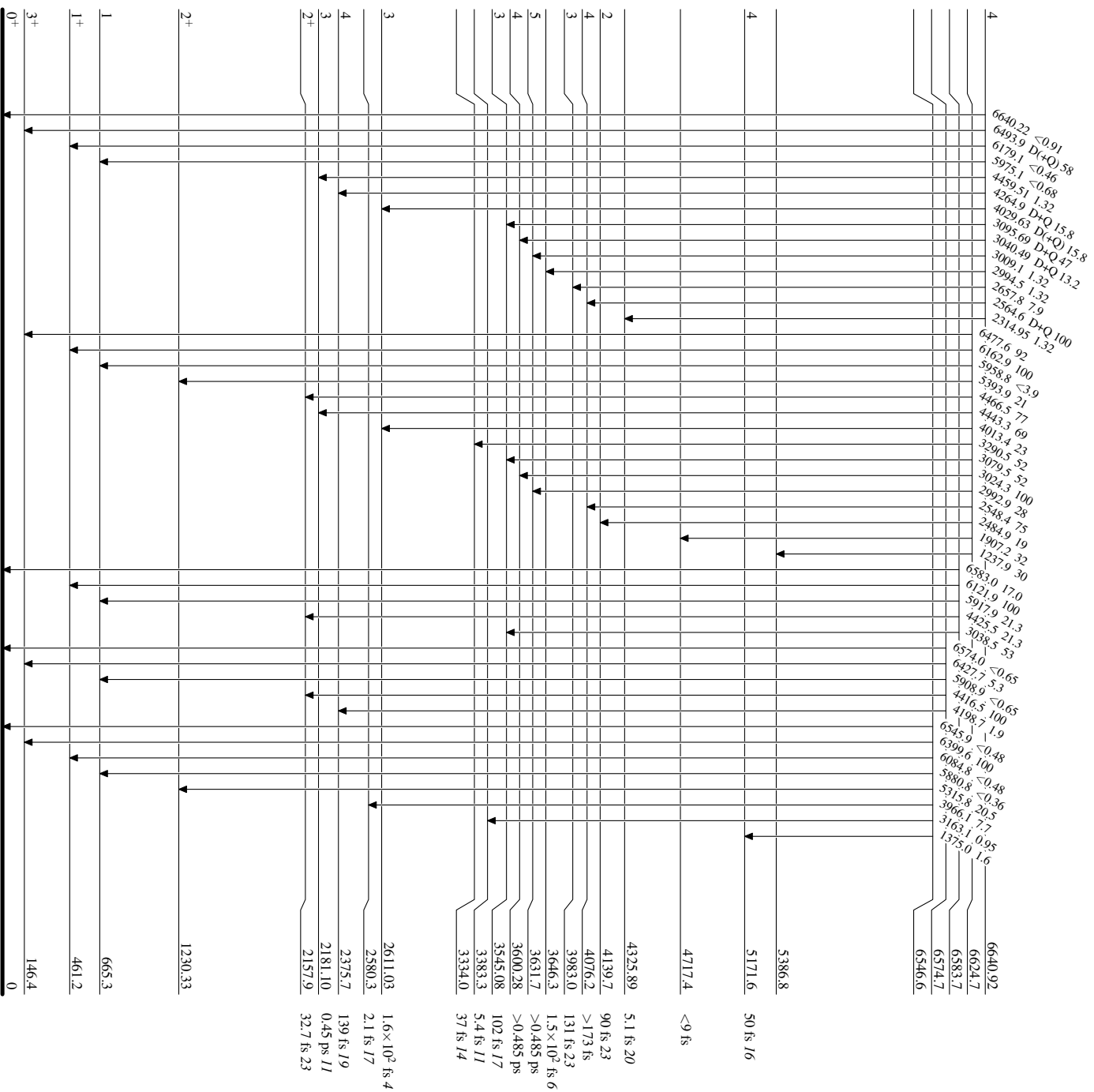
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

³³S(p,γ) 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

Intensities: Relative photon branching from each level

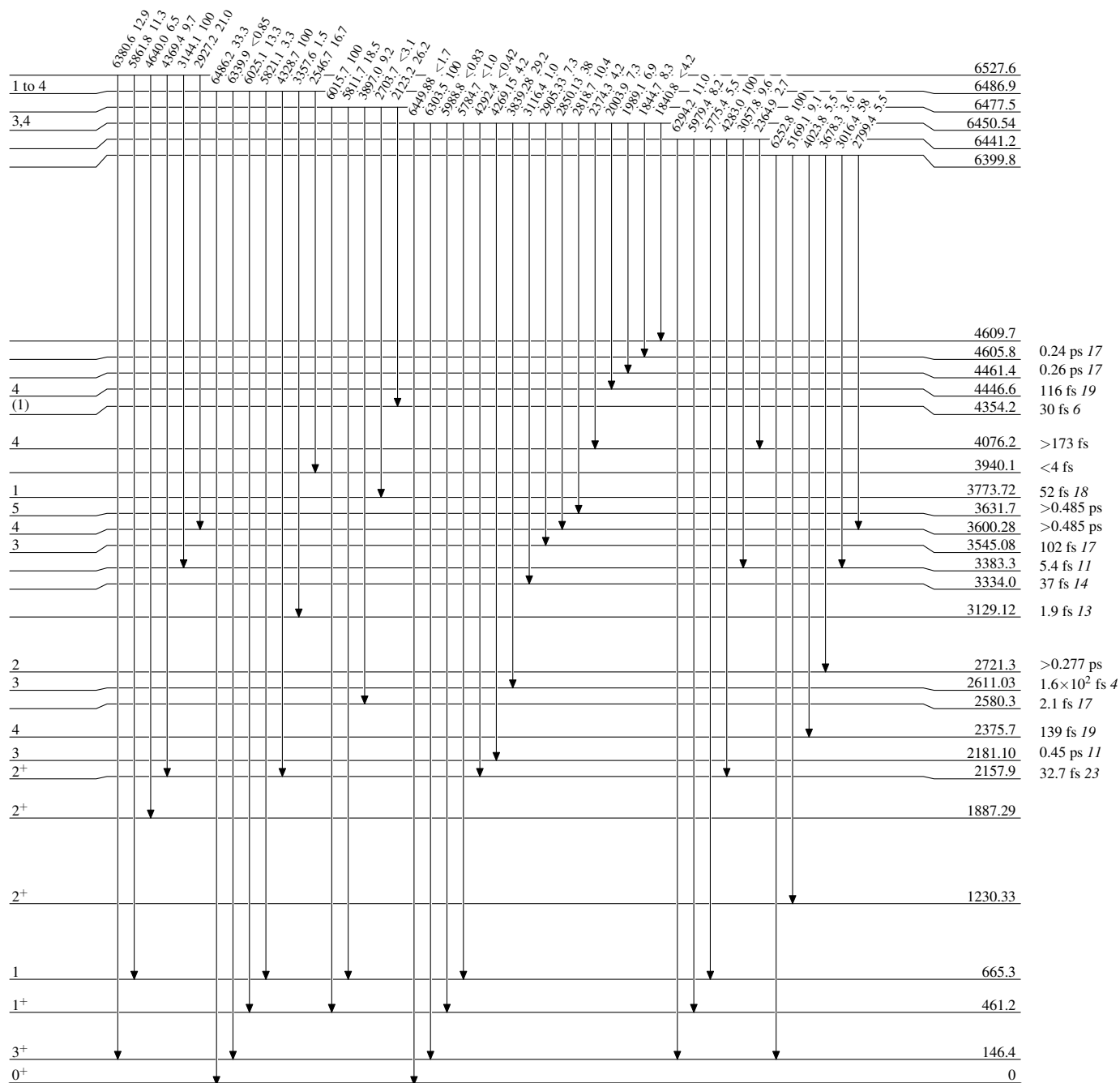


³⁴Cl₁₇

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

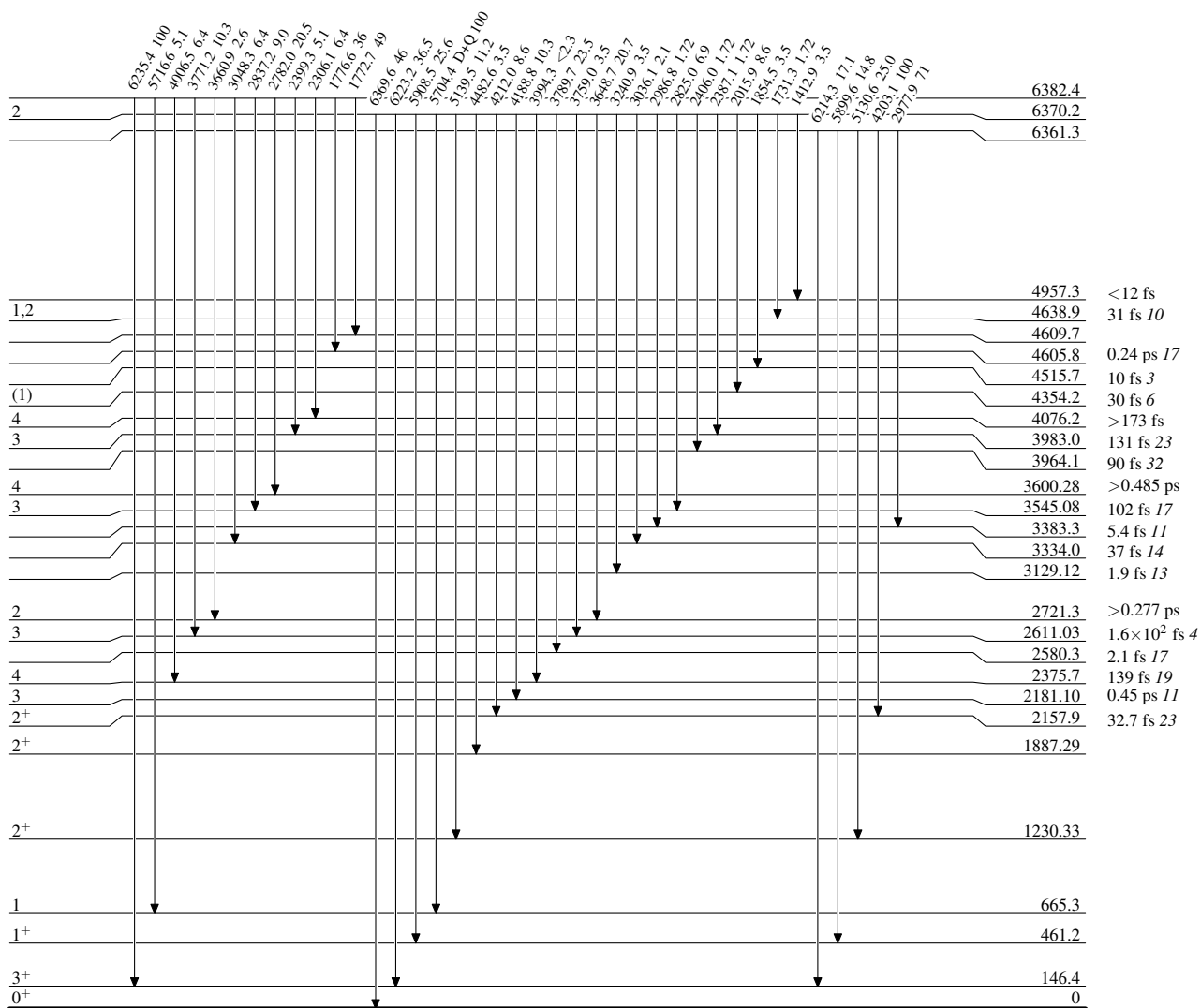
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

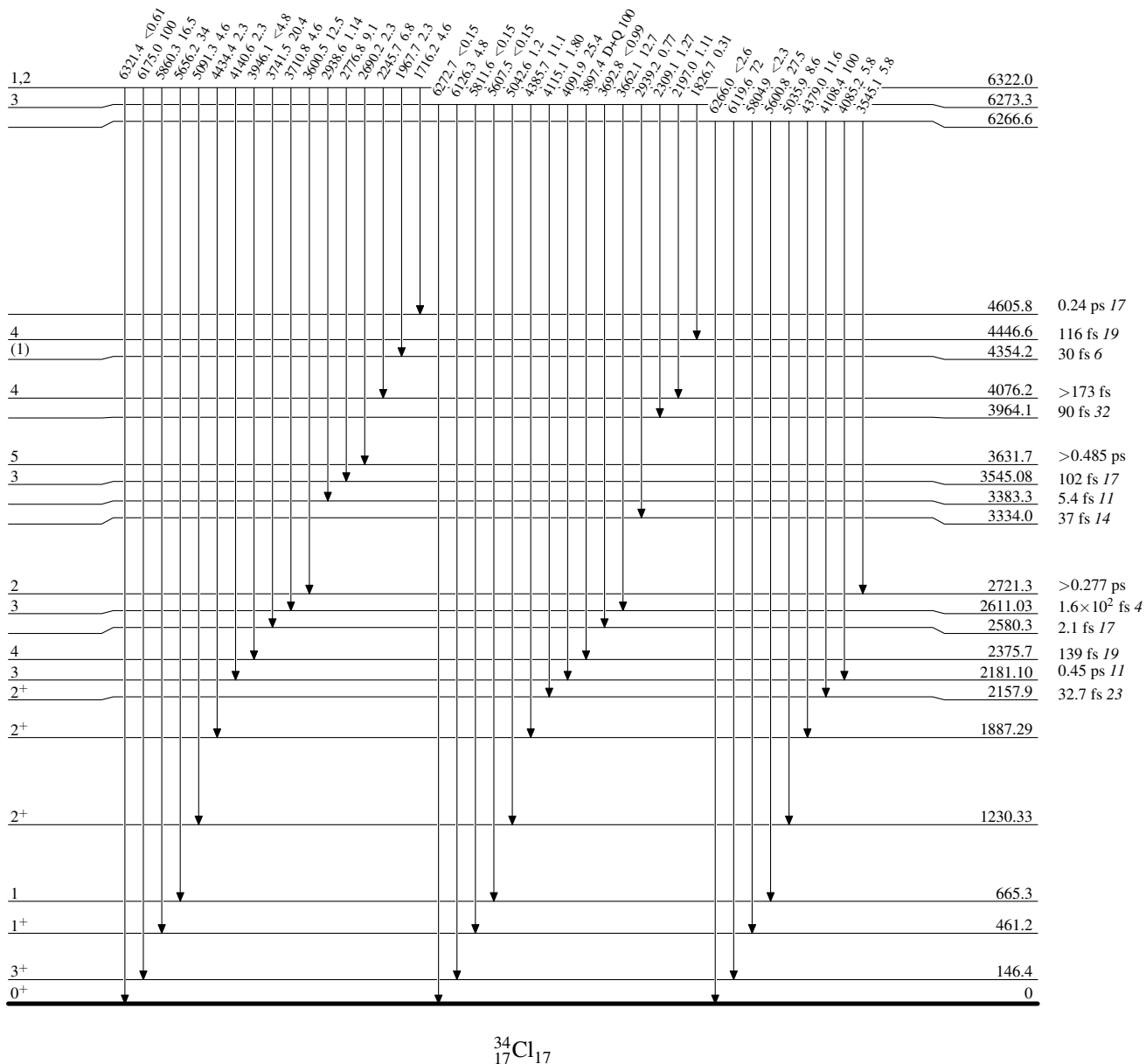
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

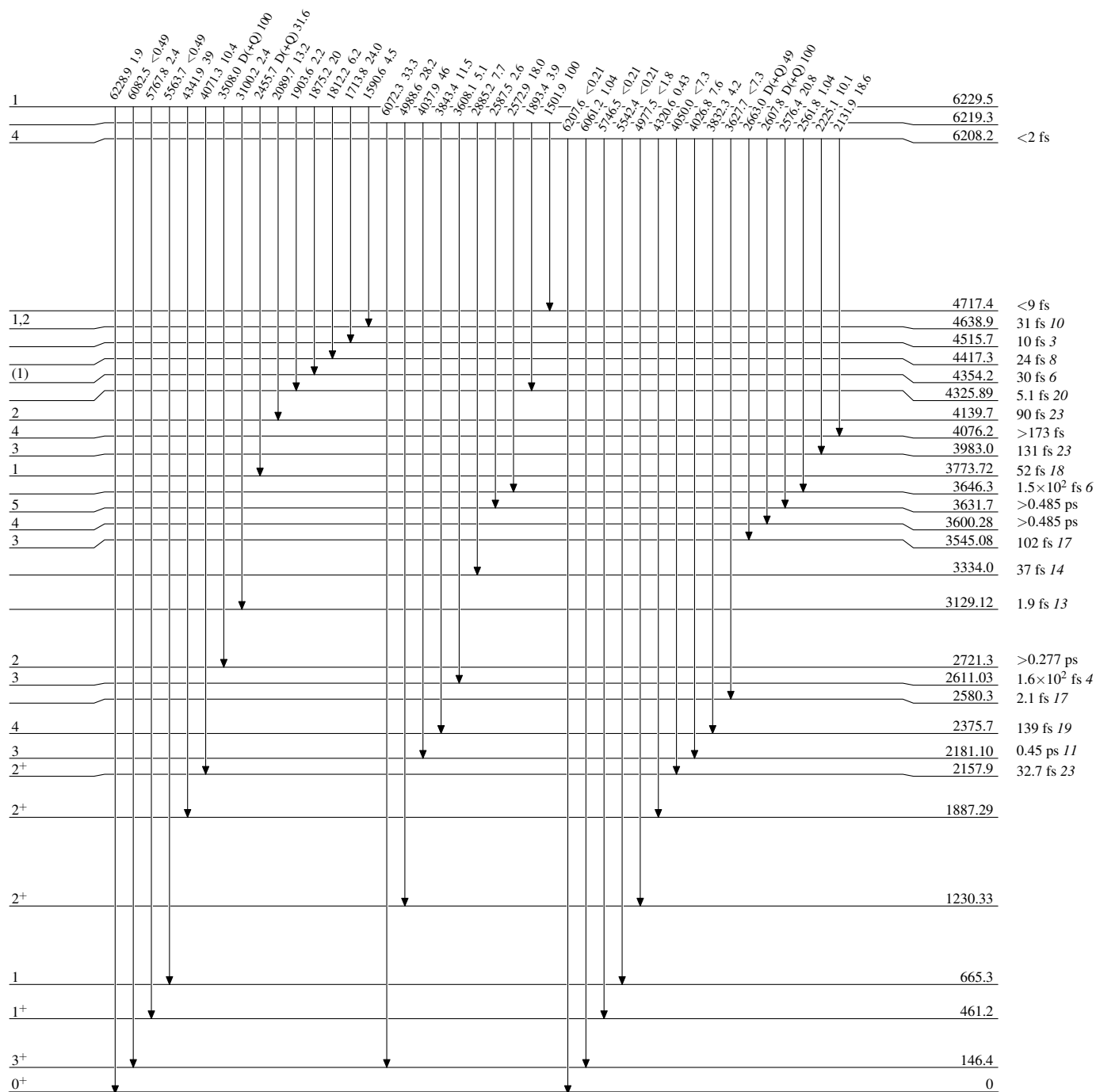
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(p,\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

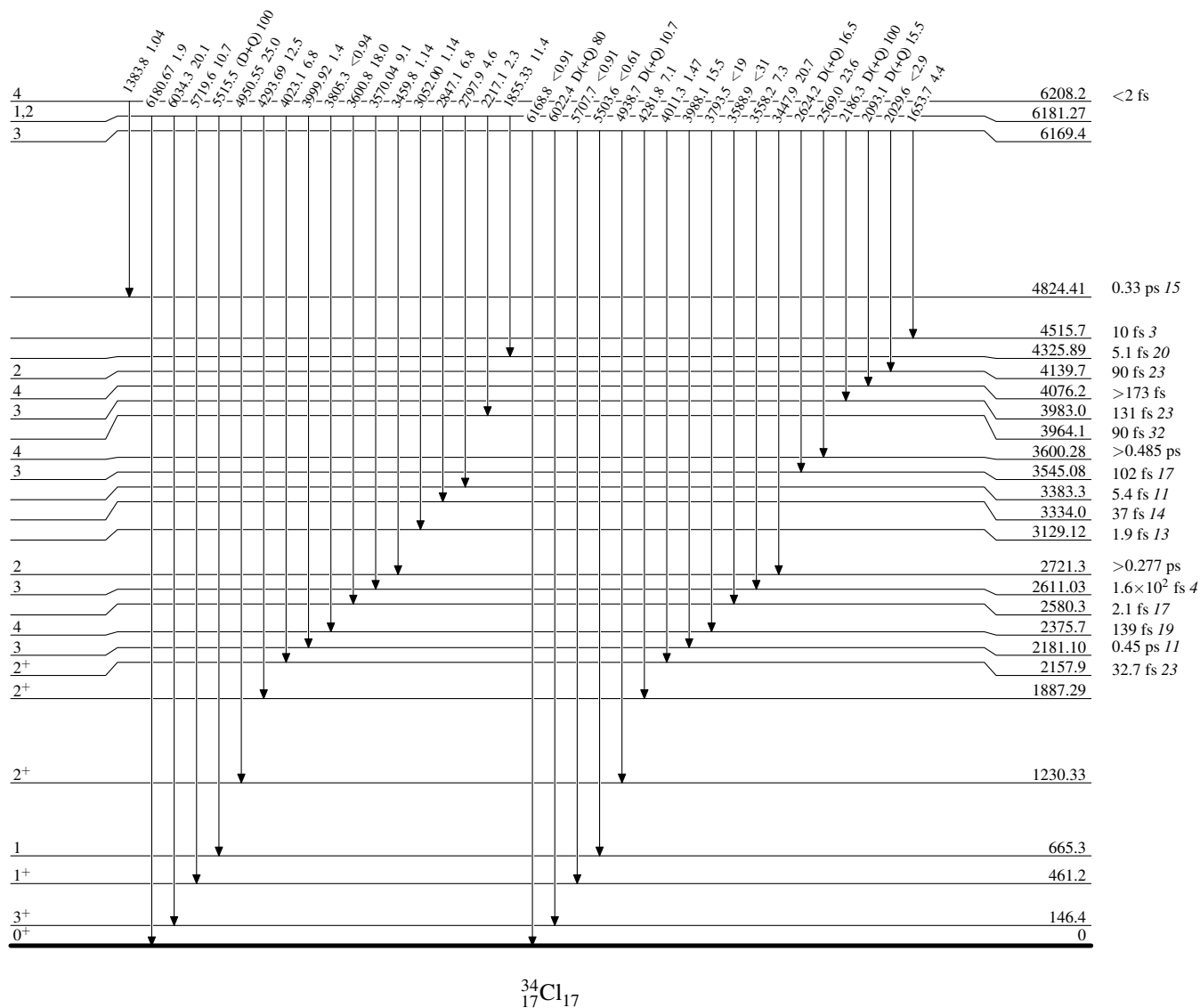
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

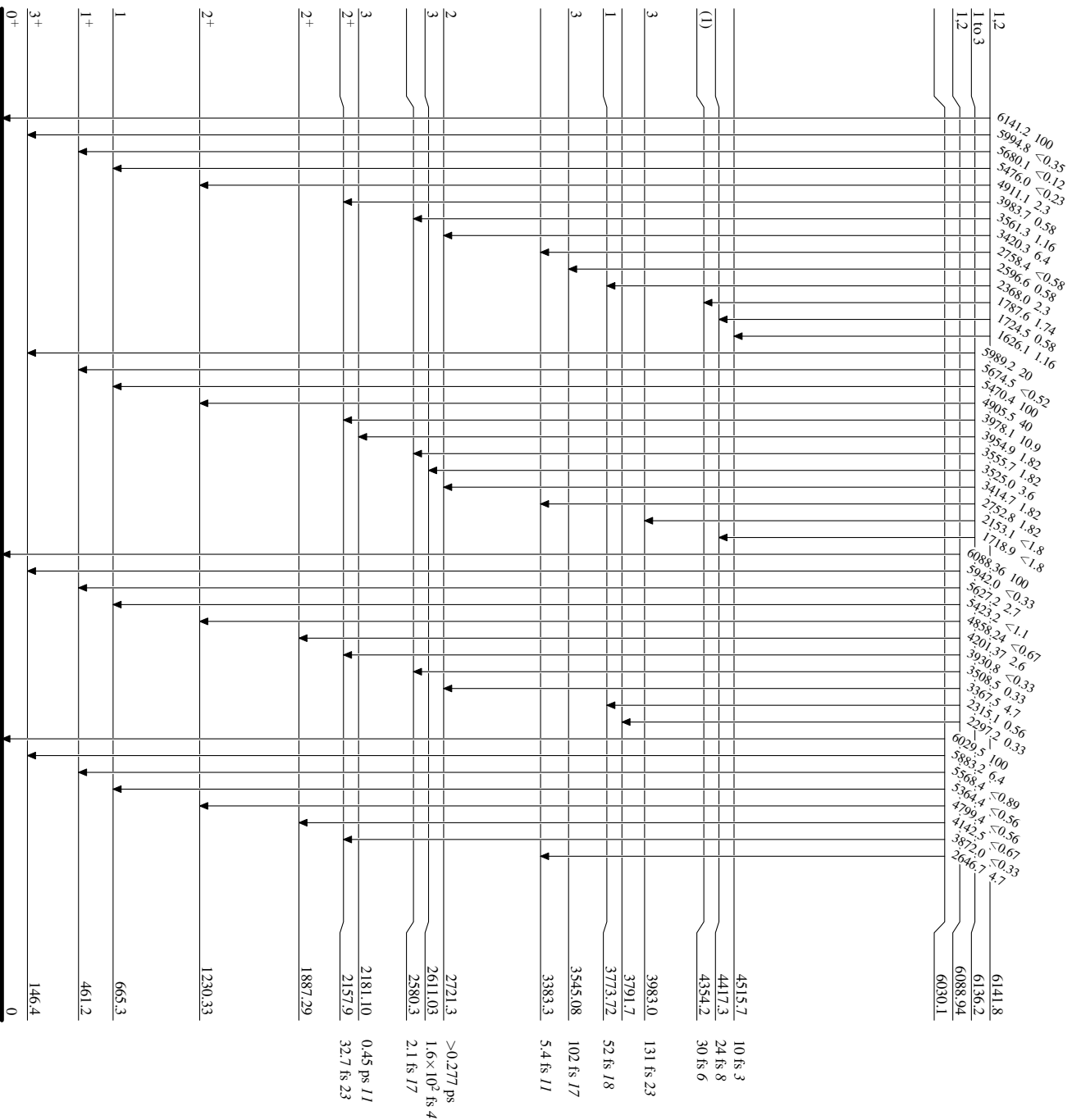
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

³³Si(p,γ) 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

Intensities: Relative photon branching from each level

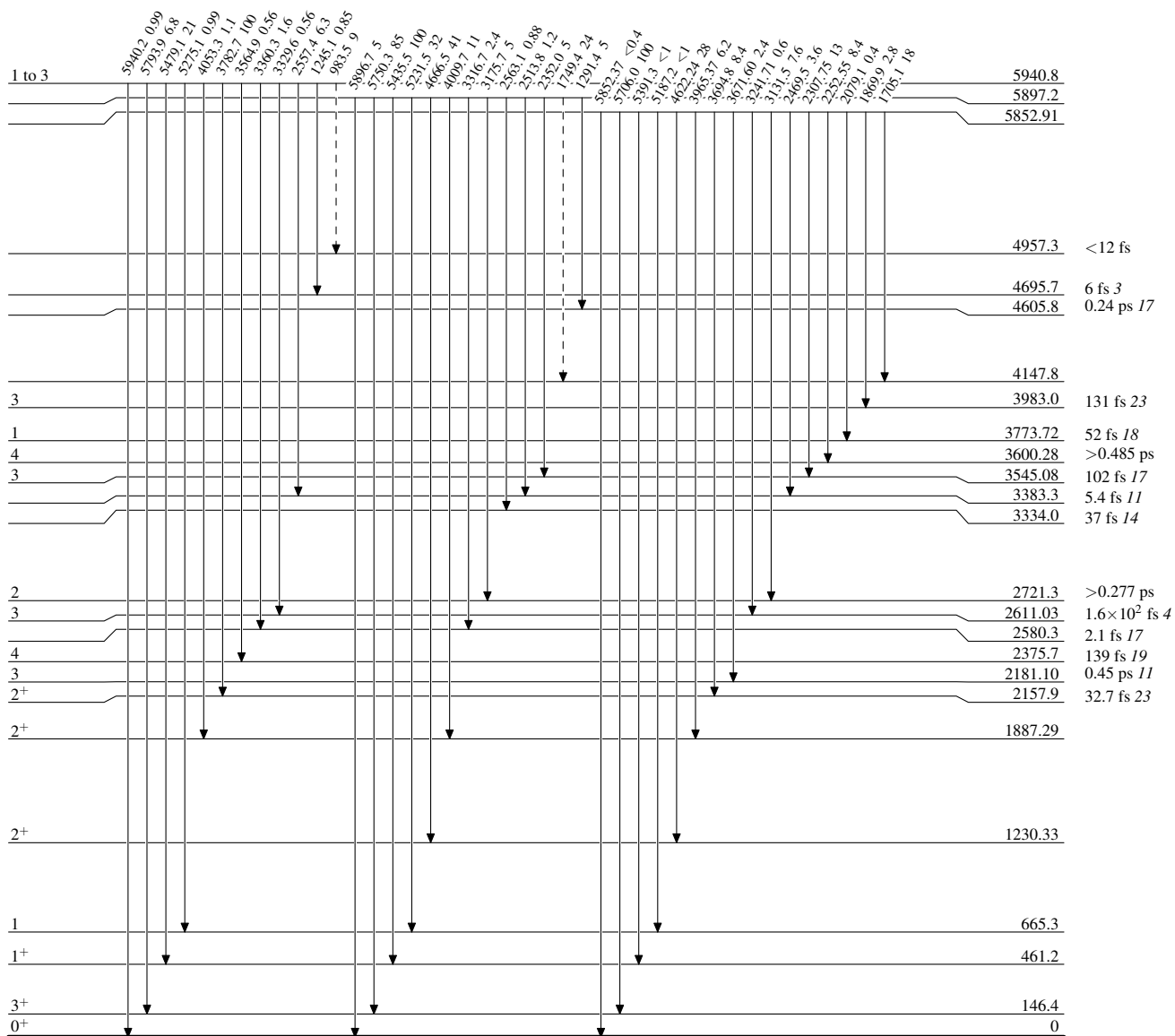


$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

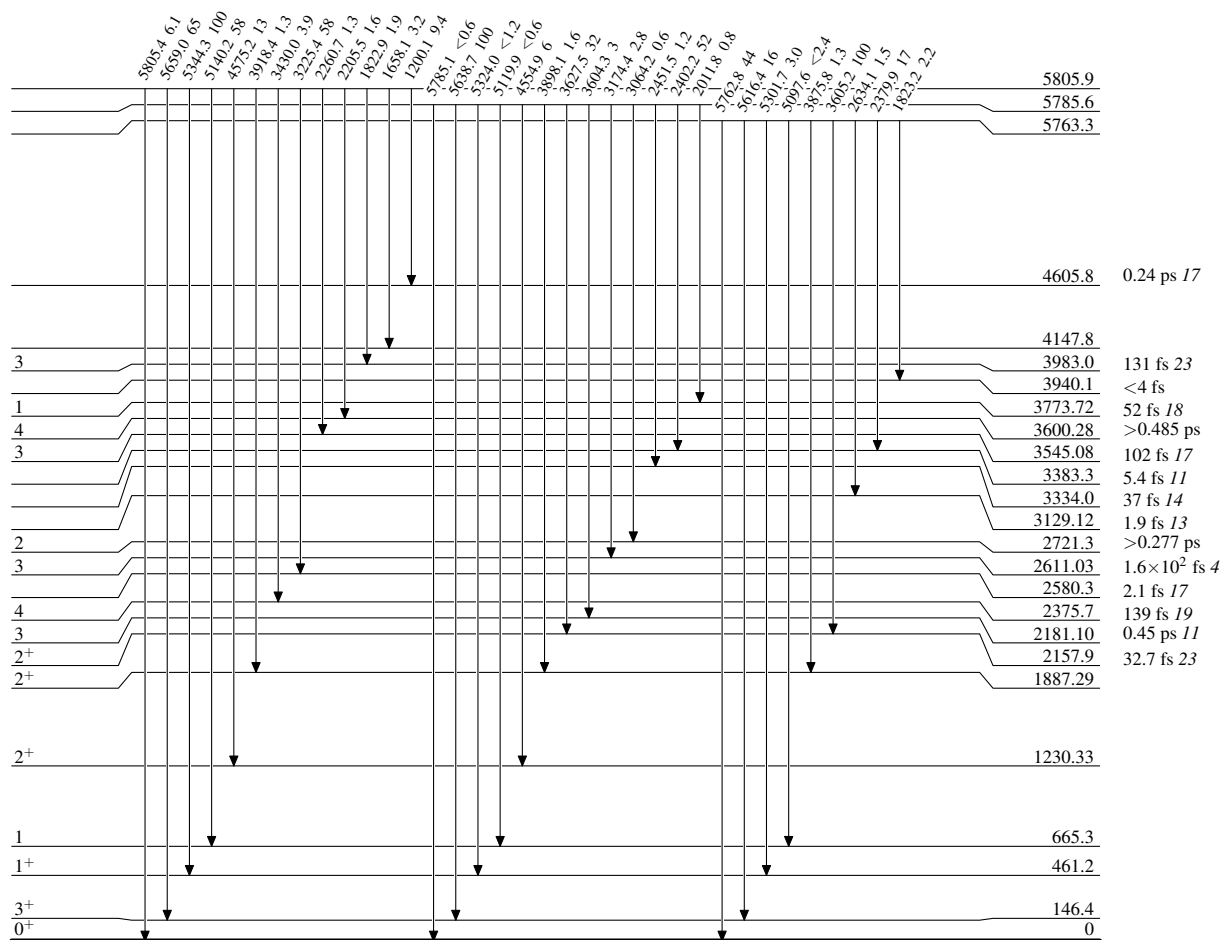
Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

Intensities: Relative photon branching from each level

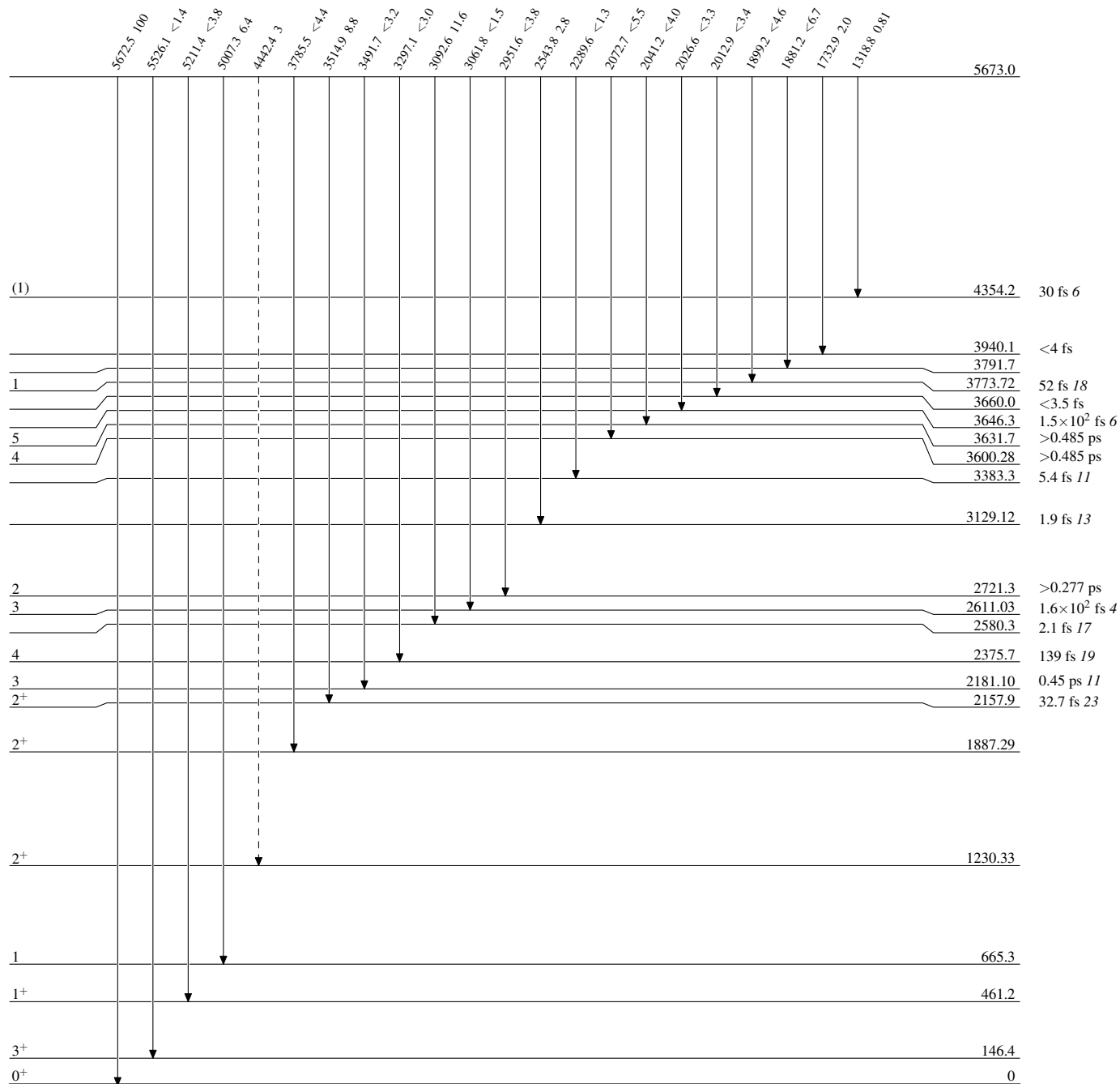
 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

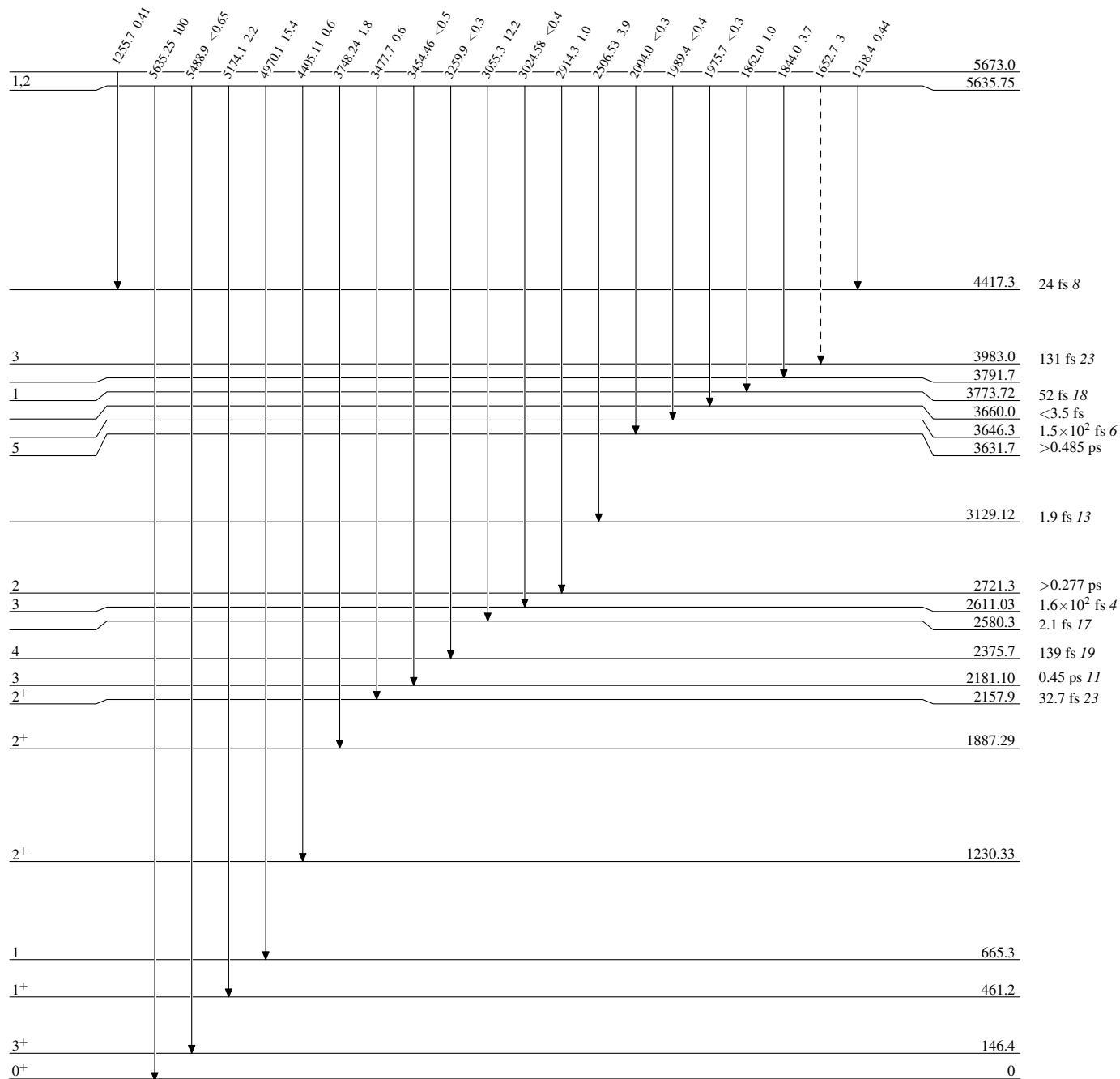
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

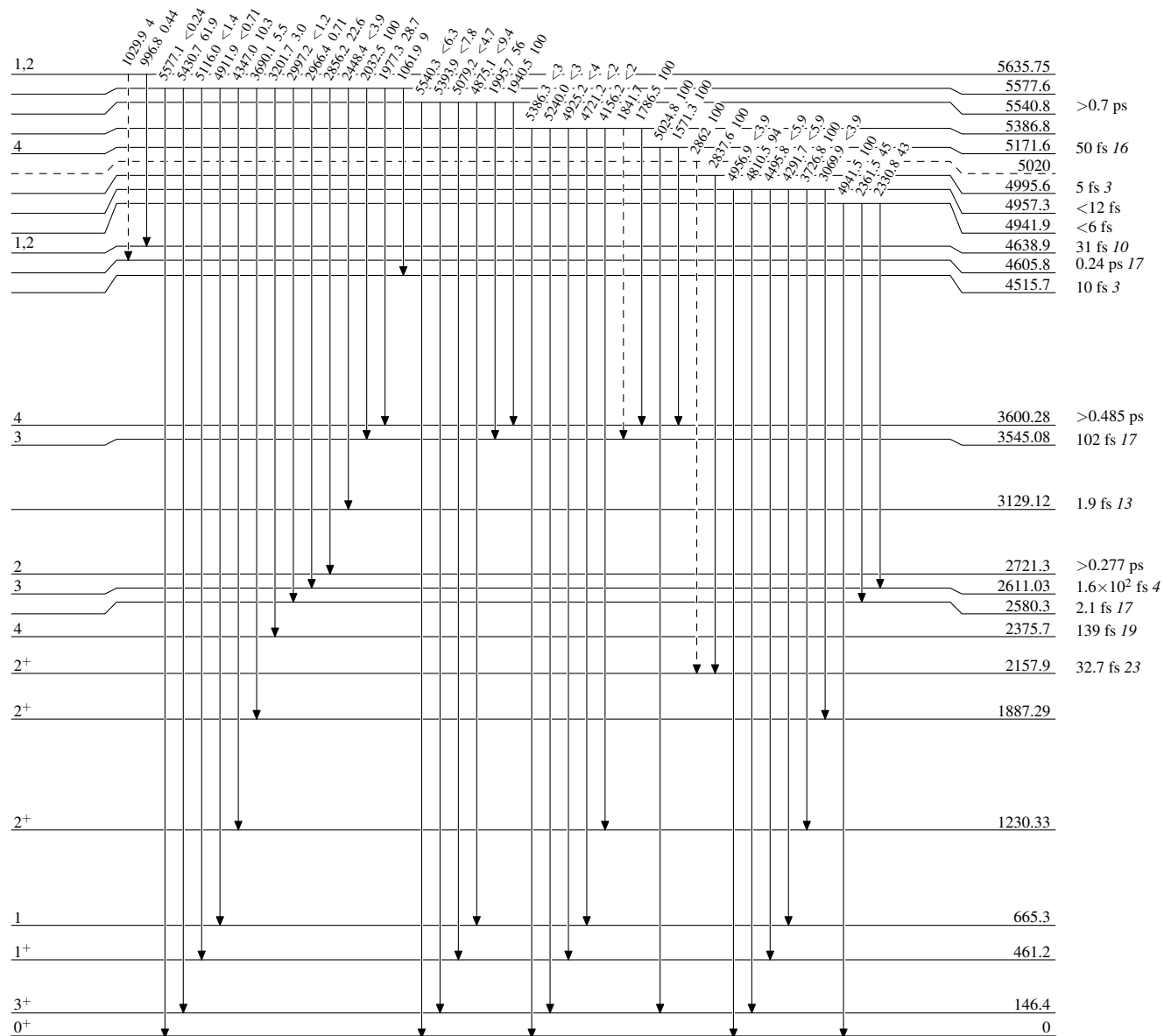
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

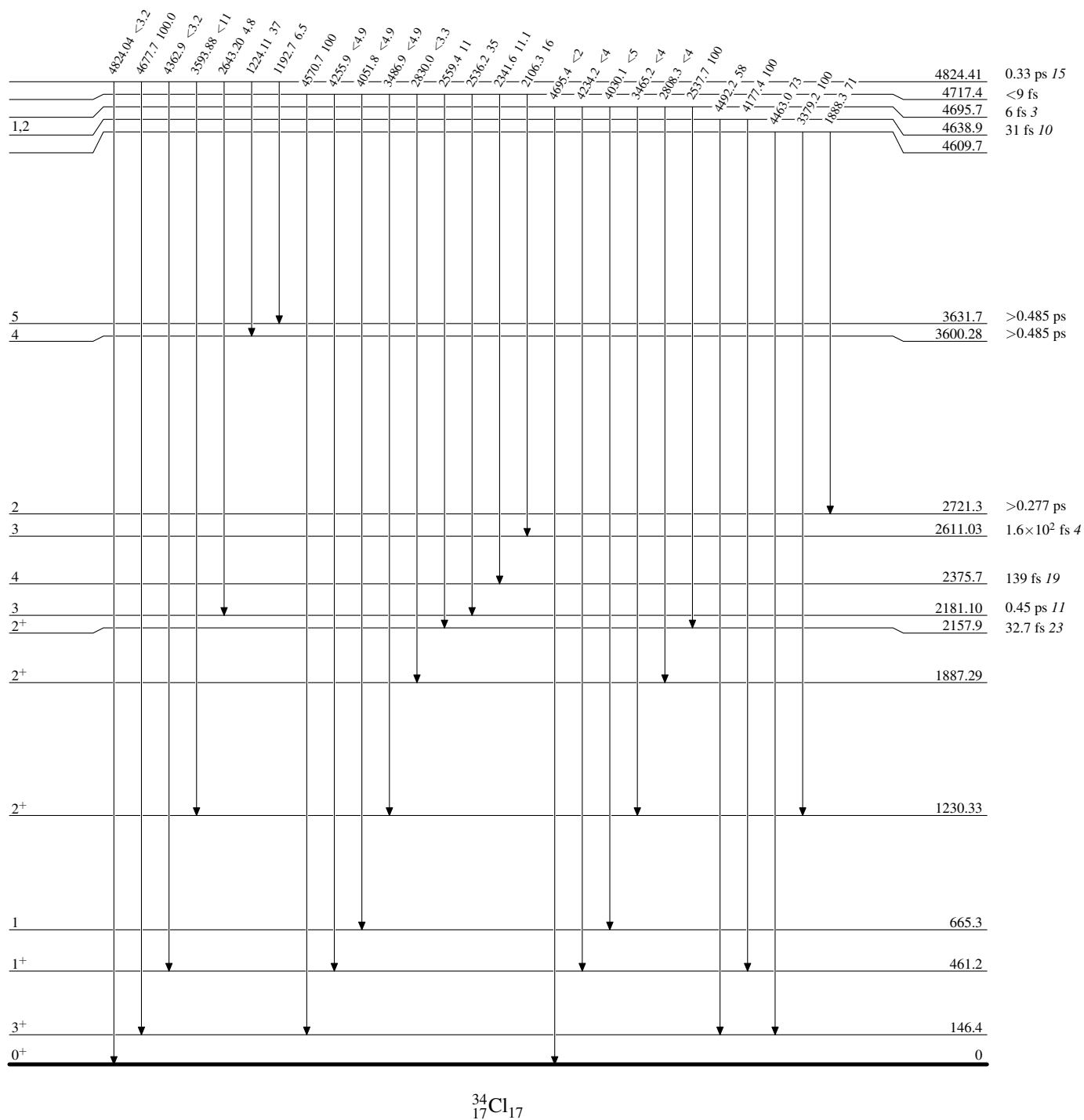
Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

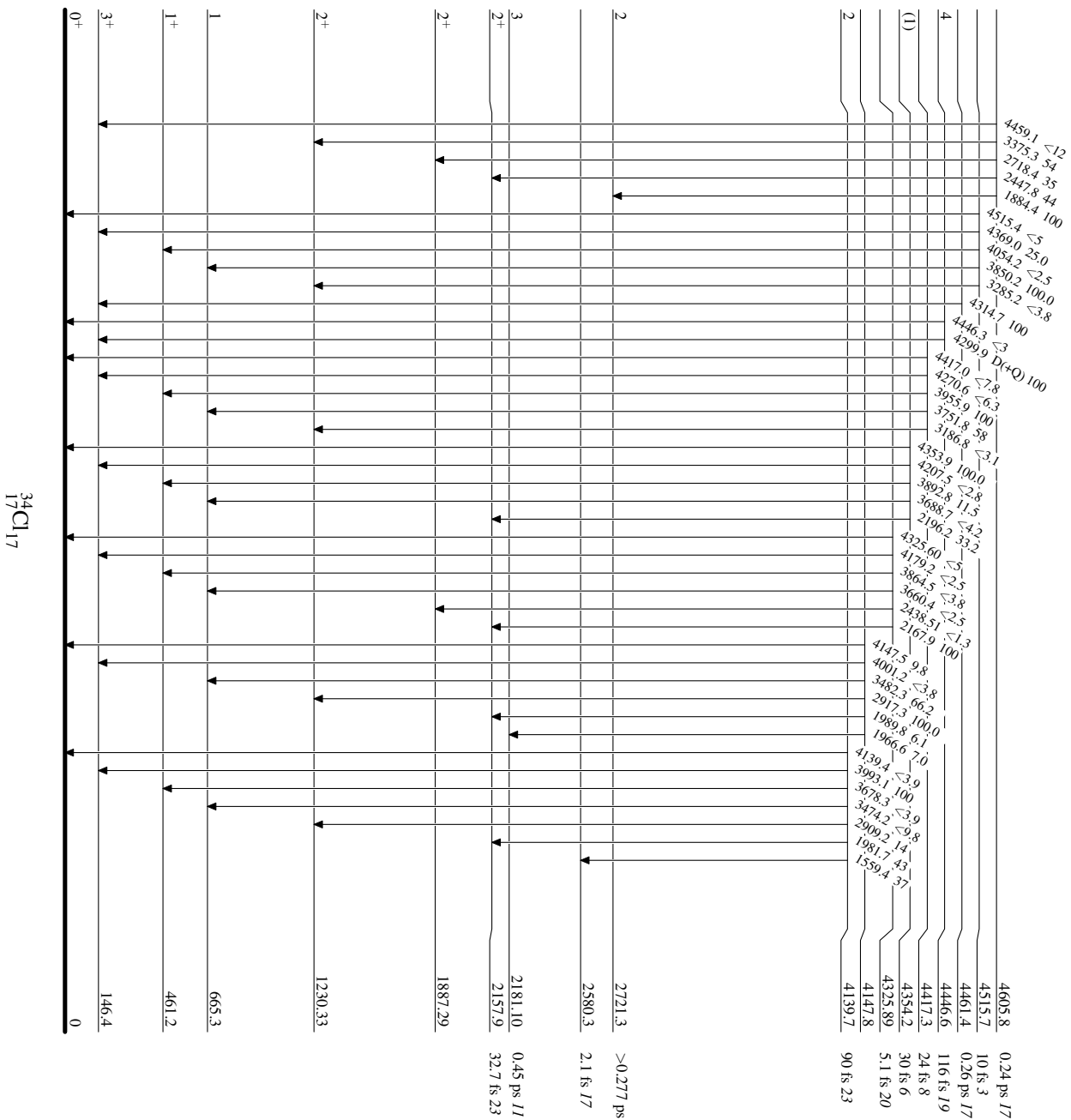
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

³³S(p,γ) 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

Intensities: Relative photon branching from each level



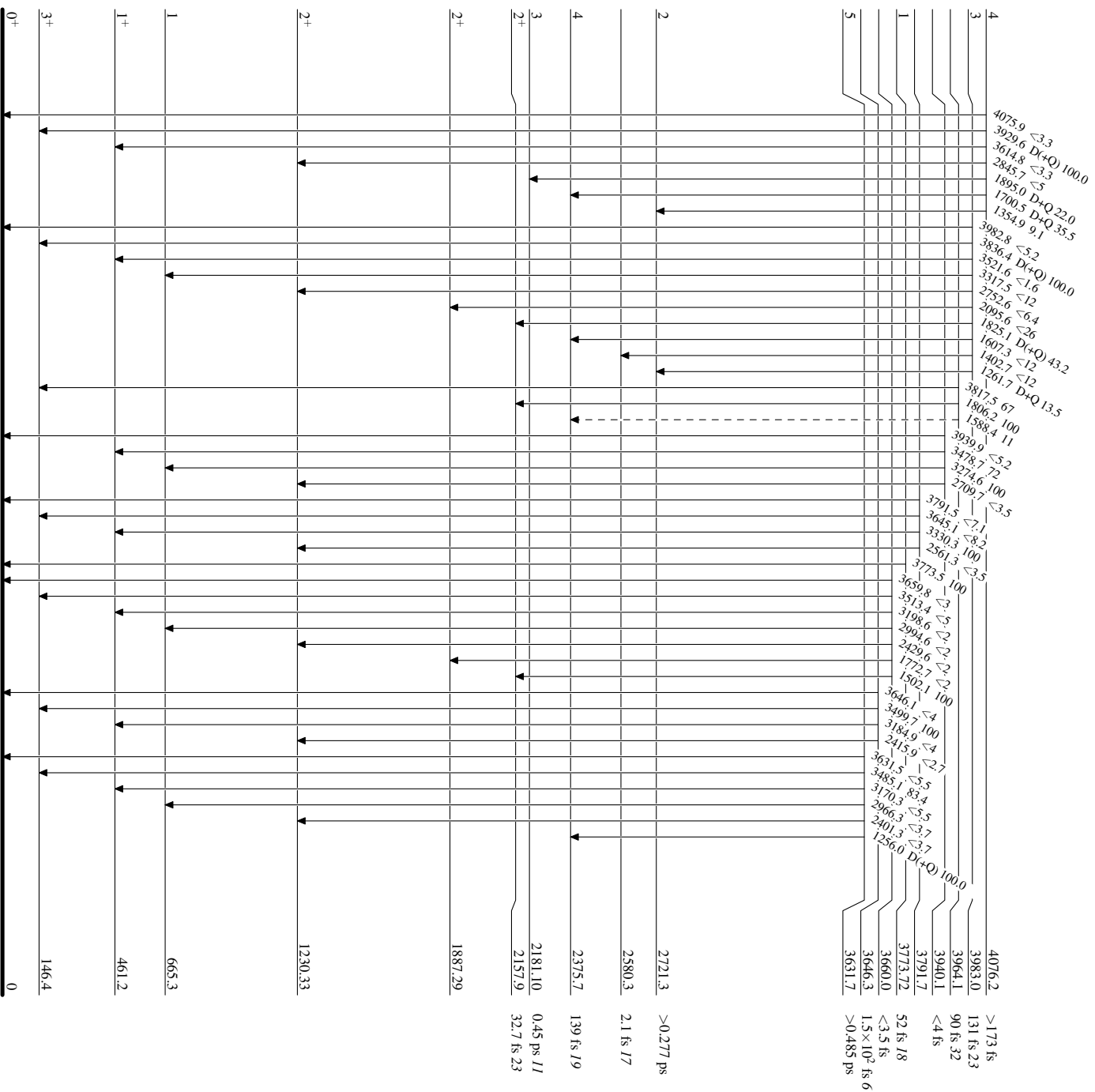
³³S(p,γ) 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----▶ γ Decay (Uncertain)



³⁴Cl₁₇

³³S(p,γ) 1983Wa27,1977Da02,1992Ka39

Level Scheme (continued)

Intensities: Relative photon branching from each level

